AI Business Management System Blueprint

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Blueprint Build Sequence

We will build this document one section at a time so the system has a solid foundation before we move into detailed design, features, architecture, and implementation.

Step 1 — Vision, Purpose, and Design Principles

Define what the system is, why it exists, what problems it solves, and what principles guide every design decision.

Step 2 — Business Goals and Success Metrics

Define the measurable business outcomes the system must improve.

Step 3 — Core Business Areas / Modules

Define the major areas of the company the system will manage.

Step 4 — Core Business Objects

Define the main records/entities the system will track, such as customers, products, orders, inventory, batches, invoices, documents, and tasks.

Step 5 — Data Model and Relationships

Define how the core objects connect to each other.

Step 6 — User Roles and Permission Model

Define who uses the system, what they can see, and what they can do.

Step 7 — Key Workflows

Define the major business processes the software must support.

Step 8 — AI Roles and Capabilities

Define where AI assists, analyzes, monitors, recommends, and automates.

Step 9 — Human Approval and Control Rules

Define what AI can do automatically and what requires human approval.

Step 10 — Dashboards and Reporting

Define the command center, daily briefing, alerts, and management reports.

Step 11 — Technical Architecture

Define the application structure, database, integrations, AI layer, security, and infrastructure.

Step 12 — MVP Scope

Define the first buildable version of the system.

Step 13 — Future Roadmap

Define later phases and advanced capabilities.

Step 1 — Vision, Purpose, and Design Principles

1.1 Product Vision

The AI Business Command Center is a centralized business management system designed to help leadership run the company with greater visibility, control, accountability, and speed.

The system will combine structured business data, operational workflows, document intelligence, task management, reporting, and AI-assisted decision support into one unified platform.

Its purpose is not merely to store information. Its purpose is to help the business understand what is happening, identify what needs attention, recommend what should happen next, and track whether the right actions are completed.

The long-term vision is to create the central nervous system of the company: a platform that connects sales, customers, orders, production, inventory, finance, compliance, legal, documents, people, and strategic projects into one intelligent operating environment.

1.2 System Purpose

The system exists to help company leadership answer the most important management questions quickly and accurately:

- What needs my attention today?
- Are we on track financially, operationally, and strategically?
- Which customers, orders, invoices, production runs, inventory items, documents, or deadlines are at risk?
- What tasks are overdue or blocked?
- Who is responsible for each important action?
- What decisions do I need to make?
- What should the business do next?
- Where are we losing time, money, margin, or opportunity?
- What risks are building before they become expensive problems?

The system should reduce the need for leadership to hunt through emails, spreadsheets, text messages, accounting systems, documents, and employee updates to understand the state of the business.

Instead, the system should provide a clear, current, and actionable view of the company.

1.3 Core Management Philosophy

The system should be built around a continuous business management loop:

Capture Data → Understand Status → Identify Risk → Recommend Action → Assign Responsibility → Track Completion → Learn From Results

Every module and feature should support this loop.

For example:

- A customer follow-up becomes a task.
- A delayed order becomes an alert.
- A low inventory item becomes a purchase recommendation.
- A production issue becomes a corrective action.
- A contract obligation becomes a compliance deadline.
- A missed payment becomes a collections priority.
- A meeting note becomes an accountability item.

The system should convert business information into action.

1.4 Primary Business Problems the System Should Solve

The system is designed to solve the following business problems:

Lack of Centralized Visibility

Important business information is often spread across multiple places. The system should create one reliable source of operational truth.

Missed Follow-Ups and Delayed Action

Customer follow-ups, vendor issues, invoice collections, production tasks, compliance deadlines, and internal commitments can slip through the cracks. The system should track responsibility and deadlines clearly.

Reactive Management

Many businesses only discover problems after they become urgent. The system should monitor early warning signals and alert leadership before issues become costly.

Manual Reporting Burden

Leadership should not have to manually gather information every day or every week to understand the state of the business. The system should generate daily and weekly management summaries automatically.

Poor Connection Between Data and Decisions

Data alone does not manage a company. The system should translate data into insights, recommendations, and next actions.

Weak Accountability

Every important action should have an owner, due date, status, priority, and related business context.

Document and Compliance Risk

Contracts, SOPs, COAs, licenses, patent matters, compliance obligations, and other important documents should be searchable, summarized, connected to business records, and monitored for deadlines or obligations.

1.5 Design Principles

The following principles should guide every decision in the design and development of the system.

1. Action-Oriented

The system should not just display information. It should help users decide and act.

Every alert, insight, or AI recommendation should be capable of becoming an assigned task, approval request, follow-up, document, order, purchase order, or report.

2. One Source of Truth

The system should centralize the most important business records and relationships. When possible, structured records should be used instead of disconnected notes or files.

3. AI-Assisted, Human-Controlled

AI should assist, analyze, summarize, monitor, recommend, and draft. However, high-impact actions should require human review and approval.

AI should not independently make legal, financial, compliance, production, or customer-facing commitments without permission.

4. Built Around Business Objects

The system should be designed around real business objects such as customers, vendors, products, orders, inventory items, batches, invoices, documents, tasks, projects, and compliance requirements.

This makes the platform scalable and prevents it from becoming a collection of disconnected tools.

5. Relationship-Driven

The value of the system comes from connecting records together.

For example, a customer connects to orders, invoices, documents, communications, tasks, and revenue. A product connects to materials, batches, inventory, sales orders, margins, compliance requirements, and documents.

The system should make these relationships visible and useful.

6. Risk-Aware

The system should continuously identify risk across the business, including cash risk, inventory risk, customer risk, production risk, compliance risk, legal risk, vendor risk, and personnel accountability risk.

7. Permission-Based

Users should only see and do what their role allows. AI access must follow the same permissions as the user asking the question.

8. Auditable

Important actions, AI-generated recommendations, approvals, record changes, and user activity should be logged.

The system should be able to answer: who did what, when, why, and based on what information?

9. Modular and Scalable

The system should be built in modules so the company can start with a strong MVP and expand over time.

Each module should be useful on its own but more powerful when connected to the rest of the platform.

10. Designed for Daily Use

The system should be practical and easy to use every day. The command center and daily briefing should become part of leadership's daily operating rhythm.

1.6 Long-Term System Outcome

When fully developed, the AI Business Command Center should allow leadership to manage the company from a single intelligent platform.

Leadership should be able to open the system and immediately understand:

- What is happening
- What has changed
- What is at risk
- What needs approval
- What needs follow-up
- What is overdue
- Who owns each priority
- What AI recommends doing next

The end goal is a system that improves decision-making, reduces missed opportunities, strengthens accountability, improves operational control, protects the business from avoidable risks, and gives leadership more time to focus on growth.

Step 2 — Business Goals and Success Metrics

2.1 Purpose of This Section

The AI Business Command Center should be judged by measurable business improvement, not by how many features it has.

This section defines the business outcomes the system must improve and the metrics that will prove whether the system is working.

The goal is to create a system that helps the company:

- Make faster decisions
- Catch problems earlier
- Improve follow-through
- Increase visibility
- Reduce missed opportunities
- Protect cash flow
- Improve operational control
- Strengthen accountability
- Reduce compliance and document risk
- Improve profitability awareness

Every major feature should connect back to at least one business goal or success metric.

2.2 Primary Business Goals

Goal 1 — Improve Daily Leadership Visibility

Leadership should be able to understand the current state of the business without manually checking multiple systems, spreadsheets, emails, documents, and employees.

The system should provide a clear daily view of what is happening, what changed, what is at risk, and what needs action.

Success Metrics

- Daily briefing available automatically
- Number of critical business areas visible from the command center
- Reduction in time spent manually gathering updates
- Number of unresolved alerts reviewed each day
- Number of decisions supported by system data

Target Outcome

Leadership should be able to understand the most important business priorities within minutes of opening the system.

Goal 2 — Improve Task Accountability and Follow-Through

The system should ensure that important work does not get lost, forgotten, or delayed without visibility.

Every important action should have an owner, due date, priority, status, and related business context.

Success Metrics

- Number of open tasks
- Number of overdue tasks
- Average task completion time
- Percentage of tasks with assigned owners
- Percentage of tasks linked to a customer, order, batch, document, invoice, project, or compliance item
- Number of blocked tasks
- Number of tasks created from meetings, emails, alerts, or AI recommendations

Target Outcome

Reduce missed follow-ups, unclear ownership, and delayed execution by making accountability visible and trackable.

Goal 3 — Improve Customer and Sales Follow-Up

The system should help the company manage leads, prospects, customers, quotes, and sales opportunities more consistently.

No valuable lead, customer conversation, quote, or follow-up should disappear into email, text messages, memory, or disconnected notes.

Success Metrics

- Number of active leads
- Number of open opportunities
- Number of stale leads with no recent activity
- Average time between customer touchpoints
- Number of follow-ups due today
- Quote turnaround time
- Quote-to-order conversion rate
- Sales pipeline value

- Lost opportunity reasons

Target Outcome

Increase revenue discipline by making every customer relationship and sales opportunity visible, prioritized, and actively managed.

Goal 4 — Improve Cash Flow Visibility and Financial Awareness

The system should help leadership understand cash position, receivables, payables, expected inflows, expected outflows, and financial risk.

The system does not replace accounting software, but it should provide management-level financial visibility and decision support.

Success Metrics

- Current cash position visibility
- Open receivables total
- Open payables total
- Number and value of overdue invoices
- Days sales outstanding
- Expected cash position over 30, 60, and 90 days
- Gross margin by product, customer, or order
- Number of financial alerts generated
- Time required to prepare weekly financial summary

Target Outcome

Leadership should be able to see short-term cash risks, collections priorities, and profitability signals before they become urgent problems.

Goal 5 — Improve Inventory Planning and Purchasing Control

The system should help prevent stockouts, overbuying, expired materials, missing packaging, delayed vendor orders, and poor purchasing visibility.

Inventory should be connected to products, production needs, sales orders, vendors, purchase orders, lead times, and reorder points.

Success Metrics

- Current inventory accuracy
- Number of low-stock alerts
- Number of stockouts

- Number of items below reorder point
- Inventory days on hand
- Vendor lead-time accuracy
- Purchase orders pending or overdue
- Inventory value by category
- Expired or aging inventory items
- Number of production delays caused by missing materials

Target Outcome

Reduce production interruptions and cash waste by improving visibility into what the company has, what it needs, when it needs it, and what should be purchased.

Goal 6 — Improve Production and Operations Control

The system should help leadership and operations teams track production activity, batch status, work orders, yield, quality checks, delays, and operational bottlenecks.

Production should be tied to customer orders, inventory availability, labor, quality control, and finished goods.

Success Metrics

- Number of active production runs
- Batch completion time
- On-time production percentage
- Yield by batch
- Waste or rework rate
- Number of production delays
- Number of quality-control issues
- Labor hours by batch or work order
- Cost variance by batch
- Orders at risk due to production status

Target Outcome

Improve production reliability, reduce operational surprises, and connect production activity directly to customer demand and profitability.

Goal 7 — Improve Compliance, Legal, and Document Control

The system should help the company manage important documents, deadlines, obligations, licenses, COAs, SOPs, contracts, patent-related matters, and compliance requirements.

The system should make documents searchable, summarized, versioned, linked to business records, and monitored for deadlines or obligations.

Success Metrics

- Number of documents stored by type
- Percentage of documents linked to relevant business records
- Number of upcoming compliance deadlines
- Number of missing or expired documents
- Number of contracts with extracted obligations
- Number of COAs linked to products, batches, or inventory
- Number of SOPs with assigned owners and review dates
- Number of legal/IP deadlines tracked
- Time required to find a key document

Target Outcome

Reduce document chaos, missed deadlines, and compliance exposure by turning important files into structured, searchable, actionable business records.

Goal 8 — Improve Decision Speed and Quality

The system should help leadership make better decisions faster by combining reliable data, business context, AI analysis, and recommended next actions.

The system should reduce decision-making based on memory, incomplete information, or scattered updates.

Success Metrics

- Time required to answer common management questions
- Number of AI-assisted analyses generated
- Number of decisions supported by linked source data
- Number of recommendations accepted, rejected, or modified
- Accuracy of AI recommendations over time
- Number of recurring reports automated
- Reduction in manual data gathering before decisions

Target Outcome

Leadership should be able to ask the system management-level questions and receive clear, sourced, actionable answers.

Goal 9 — Improve Risk Detection

The system should continuously identify risks before they become expensive, urgent, or damaging.

Risk should be monitored across sales, finance, production, inventory, vendors, compliance, legal, documents, and team execution.

Success Metrics

- Number of risks detected by category
- Number of unresolved high-priority risks
- Average time from risk detection to assignment
- Average time from assignment to resolution
- Number of risks converted into tasks
- Number of repeated risks by category
- Number of risks escalated to leadership

Target Outcome

Shift the business from reactive problem-solving to proactive risk management.

Goal 10 — Improve Management Reporting

The system should automatically generate useful business summaries for daily, weekly, monthly, and quarterly management review.

Reports should not merely show raw numbers. They should explain what changed, what matters, what is at risk, and what actions are recommended.

Success Metrics

- Daily briefing generated
- Weekly leadership report generated
- Monthly performance report generated
- Number of reports created automatically
- Time saved preparing reports
- Number of report insights converted into tasks
- Number of recurring KPIs tracked

Target Outcome

Leadership should receive consistent management reporting without relying on manual spreadsheet work or scattered updates.

2.3 Key Performance Indicator Categories

The system should track KPIs across the following categories.

Sales KPIs

- Active leads
- Open opportunities
- Pipeline value
- Quote volume
- Quote turnaround time
- Conversion rate
- Revenue by customer
- Revenue by product
- Stale opportunities
- Customer follow-ups due

Finance KPIs

- Cash position
- Receivables
- Payables
- Overdue invoices
- Gross margin
- Revenue trend
- Expense trend
- Profitability by product/customer/order
- 30/60/90-day cash forecast

Operations KPIs

- Active orders
- Orders at risk
- Production runs in progress
- Batch completion rate
- Yield
- Waste
- Rework
- On-time fulfillment
- Production bottlenecks

Inventory KPIs

- Inventory value
- Low-stock items
- Stockouts
- Reorder alerts
- Items expiring soon

- Vendor lead times
- Purchase orders in transit
- Materials required for upcoming production

Compliance and Document KPIs

- Upcoming deadlines
- Expired documents
- Missing documents
- Contracts needing review
- SOPs needing review
- COAs linked to batches/products
- License renewal status
- Patent/IP milestone status

Accountability KPIs

- Open tasks
- Overdue tasks
- Blocked tasks
- Tasks by owner
- Tasks by department
- Average completion time
- High-priority unresolved tasks
- AI-generated tasks awaiting approval

2.4 System-Wide Success Measures

The system should be considered successful if it produces measurable improvement in the following areas:

Visibility

Leadership can see the current status of the business from one command center.

Speed

Common management questions can be answered quickly without manual information gathering.

Accountability

Important tasks have owners, due dates, priorities, and status tracking.

Risk Reduction

Potential problems are identified earlier and assigned for resolution.

Cash Awareness

Leadership has better visibility into receivables, payables, and short-term cash risk.

Sales Discipline

Customer follow-ups, quotes, and opportunities are tracked consistently.

Operational Control

Production, inventory, and purchasing are connected to actual business needs.

Compliance Confidence

Important documents, deadlines, and obligations are tracked and searchable.

Decision Support

AI helps leadership understand tradeoffs, priorities, and recommended actions.

Scalability

The system structure can support more users, more modules, more data, and more automation over time.

2.5 Initial MVP Success Criteria

The first version of the system should be considered successful if it can do the following:

1. Provide a daily leadership briefing.
2. Track customers, vendors, products, inventory, orders, documents, and tasks.
3. Show overdue and upcoming tasks by owner.
4. Identify stale customer follow-ups.
5. Identify low inventory or purchasing risks.
6. Summarize open receivables and payables.
7. Store and summarize important documents.
8. Link documents to customers, vendors, products, batches, or compliance requirements.
9. Allow AI-generated recommendations to become tasks.
10. Maintain an activity log of important actions and AI outputs.

The MVP should not attempt to automate everything. Its first job is to create visibility, structure, accountability, and decision support.

2.6 Measurement Cadence

The system should support different review rhythms.

Daily Review

Focused on immediate action.

Includes:

- Today’s priorities
- Overdue tasks
- Customer follow-ups
- Order risks
- Inventory risks
- Cash alerts
- Production issues
- Compliance deadlines

Weekly Review

Focused on operating performance.

Includes:

- Sales pipeline
- Revenue progress
- Cash position
- Receivables/payables
- Production output
- Inventory status
- Team accountability
- Major risks

Monthly Review

Focused on business performance.

Includes:

- Revenue trends
- Margin trends
- Product/customer profitability
- Operational efficiency
- Sales conversion
- Inventory value
- Compliance status

- Project progress

Quarterly Review

Focused on strategic direction.

Includes:

- Business growth
- Profitability
- Strategic projects
- Hiring/resource needs
- Product roadmap
- Vendor performance
- Risk trends
- System performance and automation opportunities

2.7 Guiding Rule for New Features

Every new feature should be evaluated against this question:

Does this feature improve visibility, decision-making, accountability, cash control, operational control, compliance confidence, customer follow-through, or profitability?

If the answer is no, the feature should not be prioritized.

Step 3 — Core Business Areas / Modules

3.1 Purpose of This Section

This section defines the major business areas the AI Business Command Center must manage.

Each module should be useful on its own, but the real value of the system comes from connecting the modules together.

For example:

- Sales connects to orders.
- Orders connect to production.
- Production connects to inventory.
- Inventory connects to purchasing.
- Purchasing connects to vendors and cash flow.
- Invoices connect to customers, orders, and finance.

- Documents connect to compliance, products, customers, vendors, and batches.
- Tasks connect to every module.
- AI sits across the entire system.

The modules should not become isolated silos. They should work together as one operating system.

3.2 Module Overview

The initial system should be organized around the following core modules:

1. Command Center
2. Customers and Sales
3. Orders and Quotes
4. Production and Batches
5. Inventory and Purchasing
6. Finance and Cash Flow
7. Compliance, Legal, and IP
8. Documents and Knowledge Base
9. Tasks and Accountability
10. Projects and Strategic Initiatives
11. People and Roles
12. AI Assistant and Automation Layer
13. Reporting and Analytics
14. Settings, Permissions, and Audit Logs

3.3 Module 1 — Command Center

Purpose

The Command Center is the main leadership dashboard and the daily starting point for managing the business.

It should give leadership a clear view of what is happening, what changed, what is at risk, and what requires action.

The Command Center should not simply show charts. It should prioritize attention.

Tracks and Displays

- Daily leadership briefing
- Today’s top priorities
- Open alerts
- Overdue tasks
- Tasks waiting on leadership

- Sales follow-ups due
- Orders at risk
- Production issues
- Low inventory warnings
- Purchase order delays
- Cash-flow concerns
- Overdue invoices
- Compliance deadlines
- Document issues
- AI recommendations

AI Support

AI should generate:

- Daily executive summary
- Top risk summary
- Recommended actions
- Priority ranking
- Follow-up suggestions
- Leadership decision queue
- Plain-English explanation of business status

Key Questions Answered

- What needs my attention today?
- What is at risk?
- What changed since yesterday?
- What decisions do I need to make?
- Who is waiting on me?
- What should I follow up on first?

Connected Modules

The Command Center connects to every module in the system.

3.4 Module 2 — Customers and Sales

Purpose

This module manages relationships with leads, prospects, customers, contacts, sales opportunities, and follow-ups.

Its purpose is to make sure revenue opportunities are visible, prioritized, and consistently followed up on.

Tracks and Displays

- Leads
- Prospects
- Customers
- Contacts
- Customer status
- Sales stage
- Lead source
- Customer notes
- Communication history
- Follow-up dates
- Open opportunities
- Stale opportunities
- Customer issues
- Customer documents
- Customer lifetime value
- Revenue by customer

AI Support

AI should help with:

- Lead prioritization
- Follow-up reminders
- Drafting customer emails
- Summarizing customer history
- Identifying stale opportunities
- Suggesting next best action
- Analyzing lost opportunities
- Drafting proposals or call notes

Key Questions Answered

- Who needs to be contacted today?
- Which opportunities are going stale?
- What was the last interaction with this customer?
- Which customers are most valuable?
- Which customers are at risk?
- What should we say in the next follow-up?

Connected Modules

This module connects to:

- Orders and Quotes
- Finance and Cash Flow
- Documents

- Tasks and Accountability
- AI Assistant
- Reporting and Analytics

3.5 Module 3 — Orders and Quotes

Purpose

This module manages the commercial flow from quote to order to fulfillment to invoice.

It should connect customer demand to production, inventory, shipping, and finance.

Tracks and Displays

- Quotes
- Quote status
- Quote expiration dates
- Quote approval status
- Sales orders
- Order status
- Order due dates
- Products ordered
- Quantities
- Pricing
- Discounts
- Linked customer
- Linked production batch
- Linked invoice
- Fulfillment status
- Shipping status

AI Support

AI should help with:

- Drafting quotes
- Checking pricing or margin concerns
- Identifying orders at risk
- Summarizing order status
- Suggesting production needs
- Drafting customer updates
- Converting approved quotes into orders
- Highlighting orders blocked by inventory or production issues

Key Questions Answered

- Which quotes need approval?
- Which quotes are about to expire?
- Which orders are due soon?
- Which orders are at risk?
- Do we have inventory to fulfill this order?
- Does this order require production?
- Has this order been invoiced?

Connected Modules

This module connects to:

- Customers and Sales
- Production and Batches
- Inventory and Purchasing
- Finance and Cash Flow
- Documents
- Tasks and Accountability

3.6 Module 4 — Production and Batches

Purpose

This module manages manufacturing activity, production planning, batch records, quality checks, yield, labor, waste, and operational issues.

Its purpose is to connect customer demand, inventory availability, production capacity, quality control, and finished goods.

Tracks and Displays

- Work orders
- Production schedules
- Batch IDs
- Batch status
- Product being made
- Formula or specification reference
- Input materials
- Output quantity
- Yield
- Waste
- Rework
- Labor hours
- Equipment used

- Quality-control status
- Deviations or issues
- COAs or lab documents
- Responsible person
- Completion dates

AI Support

AI should help with:

- Production planning recommendations
- Batch issue summaries
- Yield analysis
- Cost variance analysis
- Identifying bottlenecks
- Flagging missing materials
- Drafting production notes
- Creating corrective action tasks
- Summarizing production performance

Key Questions Answered

- What is currently in production?
- Which batches are delayed?
- What materials are required?
- What is the yield for this batch?
- Were there quality issues?
- Which orders depend on this batch?
- What caused a production delay?

Connected Modules

This module connects to:

- Orders and Quotes
- Inventory and Purchasing
- Products
- Documents
- Compliance, Legal, and IP
- Finance and Cash Flow
- Tasks and Accountability

3.7 Module 5 — Inventory and Purchasing

Purpose

This module manages raw materials, packaging, finished goods, inventory levels, reorder points, purchase orders, vendors, and receiving.

Its purpose is to prevent stockouts, reduce waste, improve purchasing decisions, and connect inventory to actual business demand.

Tracks and Displays

- Raw materials
- Packaging
- Finished goods
- Inventory locations
- Current quantities
- Reorder points
- Units of measure
- Lot numbers
- Expiration dates
- Inventory value
- Purchase orders
- Receiving status
- Vendor lead times
- Vendor pricing
- Items below reorder point
- Items needed for upcoming production

AI Support

AI should help with:

- Low-stock alerts
- Reorder recommendations
- Purchase order drafts
- Demand forecasting
- Expiration warnings
- Vendor performance summaries
- Identifying overstocked items
- Identifying materials needed for scheduled production
- Suggesting purchase priorities based on orders and lead times

Key Questions Answered

- What are we low on?
- What should we reorder now?

- What inventory is aging or expiring?
- Which purchase orders are late?
- Do we have materials for upcoming production?
- Which vendors are causing delays?
- How much cash is tied up in inventory?

Connected Modules

This module connects to:

- Vendors
- Production and Batches
- Orders and Quotes
- Finance and Cash Flow
- Documents
- Compliance, Legal, and IP
- Tasks and Accountability

3.8 Module 6 — Finance and Cash Flow

Purpose

This module provides management-level financial visibility.

It should not necessarily replace accounting software, but it should help leadership understand receivables, payables, cash position, margins, profitability, and short-term financial risk.

Tracks and Displays

- Cash position
- Receivables
- Payables
- Customer invoices
- Vendor bills
- Payment status
- Overdue invoices
- Days sales outstanding
- Revenue
- Expenses
- Gross margin
- Product profitability
- Customer profitability
- Order profitability
- 30/60/90-day cash forecast
- Financial alerts

AI Support

AI should help with:

- Cash-flow summaries
- Collections prioritization
- Margin analysis
- Profitability explanations
- Expense anomaly detection
- Drafting customer payment reminders
- Weekly financial summaries
- Forecasting cash risk

Key Questions Answered

- How much cash do we have?
- What money is owed to us?
- What money do we owe?
- Which invoices are overdue?
- What does cash look like 30, 60, and 90 days out?
- Which products or customers are most profitable?
- Where are margins shrinking?

Connected Modules

This module connects to:

- Customers and Sales
- Orders and Quotes
- Inventory and Purchasing
- Production and Batches
- Documents
- Reporting and Analytics
- AI Assistant

3.9 Module 7 — Compliance, Legal, and IP

Purpose

This module manages obligations, deadlines, records, documents, legal matters, intellectual property matters, and compliance requirements.

Its purpose is to reduce legal, regulatory, documentation, and deadline risk.

Tracks and Displays

- Compliance requirements
- Licenses
- Renewal dates
- COAs
- SOPs
- Contracts
- NDAs
- Insurance documents
- Patent/IP matters
- Filing deadlines
- Legal tasks
- Required evidence
- Missing documents
- Expired documents
- Regulatory notes

AI Support

AI should help with:

- Document summarization
- Obligation extraction
- Deadline identification
- Compliance checklist generation
- Contract risk flagging
- SOP review reminders
- Linking COAs to products or batches
- Creating legal/compliance tasks
- Summarizing IP status

Key Questions Answered

- What compliance deadlines are coming up?
- Which documents are missing or expired?
- What obligations exist in this contract?
- Which products or batches need COAs?
- What patent/IP matters need attention?
- Which SOPs need review?

Connected Modules

This module connects to:

- Documents and Knowledge Base
- Products
- Production and Batches

- Vendors
- Customers
- Tasks and Accountability
- Command Center

3.10 Module 8 — Documents and Knowledge Base

Purpose

This module stores, organizes, summarizes, and connects important business documents to relevant records.

Its purpose is to turn documents into usable business intelligence rather than disconnected files.

Tracks and Displays

- Contracts
- SOPs
- COAs
- Licenses
- Insurance documents
- Vendor documents
- Customer documents
- Product specifications
- Batch records
- Patent documents
- Compliance documents
- Meeting notes
- Policies
- Training materials
- Uploaded files
- Document versions
- Review dates

AI Support

AI should help with:

- Document summarization
- Search and question answering
- Extracting key dates
- Extracting obligations
- Identifying missing information
- Creating tasks from documents
- Linking documents to records
- Comparing document versions

- Creating plain-English explanations

Key Questions Answered

- Where is this document?
- What does this document say?
- What deadlines or obligations are inside it?
- Which customer, vendor, product, batch, or compliance requirement does it relate to?
- Has this document expired?
- Has it been reviewed recently?

Connected Modules

This module connects to every other module.

3.11 Module 9 — Tasks and Accountability

Purpose

This module manages the execution layer of the business.

Every important action should become a task with an owner, due date, priority, status, and related business context.

Tracks and Displays

- Tasks
- Owners
- Due dates
- Priorities
- Status
- Blockers
- Related records
- AI-generated tasks
- Approval status
- Recurring tasks
- Overdue tasks
- Tasks by person
- Tasks by department
- Tasks waiting on leadership

AI Support

AI should help with:

- Creating tasks from emails, meetings, documents, alerts, and recommendations

- Summarizing overdue tasks
- Identifying blocked work
- Suggesting priority levels
- Creating recurring task schedules
- Producing weekly accountability reports
- Detecting repeated missed deadlines

Key Questions Answered

- What is overdue?
- Who owns this?
- What is blocked?
- What needs my approval?
- What did AI recommend that has not been acted on?
- Which tasks are connected to this customer, order, batch, invoice, document, or project?

Connected Modules

This module connects to every other module.

3.12 Module 10 — Projects and Strategic Initiatives

Purpose

This module manages larger initiatives that require planning, milestones, tasks, documents, budgets, and leadership oversight.

Examples may include product launches, facility improvements, patent filings, new customer programs, equipment purchases, hiring plans, technology projects, and expansion initiatives.

Tracks and Displays

- Projects
- Objectives
- Owners
- Milestones
- Budgets
- Deadlines
- Project status
- Risks
- Linked tasks
- Linked documents
- Linked customers/vendors/products
- Strategic priority
- Progress updates

AI Support

AI should help with:

- Project summaries
- Milestone tracking
- Risk identification
- Drafting project plans
- Creating tasks from project goals
- Summarizing progress
- Identifying blocked projects
- Preparing leadership updates

Key Questions Answered

- What strategic projects are active?
- Which projects are behind schedule?
- What milestones are coming up?
- What risks could delay this project?
- Who owns the next action?
- What documents or tasks are linked to this initiative?

Connected Modules

This module connects to:

- Tasks and Accountability
- Documents and Knowledge Base
- Finance and Cash Flow
- Compliance, Legal, and IP
- People and Roles
- Command Center

3.13 Module 11 — People and Roles

Purpose

This module manages users, employees, advisors, roles, responsibilities, permissions, and accountability structure.

Its purpose is to make clear who is responsible for what and what each user can access or change.

Tracks and Displays

- Users
- Employees

- Advisors
- Roles
- Departments
- Responsibilities
- Permission levels
- Assigned tasks
- Activity history
- Approval authority
- Workload indicators

AI Support

AI should help with:

- Workload summaries
- Task distribution insights
- Accountability reports
- Identifying overloaded users
- Identifying unclear ownership
- Suggesting task assignments based on role
- Summarizing team performance signals

Key Questions Answered

- Who owns this responsibility?
- Who is overloaded?
- Who has overdue tasks?
- Who can approve this action?
- What can this user access?
- What work is assigned to this person?

Connected Modules

This module connects to:

- Tasks and Accountability
- Settings, Permissions, and Audit Logs
- Command Center
- Projects and Strategic Initiatives
- AI Assistant

3.14 Module 12 — AI Assistant and Automation Layer

Purpose

This module provides AI capabilities across the entire system.

AI should not be treated as a separate chatbot only. It should be embedded into the workflows, dashboards, records, documents, alerts, and reports throughout the platform.

AI Modes

The AI layer should support four main modes:

AI Analyst

Explains data, trends, performance, and risk.

AI Assistant

Drafts emails, summaries, notes, reports, tasks, and documents.

AI Monitor

Watches for deadlines, thresholds, missing information, stale activity, and risk signals.

AI Operator

Performs controlled actions, usually with human approval.

Tracks and Displays

- AI recommendations
- AI-generated summaries
- AI-created tasks
- AI-detected risks
- AI-drafted communications
- User feedback on AI outputs
- Approval status
- Source records used
- AI action history

Key Questions Answered

- What does AI recommend?
- Why did AI recommend this?
- What data was used?
- Has a human approved this?
- What AI-generated tasks are open?
- Which recommendations were accepted or rejected?

Connected Modules

The AI layer connects to every module, while respecting permissions and approval rules.

3.15 Module 13 — Reporting and Analytics

Purpose

This module generates daily, weekly, monthly, quarterly, and custom reports.

Reports should combine numbers, explanations, risks, and recommended actions.

Tracks and Displays

- Daily leadership briefing
- Weekly operating report
- Monthly performance report
- Quarterly strategic report
- Sales reports
- Finance reports
- Inventory reports
- Production reports
- Compliance reports
- Accountability reports
- Custom KPI dashboards

AI Support

AI should help with:

- Report generation
- Plain-English summaries
- Trend explanation
- Risk commentary
- Recommended actions
- Comparing performance across periods
- Highlighting unusual changes
- Creating executive summaries

Key Questions Answered

- What happened this week?
- What changed from last month?
- What are the biggest risks?
- What should leadership discuss?
- Which KPIs are improving or declining?
- What actions came from this report?

Connected Modules

This module connects to every module.

3.16 Module 14 — Settings, Permissions, and Audit Logs

Purpose

This module controls system configuration, user permissions, approval rules, security settings, and audit history.

Its purpose is to protect the business, maintain accountability, and ensure AI and users operate within approved boundaries.

Tracks and Displays

- Company settings
- User roles
- Permission groups
- Approval rules
- Automation rules
- Notification settings
- AI access controls
- Activity logs
- Record change history
- Login/activity history
- Approval history
- Deleted or archived records

AI Support

AI should help with:

- Explaining audit history
- Identifying unusual activity
- Summarizing approval queues
- Suggesting permission review items
- Flagging risky automations
- Creating system health summaries

Key Questions Answered

- Who changed this record?
- Who approved this action?
- What did AI generate?
- What data did AI use?
- Who has access to this module?
- What automations are active?
- What sensitive actions happened recently?

Connected Modules

This module connects to every module and governs user and AI behavior throughout the system.

3.17 Module Priority for MVP

Not every module should be fully built at the same time.

The MVP should focus on the modules that create the strongest foundation and fastest management value.

MVP Priority Modules

1. Command Center
2. Customers and Sales
3. Orders and Quotes
4. Inventory and Purchasing
5. Tasks and Accountability
6. Documents and Knowledge Base
7. AI Assistant and Automation Layer
8. Settings, Permissions, and Audit Logs

Phase 2 Modules

1. Production and Batches
2. Finance and Cash Flow
3. Compliance, Legal, and IP
4. Reporting and Analytics
5. People and Roles
6. Projects and Strategic Initiatives

Reasoning

The MVP should first establish the core business record system, daily visibility, task accountability, customer follow-up, inventory awareness, document intelligence, and AI support.

Once those foundations are stable, the system can deepen into production control, finance intelligence, compliance management, strategic projects, and more advanced reporting.

3.18 Module Design Rule

Every module should follow the same basic design structure:

1. Records
2. Relationships

3. Status
4. Alerts
5. Tasks
6. Documents
7. AI insights
8. Reports
9. Permissions
10. Audit history

This keeps the system consistent as it grows.

Step 4 — Core Business Objects

4.1 Purpose of This Section

Core business objects are the foundation of the system.

A business object is a primary record the company needs to track, manage, connect, search, report on, or use for AI-assisted decision-making.

The system should not be built around disconnected notes, spreadsheets, or files. It should be built around structured objects that represent how the business actually operates.

For example:

- Customers place orders.
- Orders require products.
- Products require materials.
- Materials come from vendors.
- Production batches consume materials and create finished goods.
- Orders create invoices.
- Invoices affect cash flow.
- Documents support products, vendors, customers, batches, compliance, and legal obligations.
- Tasks create accountability across all business areas.
- Alerts identify risks that require attention.
- Activity logs preserve history and accountability.

These objects create the backbone of the AI Business Command Center.

4.2 Core Object List

The initial system should include the following core business objects:

1. Company

2. User
3. Role
4. Customer
5. Contact
6. Vendor
7. Product
8. Material
9. Inventory Item
10. Inventory Transaction
11. Quote
12. Sales Order
13. Purchase Order
14. Work Order
15. Batch / Production Run
16. Invoice
17. Payment
18. Expense / Bill
19. Task
20. Document
21. Compliance Requirement
22. Project
23. Alert
24. Note / Communication
25. Approval Request
26. Report
27. Activity Log
28. AI Recommendation

4.3 Object: Company

Purpose

Represents the business entity using the system.

This object stores company-level information, settings, identifiers, and configuration data.

Key Fields

- Company ID
- Legal name
- DBA / trade name
- Entity type
- EIN / tax ID
- Primary address
- Operating locations
- Phone number

- Website
- Industry
- Business licenses
- Insurance information
- Default currency
- Default timezone
- Company logo
- Primary administrator
- Company settings

Key Relationships

- Has many users
- Has many customers
- Has many vendors
- Has many products
- Has many documents
- Has many locations
- Has many reports
- Has many audit logs

AI Use

AI can use company context to tailor summaries, reports, recommendations, document drafts, and business analysis.

4.4 Object: User

Purpose

Represents a person who accesses the system.

Users may include owners, executives, managers, employees, operators, outside advisors, accountants, attorneys, consultants, or other approved participants.

Key Fields

- User ID
- Name
- Email
- Phone
- Role
- Department
- Job title
- Status
- Permission group

- Approval authority
- Assigned tasks
- Notification preferences
- Last login
- Activity status

Key Relationships

- Belongs to company
- Has one or more roles
- Owns tasks
- Creates records
- Updates records
- Approves or rejects approval requests
- Appears in activity logs

AI Use

AI can summarize a user's workload, overdue tasks, approvals waiting on them, and responsibilities.

AI must respect each user's permissions.

4.5 Object: Role

Purpose

Defines what a user can access, view, edit, approve, or administer.

Roles are the foundation of the permission model.

Key Fields

- Role ID
- Role name
- Description
- Department
- Permission set
- Approval authority level
- Sensitive data access
- Module access
- Active/inactive status

Example Roles

- Owner / CEO
- Administrator

- Operations Manager
- Sales Manager
- Finance Manager
- Compliance Manager
- Production Operator
- Warehouse / Inventory User
- Outside Advisor
- Read-Only User

Key Relationships

- Assigned to users
- Controls module access
- Controls approval authority
- Controls AI visibility
- Controls reporting access

AI Use

AI uses role permissions to determine what information a user can ask about, summarize, or act on.

4.6 Object: Customer

Purpose

Represents a business or individual that buys from, may buy from, or has a commercial relationship with the company.

Key Fields

- Customer ID
- Company/customer name
- Customer type
- Status
- Lead source
- Sales stage
- Primary contact
- Billing address
- Shipping address
- Email
- Phone
- Website
- Tax/resale information
- Payment terms
- Credit limit
- Lifetime revenue

- Open balance
- Last contact date
- Next follow-up date
- Account owner
- Notes

Key Relationships

- Has many contacts
- Has many quotes
- Has many sales orders
- Has many invoices
- Has many payments
- Has many documents
- Has many tasks
- Has many communications
- May have related projects

AI Use

AI can summarize customer history, identify stale follow-ups, draft emails, prioritize opportunities, detect customer risk, and explain customer value.

4.7 Object: Contact

Purpose

Represents an individual person associated with a customer, vendor, partner, advisor, or other organization.

Key Fields

- Contact ID
- First name
- Last name
- Organization
- Contact type
- Job title
- Email
- Phone
- Mobile phone
- Address
- Preferred communication method
- Notes
- Last contacted date
- Relationship owner

Key Relationships

- Belongs to customer, vendor, or other organization
- Can be linked to communications
- Can be linked to tasks
- Can be linked to documents
- Can be linked to meetings or notes

AI Use

AI can help draft messages, summarize recent communication, recommend follow-ups, and identify neglected relationships.

4.8 Object: Vendor

Purpose

Represents suppliers, service providers, contractors, labs, logistics providers, advisors, or other companies the business buys from or relies on.

Key Fields

- Vendor ID
- Vendor name
- Vendor type
- Primary contact
- Email
- Phone
- Website
- Address
- Products/services supplied
- Payment terms
- Lead time
- Pricing notes
- Reliability score
- Compliance documents
- Insurance documents
- Tax documents
- Status
- Notes

Key Relationships

- Has many contacts
- Has many purchase orders
- Supplies materials or services

- Has many documents
- Has many tasks
- May be linked to expenses/bills
- May be linked to compliance requirements

AI Use

AI can summarize vendor performance, flag late purchase orders, identify missing vendor documents, compare pricing, and recommend follow-up actions.

4.9 Object: Product

Purpose

Represents an item or service the company sells, manufactures, packages, distributes, or manages commercially.

Key Fields

- Product ID
- Product name
- SKU
- Product category
- Description
- Status
- Unit of measure
- Standard cost
- Sale price
- Target margin
- Packaging type
- Formula/specification reference
- Required materials
- Compliance requirements
- Labeling requirements
- Storage requirements
- Active/inactive status

Key Relationships

- Appears on quotes
- Appears on sales orders
- Connected to materials
- Connected to batches
- Connected to inventory items
- Connected to documents
- Connected to compliance requirements

- Connected to revenue and margin reporting

AI Use

AI can analyze product profitability, identify margin concerns, summarize product requirements, flag missing documentation, and recommend production or purchasing actions.

4.10 Object: Material

Purpose

Represents a raw material, ingredient, packaging item, label, supply, component, or other input needed for production, fulfillment, or operations.

Key Fields

- Material ID
- Material name
- SKU/internal code
- Category
- Unit of measure
- Default vendor
- Standard cost
- Current cost
- Lead time
- Reorder point
- Minimum stock level
- Storage requirements
- Expiration tracking required
- Lot tracking required
- Compliance documents required
- Active/inactive status

Key Relationships

- Supplied by vendors
- Used in products
- Used in batches
- Represented by inventory items/lots
- Appears on purchase orders
- Has related documents
- May have compliance requirements

AI Use

AI can recommend reorders, identify material shortages, forecast needs based on orders and production, and flag aging or expiring materials.

4.11 Object: Inventory Item

Purpose

Represents the current stock of a material, component, packaging item, or finished good.

Inventory may be tracked by item, location, lot, batch, expiration date, and quantity.

Key Fields

- Inventory Item ID
- Item type
- Related product or material
- SKU
- Location
- Quantity on hand
- Quantity available
- Quantity reserved
- Unit of measure
- Lot number
- Batch number
- Expiration date
- Cost basis
- Status
- Last counted date
- Reorder point

Key Relationships

- Belongs to product or material
- May be linked to vendor
- May be linked to purchase order
- May be linked to batch
- May be reserved for sales order
- Has inventory transactions
- Has related documents

AI Use

AI can flag low stock, reserved stock conflicts, expiring inventory, inventory needed for production, and overstocked items.

4.12 Object: Inventory Transaction

Purpose

Records every movement or adjustment of inventory.

This object is necessary for traceability, auditability, and accurate inventory history.

Key Fields

- Transaction ID
- Inventory item
- Transaction type
- Quantity change
- Previous quantity
- New quantity
- Unit of measure
- Location
- Related record
- Reason
- Performed by
- Date/time
- Notes

Transaction Types

- Receive
- Consume
- Produce
- Reserve
- Release reservation
- Transfer
- Adjust
- Scrap/waste
- Return
- Ship

Key Relationships

- Belongs to inventory item
- May relate to purchase order
- May relate to sales order
- May relate to batch
- May relate to work order
- Appears in activity log

AI Use

AI can analyze inventory movement, explain stock changes, detect unusual adjustments, and identify causes of shortages or waste.

4.13 Object: Quote

Purpose

Represents a proposed sale to a customer before it becomes an order.

Key Fields

- Quote ID
- Customer
- Contact
- Quote date
- Expiration date
- Quote status
- Products/services quoted
- Quantities
- Pricing
- Discounts
- Estimated margin
- Payment terms
- Delivery terms
- Notes
- Approval status
- Created by

Key Relationships

- Belongs to customer
- May convert to sales order
- Has products
- May have documents
- May create tasks
- May require approval
- May link to communications

AI Use

AI can draft quotes, check margins, summarize quote status, identify expiring quotes, and suggest follow-up messages.

4.14 Object: Sales Order

Purpose

Represents a confirmed customer order that must be fulfilled, produced, shipped, invoiced, and paid.

Key Fields

- Sales Order ID
- Customer
- Contact
- Order date
- Due date
- Status
- Products ordered
- Quantities
- Pricing
- Discounts
- Total value
- Required delivery date
- Fulfillment status
- Production required
- Inventory reserved
- Shipping status
- Invoice status
- Payment status
- Notes

Key Relationships

- Belongs to customer
- May originate from quote
- Has products
- May require work orders
- May require batches
- Reserves inventory
- Creates invoice
- Has documents
- Has tasks
- Has communications

AI Use

AI can identify at-risk orders, check inventory availability, recommend production actions, draft customer updates, and summarize fulfillment status.

4.15 Object: Purchase Order

Purpose

Represents an order placed with a vendor for materials, products, packaging, equipment, or services.

Key Fields

- Purchase Order ID
- Vendor
- Order date
- Expected delivery date
- Status
- Items ordered
- Quantities
- Unit costs
- Total cost
- Payment terms
- Receiving status
- Linked production need
- Linked inventory items
- Approval status
- Notes

Key Relationships

- Belongs to vendor
- Contains materials or services
- Creates inventory upon receiving
- May be linked to production/work orders
- May be linked to expense/bill
- Has documents
- Has tasks
- May require approval

AI Use

AI can draft purchase orders, flag late vendor deliveries, recommend purchases, compare vendor lead times, and identify purchasing risks.

4.16 Object: Work Order

Purpose

Represents planned production work required to make a product, fulfill demand, or complete an operational process.

A work order may result in one or more batches or production runs.

Key Fields

- Work Order ID
- Product
- Quantity to produce
- Requested start date
- Required completion date
- Status
- Priority
- Related sales order
- Required materials
- Assigned team/person
- Production instructions
- Quality requirements
- Notes

Key Relationships

- May be created from sales order
- May create batch/production run
- Requires materials
- Uses documents/SOPs
- Has tasks
- May trigger inventory reservations

AI Use

AI can recommend work order priority, identify missing materials, create production task lists, and summarize work order readiness.

4.17 Object: Batch / Production Run

Purpose

Represents a specific production run or batch that consumes inputs and creates output.

This is essential for manufacturing traceability, quality control, costing, and inventory accuracy.

Key Fields

- Batch ID
- Work order
- Product
- Batch number

- Start date
- Completion date
- Status
- Formula/specification used
- Input materials
- Input quantities
- Output quantity
- Yield
- Waste
- Rework
- Labor hours
- Equipment used
- Quality-control status
- COA/lab result reference
- Deviations/issues
- Responsible person
- Notes

Key Relationships

- Created from work order
- Uses materials/inventory
- Creates finished goods inventory
- May fulfill sales orders
- Has documents
- Has quality records
- Has tasks
- May have compliance requirements
- Impacts product costing and margin

AI Use

AI can summarize batch performance, identify yield issues, detect cost variance, flag missing quality records, and create corrective action tasks.

4.18 Object: Invoice

Purpose

Represents an amount owed by a customer or, in some contexts, a payable invoice from a vendor.

For clarity, customer invoices and vendor bills may be separated in the data model, but both affect cash flow.

Key Fields

- Invoice ID
- Invoice type
- Customer or vendor
- Related order
- Invoice date
- Due date
- Amount
- Balance due
- Status
- Payment terms
- Days overdue
- Payment status
- Notes

Key Relationships

- Belongs to customer or vendor
- May be created from sales order or purchase order
- Has payments
- Has documents
- Has tasks
- Appears in finance reports

AI Use

AI can summarize overdue invoices, prioritize collections, draft payment reminders, and support cash-flow forecasting.

4.19 Object: Payment

Purpose

Represents money received from customers or paid to vendors.

Key Fields

- Payment ID
- Payment type
- Related invoice/bill
- Customer or vendor
- Payment date
- Amount
- Payment method
- Reference number

- Status
- Notes

Key Relationships

- Applies to invoice or bill
- Belongs to customer or vendor
- Impacts cash flow
- Appears in financial reports
- May have documents/receipts

AI Use

AI can reconcile payment status, summarize cash movement, detect unpaid balances, and assist with cash-flow projections.

4.20 Object: Expense / Bill

Purpose

Represents an amount owed by the company to a vendor, supplier, employee, contractor, landlord, utility, lender, or other party.

Key Fields

- Bill/Expense ID
- Vendor/payee
- Expense category
- Related purchase order
- Bill date
- Due date
- Amount
- Status
- Payment terms
- Approval status
- Payment status
- Notes

Key Relationships

- Belongs to vendor/payee
- May be linked to purchase order
- May have payment
- May have document/receipt
- Appears in cash-flow reports
- May require approval

AI Use

AI can identify upcoming payables, flag unusual expenses, summarize spending by category, and forecast cash outflows.

4.21 Object: Task

Purpose

Represents an action that needs to be completed.

Tasks are the accountability layer of the entire system.

Key Fields

- Task ID
- Title
- Description
- Owner
- Due date
- Priority
- Status
- Related module
- Related record
- Created by
- AI-generated flag
- Approval required flag
- Blocker status
- Completion date
- Notes

Key Relationships

- Can link to any object
- Belongs to owner/user
- May be created from alert
- May be created from AI recommendation
- May be created from document
- May be created from communication
- Appears in command center

AI Use

AI can create, prioritize, summarize, assign, and monitor tasks, while identifying overdue or blocked work.

4.22 Object: Document

Purpose

Represents a stored file, document, record, contract, SOP, COA, patent file, license, policy, note, image, or other business document.

Key Fields

- Document ID
- Document name
- Document type
- File location
- Related module
- Related record
- Version
- Status
- Owner
- Upload date
- Expiration date
- Review date
- Summary
- Extracted obligations
- Extracted dates
- Tags
- Access level

Key Relationships

- Can link to customers
- Can link to vendors
- Can link to products
- Can link to batches
- Can link to compliance requirements
- Can link to projects
- Can create tasks
- Can require review or approval

AI Use

AI can summarize documents, answer questions, extract obligations, identify deadlines, compare versions, and create tasks from document contents.

4.23 Object: Compliance Requirement

Purpose

Represents a legal, regulatory, quality, documentation, licensing, contractual, safety, tax, insurance, or internal compliance obligation.

Key Fields

- Compliance Requirement ID
- Requirement name
- Requirement type
- Jurisdiction
- Description
- Applies to
- Frequency
- Due date
- Responsible person
- Required evidence
- Status
- Risk level
- Related documents
- Review date
- Notes

Key Relationships

- May apply to product
- May apply to batch
- May apply to vendor
- May apply to customer
- May apply to company
- Has documents
- Has tasks
- Appears in alerts
- Appears in compliance reports

AI Use

AI can extract compliance obligations from documents, generate checklists, identify upcoming deadlines, and flag missing evidence.

4.24 Object: Project

Purpose

Represents a larger initiative that requires coordinated tasks, milestones, documents, budget, people, and progress tracking.

Key Fields

- Project ID
- Project name
- Objective
- Owner
- Status
- Priority
- Start date
- Target completion date
- Budget
- Milestones
- Risks
- Related records
- Notes

Key Relationships

- Has tasks
- Has documents
- Has owner/user
- May relate to customers
- May relate to products
- May relate to compliance/IP
- May relate to finance
- Appears in reports

AI Use

AI can summarize project status, identify stalled milestones, create task plans, flag risks, and generate leadership updates.

4.25 Object: Alert

Purpose

Represents a warning, risk, notification, exception, deadline, or business condition requiring attention.

Alerts should be actionable and linked to records.

Key Fields

- Alert ID
- Alert type
- Severity
- Title
- Description
- Related module
- Related record
- Trigger condition
- Status
- Assigned owner
- Created date
- Due date
- Resolution notes

Example Alert Types

- Low inventory
- Overdue invoice
- Stale customer follow-up
- Delayed order
- Late purchase order
- Production delay
- Missing document
- Expiring document
- Compliance deadline
- Overdue task
- Margin concern
- Cash-flow risk

Key Relationships

- Can link to any object
- Can create task
- Can create approval request
- Appears in command center
- Appears in reports

AI Use

AI can prioritize alerts, summarize risk, recommend actions, and convert alerts into tasks or approval requests.

4.26 Object: Note / Communication

Purpose

Represents communication history, meeting notes, call notes, emails, text summaries, internal updates, or general record notes.

Key Fields

- Note/Communication ID
- Type
- Subject
- Body/summary
- Related record
- Participants
- Date/time
- Created by
- Follow-up required flag
- Sent/received status
- Source
- Attachments

Key Relationships

- Can link to customers
- Can link to contacts
- Can link to vendors
- Can link to orders
- Can link to projects
- Can create tasks
- Can link to documents

AI Use

AI can summarize communications, extract action items, detect follow-up needs, draft replies, and update customer/vendor history.

4.27 Object: Approval Request

Purpose

Represents an action that requires human approval before it can be completed.

This object is critical for safe AI-assisted workflows.

Key Fields

- Approval Request ID
- Request type
- Requested action
- Related record
- Requested by
- Approver
- Status
- Priority
- Created date
- Due date
- Decision date
- Approval notes
- AI-generated flag
- Source/reason

Example Approval Types

- Send customer email
- Issue quote
- Approve purchase order
- Approve price discount
- Change production specification
- Update compliance status
- Delete/archive record
- Grant permission
- Approve AI-generated document

Key Relationships

- Can link to any object
- Belongs to requester
- Belongs to approver
- May originate from AI recommendation
- May create activity log entry
- May trigger task or workflow

AI Use

AI can prepare approval summaries, explain the reason for approval, list supporting data, and recommend approve/reject/escalate options.

4.28 Object: Report

Purpose

Represents a generated business report, dashboard summary, executive briefing, or recurring management review.

Key Fields

- Report ID
- Report type
- Title
- Period covered
- Generated date
- Generated by
- Related modules
- KPI summary
- Insights
- Risks
- Recommendations
- Linked tasks
- Distribution list
- Status

Report Types

- Daily briefing
- Weekly leadership report
- Monthly performance report
- Quarterly strategy report
- Sales report
- Finance report
- Inventory report
- Production report
- Compliance report
- Accountability report

Key Relationships

- Pulls data from multiple modules
- May create tasks
- May include AI recommendations
- May be linked to users or departments
- May be archived in documents

AI Use

AI can generate reports, explain trends, highlight risks, recommend actions, and convert report items into tasks.

4.29 Object: Activity Log

Purpose

Records important system activity for accountability, traceability, security, and auditability.

This object should quietly support the entire platform.

Key Fields

- Activity Log ID
- User
- Action type
- Related module
- Related record
- Previous value
- New value
- Timestamp
- Source
- IP/device information
- AI-generated flag
- Notes

Activity Types

- Record created
- Record updated
- Record deleted/archived
- Task completed
- Approval submitted
- Approval accepted/rejected
- Document uploaded
- AI recommendation generated
- AI output edited
- Alert triggered
- Permission changed
- Login/session activity

Key Relationships

- Can link to any object

- Belongs to user/system/AI
- Supports audit reports
- Supports security review

AI Use

AI can summarize activity history, explain what changed, detect unusual activity, and help reconstruct decision history.

4.30 Object: AI Recommendation

Purpose

Represents a recommendation, insight, warning, draft, forecast, or suggested action generated by AI.

AI recommendations should be stored so the company can track what AI suggested, what data it used, whether a human accepted it, and what happened next.

Key Fields

- AI Recommendation ID
- Recommendation type
- Title
- Description
- Related module
- Related record
- Source data used
- Confidence level
- Recommended action
- Risk level
- Human decision
- Status
- Created date
- Reviewed by
- Review notes
- Converted to task flag
- Converted to approval request flag

Recommendation Types

- Follow-up recommendation
- Purchasing recommendation
- Production recommendation
- Cash-flow warning
- Margin warning
- Compliance warning

- Document obligation
- Task suggestion
- Risk alert
- Report insight

Key Relationships

- Can link to any object
- May create task
- May create alert
- May create approval request
- Appears in command center
- Appears in reports
- Appears in activity log

AI Use

This object is created by AI and reviewed by humans.

It allows the system to learn over time which recommendations are useful, ignored, accepted, or rejected.

4.31 Universal Fields for Most Objects

Most objects should include a common set of system fields.

Standard System Fields

- ID
- Company ID
- Created date
- Created by
- Last updated date
- Last updated by
- Status
- Owner
- Tags
- Notes
- Related documents
- Related tasks
- Activity history
- Access permissions
- Archive/delete status

These fields help keep the system consistent, searchable, permission-aware, and auditable.

4.32 Object Relationship Principles

The system should be designed so that important records are connected rather than isolated.

Relationship Rules

1. Every task should link to a relevant business object whenever possible.
2. Every document should be linked to the record it supports.
3. Every alert should link to the record that triggered it.
4. Every AI recommendation should show its source data and related record.
5. Every approval request should link to the action or record being approved.
6. Every sales order should connect to customer, product, inventory, production, invoice, and payment records when applicable.
7. Every production batch should connect to inputs, outputs, documents, quality records, tasks, and costing.
8. Every invoice should connect to customer/order and payment status.
9. Every compliance requirement should connect to evidence documents and responsible owners.
10. Every major action should create an activity log entry.

4.33 MVP Object Priority

The MVP does not need every object fully built at launch.

Required for MVP

- Company
- User
- Role
- Customer
- Contact
- Vendor
- Product
- Material
- Inventory Item
- Quote
- Sales Order
- Purchase Order
- Task
- Document
- Alert
- AI Recommendation
- Approval Request
- Activity Log

Phase 2 Objects

- Inventory Transaction
- Work Order
- Batch / Production Run
- Invoice
- Payment
- Expense / Bill
- Compliance Requirement
- Project
- Report
- Note / Communication

Reasoning

The MVP should create the foundational record system first: customers, vendors, products, inventory, quotes, orders, purchasing, tasks, documents, alerts, AI recommendations, approvals, and activity history.

Once the foundation is reliable, the system can deepen into production traceability, finance, compliance, reporting, and strategic project management.

4.34 Design Rule for Objects

Every core object should answer the following questions:

1. What is this record?
2. Who owns it?
3. What status is it in?
4. What other records is it connected to?
5. What tasks are attached to it?
6. What documents support it?
7. What risks or alerts are associated with it?
8. What has changed over time?
9. What can AI summarize or recommend about it?
10. Who is allowed to view, edit, approve, or delete it?

This rule ensures that every object supports visibility, accountability, AI assistance, and scale.

Step 5 — Data Model and Relationships

5.1 Purpose of This Section

This section defines how the core business objects connect to one another.

The data model is the foundation that allows the system to operate as one connected business management platform instead of a collection of separate tools.

A strong data model allows the system to answer questions such as:

- Which customer orders are at risk because inventory is low?
- Which production batches are tied to which customer orders?
- Which invoices are unpaid for which customers?
- Which documents support a product, vendor, batch, or compliance requirement?
- Which tasks are overdue for a customer, order, batch, invoice, project, or document?
- Which AI recommendations were created from which data?
- Which actions were approved, by whom, and when?

The goal is to make every important record traceable, connected, searchable, reportable, and usable by AI.

5.2 Data Model Design Philosophy

The system should use a relational business data model.

This means records should be connected through clear relationships instead of duplicated across disconnected spreadsheets or notes.

The basic design philosophy is:

Every important business action should be connected to the object it affects.

Examples:

- A customer follow-up task should link to the customer.
- A purchase recommendation should link to the material and vendor.
- A low-stock alert should link to the inventory item.
- A production issue should link to the work order or batch.
- A COA should link to the product, batch, or inventory lot it supports.
- A payment reminder should link to the invoice and customer.
- A compliance deadline should link to the requirement, document, and responsible person.
- An AI recommendation should link to the source records it used.

This structure creates accountability, context, and traceability.

5.3 Core Relationship Types

The system should support the following relationship types.

One-to-One

One record is linked to one other record.

Examples:

- One user profile belongs to one user account.
- One invoice may have one final payment record if paid in full at once.
- One document may represent one specific signed contract version.

One-to-Many

One record can have many related records.

Examples:

- One customer can have many contacts.
- One customer can have many quotes.
- One customer can have many sales orders.
- One vendor can have many purchase orders.
- One product can have many batches.
- One invoice can have many payment attempts or partial payments.

Many-to-Many

Many records can be linked to many other records.

Examples:

- Products can use many materials, and materials can be used in many products.
- Documents can relate to many records, and records can have many documents.
- Tasks can relate to multiple records when necessary.
- Compliance requirements can apply to multiple products, vendors, or locations.
- Users can participate in many projects, and projects can include many users.

Many-to-many relationships should usually be managed through join/link tables so the system remains clean and reportable.

5.4 Primary Data Flow: Customer Demand to Cash

The most important business flow connects customer demand to fulfillment and payment.

Flow

Customer → Quote → Sales Order → Inventory / Production → Fulfillment → Invoice → Payment

Relationship Logic

1. A customer may request a quote.
2. A quote contains one or more products or services.
3. An approved quote may become a sales order.
4. A sales order may be fulfilled from existing inventory or may require production.
5. If production is required, the sales order may create one or more work orders.
6. Work orders may create one or more batches or production runs.
7. Batches consume materials and create finished goods inventory.
8. Inventory is reserved or shipped against the sales order.
9. The sales order creates an invoice.
10. The invoice is paid through one or more payments.
11. Tasks, documents, alerts, and AI recommendations may attach at any stage.

Required Relationships

- Sales Order must link to Customer.
- Sales Order must include at least one Product or Service.
- Invoice must link to Customer.
- Invoice should link to Sales Order when applicable.
- Payment should link to Invoice.

Optional Relationships

- Quote may link to Sales Order.
- Sales Order may link to Work Order.
- Work Order may link to Batch.
- Batch may link to Sales Order.
- Sales Order may link to Shipment/Fulfillment record in a future version.

AI Value

This relationship chain allows AI to answer:

- Which orders are at risk?
- Why is this order delayed?
- Can we fulfill this order from inventory?
- Does this order need production?
- Has this customer been invoiced?
- Has the invoice been paid?
- What is the total customer value?

5.5 Product, Material, and Inventory Relationships

Products, materials, and inventory must be carefully separated.

Product

A product is something the company sells or produces.

Material

A material is an input, component, ingredient, packaging item, label, or supply used to make, package, or deliver a product.

Inventory Item

An inventory item represents actual stock on hand, either material inventory or finished goods inventory.

Core Flow

Product ↔ Materials → Inventory Items → Production / Orders

Relationship Logic

1. A product may require multiple materials.
2. A material may be used in multiple products.
3. Inventory items track the quantity of materials or finished goods on hand.
4. Materials are purchased from vendors through purchase orders.
5. Materials are consumed by batches.
6. Batches produce finished goods inventory.
7. Finished goods inventory fulfills sales orders.

Required Relationships

- Inventory Item must link to either Product or Material.
- Material should have a default unit of measure.
- Product should have a SKU or internal identifier.
- Purchase Order line item should link to Material, Product, or Service.

Optional Relationships

- Material may link to default Vendor.
- Product may link to required Materials.
- Inventory Item may link to Lot Number.
- Inventory Item may link to Expiration Date.
- Inventory Item may link to Batch.
- Inventory Item may link to Location.

AI Value

This allows AI to answer:

- What materials are needed for this product?

- Do we have enough materials for upcoming production?
- Which materials are below reorder point?
- Which inventory items are expiring?
- Which products are affected by a material shortage?
- How much finished goods inventory is available for sale?

5.6 Vendor and Purchasing Relationships

Vendors supply the goods and services the company depends on.

Flow

Vendor → Purchase Order → Receiving → Inventory / Expense

Relationship Logic

1. A vendor may supply materials, products, services, equipment, documents, or compliance support.
2. A purchase order is created for a vendor.
3. Purchase order line items specify what is being purchased.
4. When physical goods are received, inventory is increased.
5. When services or expenses are billed, bills/expenses are created.
6. Purchase orders may be linked to production needs, inventory alerts, or AI recommendations.

Required Relationships

- Purchase Order must link to Vendor.
- Purchase Order must contain at least one line item.
- Purchase Order line item should define item type: material, product, service, equipment, or other.

Optional Relationships

- Purchase Order may link to Material.
- Purchase Order may link to Inventory Item after receiving.
- Purchase Order may link to Work Order or Production Need.
- Purchase Order may link to Expense/Bill.
- Purchase Order may require Approval Request.

AI Value

This allows AI to answer:

- Which purchase orders are late?
- Which vendors are causing delays?
- What should we reorder now?
- What purchases are needed for upcoming production?
- Which vendor documents are missing?

- How much cash is committed to open purchase orders?

5.7 Production and Batch Relationships

Production connects customer demand, product specifications, materials, inventory, labor, quality, and finished goods.

Flow

Sales Order / Forecast → Work Order → Batch → Material Consumption → Finished Goods Inventory → Fulfillment

Relationship Logic

1. A sales order, forecast, or internal need may create a work order.
2. A work order specifies what product needs to be produced and in what quantity.
3. A work order references required product specifications, SOPs, and materials.
4. A work order may create one or more batches.
5. A batch consumes material inventory.
6. A batch creates finished goods inventory.
7. A batch may require quality checks, COAs, lab results, and compliance documents.
8. Finished goods inventory may be used to fulfill sales orders.

Required Relationships

- Work Order must link to Product.
- Batch must link to Product.
- Batch should link to Work Order when applicable.
- Batch should record input materials and output quantity.

Optional Relationships

- Work Order may link to Sales Order.
- Batch may link to Sales Order.
- Batch may link to COA Document.
- Batch may link to Compliance Requirement.
- Batch may link to Equipment.
- Batch may link to Labor Records in a future version.

AI Value

This allows AI to answer:

- What production is needed to fulfill orders?
- What materials are missing for this work order?
- Which batches are delayed?

- What was the yield for this batch?
- What quality documents are missing?
- Which customer orders depend on this batch?
- What caused the production issue?

5.8 Finance Relationships

Finance records should connect directly to business activity.

Customer-Side Flow

Customer → Sales Order → Invoice → Payment

Vendor-Side Flow

Vendor → Purchase Order / Expense → Bill → Payment

Relationship Logic

1. Customer invoices should link to customers and sales orders.
2. Payments should link to invoices.
3. Vendor bills should link to vendors and, when applicable, purchase orders.
4. Vendor payments should link to bills.
5. Financial records should support cash-flow forecasting.
6. Margin analysis should connect revenue, product cost, inventory cost, production cost, and expenses where possible.

Required Relationships

- Customer Invoice must link to Customer.
- Customer Invoice should link to Sales Order when applicable.
- Payment must link to Invoice or Bill.
- Expense/Bill must link to Vendor or Payee.

Optional Relationships

- Expense/Bill may link to Purchase Order.
- Expense/Bill may link to Project.
- Invoice may link to Document.
- Payment may link to bank transaction in a future integration.

AI Value

This allows AI to answer:

- What is owed to us?

- What do we owe?
- Which invoices are overdue?
- Which customers are slow to pay?
- What is our expected cash position?
- Which products or customers are most profitable?
- What expenses are unusual?

5.9 Document Relationships

Documents should not sit in an isolated file vault.

Every important document should be linked to the business record it supports.

Document Can Link To

- Company
- Customer
- Contact
- Vendor
- Product
- Material
- Inventory Item
- Quote
- Sales Order
- Purchase Order
- Work Order
- Batch
- Invoice
- Expense/Bill
- Compliance Requirement
- Project
- Task
- Approval Request
- Report

Relationship Logic

1. A document may support one or many records.
2. A record may have one or many documents.
3. Documents may have versions.
4. Documents may have review dates, expiration dates, obligations, or extracted tasks.
5. Documents may trigger alerts or compliance requirements.

Required Relationships

- Every document should have a document type.

- Every document should have an owner.
- Every document should have access permissions.

Optional Relationships

- Document may link to multiple business records.
- Document may create tasks.
- Document may create alerts.
- Document may create compliance requirements.
- Document may require approval.

AI Value

This allows AI to answer:

- What does this document say?
- Which records does this document support?
- What deadlines are inside this document?
- What obligations are inside this contract?
- Which documents are missing or expired?
- Which documents need review?

5.10 Task Relationships

Tasks are the execution layer across the entire business.

Relationship Logic

A task should almost always link to a business record.

Examples:

- Follow up with a customer → Customer
- Send revised quote → Quote
- Check order status → Sales Order
- Reorder material → Material or Purchase Order
- Review batch issue → Batch
- Collect payment → Invoice
- Upload COA → Document or Batch
- Review contract → Document
- Complete license renewal → Compliance Requirement
- Approve purchase → Approval Request

Task Can Link To

- Customer

- Contact
- Vendor
- Product
- Material
- Inventory Item
- Quote
- Sales Order
- Purchase Order
- Work Order
- Batch
- Invoice
- Expense/Bill
- Document
- Compliance Requirement
- Project
- Alert
- AI Recommendation
- Approval Request

Required Relationships

- Task must have an owner.
- Task must have a status.
- Task should have a due date unless intentionally unscheduled.
- Task should link to a related record whenever possible.

Optional Relationships

- Task may have watchers.
- Task may have subtasks.
- Task may be recurring.
- Task may require approval.
- Task may originate from AI.
- Task may originate from alert, document, communication, report, or meeting.

AI Value

This allows AI to answer:

- What tasks are overdue?
- Who owns this action?
- What is blocked?
- What tasks relate to this customer or order?
- What recommendations have not been acted on?
- What should be prioritized today?

5.11 Alert Relationships

Alerts identify conditions that require attention.

Alerts should not be generic notifications. They should link to the record causing the issue and should be convertible into tasks or approval requests.

Alert Can Link To

- Inventory Item
- Material
- Product
- Customer
- Quote
- Sales Order
- Purchase Order
- Work Order
- Batch
- Invoice
- Document
- Compliance Requirement
- Project
- User
- Task
- AI Recommendation

Relationship Logic

1. A trigger condition creates an alert.
2. The alert links to the record that caused it.
3. The alert is assigned a severity.
4. The alert may be assigned to an owner.
5. The alert may become a task.
6. The alert may be resolved, dismissed, escalated, or converted into an approval request.

Required Relationships

- Alert must have a type.
- Alert must have a severity.
- Alert must have a related module.
- Alert should have a related record.

AI Value

This allows AI to answer:

- What risks need attention?
- Which alerts are high priority?

- What should be done about this alert?
- Which alerts are unresolved?
- Which risk categories keep recurring?

5.12 Approval Request Relationships

Approval requests create human control over sensitive or high-impact actions.

Approval Request Can Link To

- Quote
- Sales Order
- Purchase Order
- Invoice
- Expense/Bill
- Document
- Product
- Material
- Batch
- Compliance Requirement
- Task
- AI Recommendation
- User/Role
- System Setting

Relationship Logic

1. A user or AI-generated workflow requests approval.
2. The approval request links to the action or record involved.
3. An approver reviews the request.
4. The approver accepts, rejects, requests changes, or escalates.
5. The decision is logged.
6. Approved actions may trigger follow-up workflows.

Required Relationships

- Approval Request must have requester.
- Approval Request must have approver or approval group.
- Approval Request must have status.
- Approval Request must link to the relevant record/action.

AI Value

This allows AI to answer:

- What approvals are waiting?
- Who needs to approve this?

- Why is approval needed?
- What data supports this request?
- What approvals are overdue?

5.13 AI Recommendation Relationships

AI recommendations should be stored as first-class records.

This is important because the company needs to know what AI suggested, what data it used, and whether humans accepted or rejected the recommendation.

AI Recommendation Can Link To

- Any core business object
- Alert
- Task
- Approval Request
- Report
- Activity Log

Relationship Logic

1. AI analyzes source data.
2. AI creates a recommendation.
3. The recommendation links to the source records used.
4. A human may accept, reject, modify, or ignore it.
5. Accepted recommendations may become tasks, alerts, drafts, reports, or approval requests.
6. The outcome is stored for future evaluation.

Required Relationships

- AI Recommendation must have a type.
- AI Recommendation must include source data references when possible.
- AI Recommendation must have status.
- AI Recommendation must record human decision when reviewed.

AI Value

This allows the business to track AI usefulness over time and improve trust, accuracy, and adoption.

5.14 Activity Log Relationships

Activity logs support traceability and auditability across the entire platform.

Activity Log Can Link To

- Any core business object
- User
- AI Recommendation
- Approval Request
- Document
- System Setting

Relationship Logic

1. Important activity occurs.
2. The system creates a log entry.
3. The log entry records user/system/AI, action, timestamp, related record, and relevant before/after values.
4. Logs can be reviewed for audits, security, accountability, and troubleshooting.

Required Relationships

- Activity Log must have actor: user, system, or AI.
- Activity Log must have action type.
- Activity Log must have timestamp.
- Activity Log should link to related record.

AI Value

This allows AI to answer:

- Who changed this?
- When was this approved?
- What happened before this issue?
- What did AI generate?
- Which user completed this action?

5.15 Parent-Child Relationship Summary

The following relationships should be treated as major parent-child patterns.

Company

Company has many:

- Users
- Customers
- Vendors
- Products

- Materials
- Inventory Items
- Orders
- Documents
- Tasks
- Alerts
- Reports
- Activity Logs

Customer

Customer has many:

- Contacts
- Quotes
- Sales Orders
- Invoices
- Payments
- Documents
- Tasks
- Notes/Communications

Vendor

Vendor has many:

- Contacts
- Purchase Orders
- Bills/Expenses
- Documents
- Tasks
- Notes/Communications

Product

Product has many:

- Sales Order Lines
- Quote Lines
- Work Orders
- Batches
- Inventory Items
- Documents
- Compliance Requirements

Material

Material has many:

- Inventory Items
- Purchase Order Lines
- Batch Inputs
- Vendor Relationships
- Documents

Sales Order

Sales Order has many:

- Line Items
- Tasks
- Documents
- Work Orders
- Invoices
- Alerts

Work Order

Work Order has many:

- Batches
- Tasks
- Documents
- Required Materials
- Alerts

Batch

Batch has many:

- Material Inputs
- Inventory Outputs
- Quality Documents
- Tasks
- Alerts
- Activity Logs

Document

Document may have many:

- Linked Records
- Extracted Obligations

- Tasks
- Alerts
- Versions

5.16 Many-to-Many Relationship Tables

Many-to-many relationships should be managed with link tables.

Recommended Link Tables

Product Materials

Connects products to the materials required to produce them.

Fields:

- Product ID
- Material ID
- Quantity required
- Unit of measure
- Waste factor
- Optional/required flag

Vendor Materials

Connects vendors to the materials they can supply.

Fields:

- Vendor ID
- Material ID
- Vendor SKU
- Unit cost
- Lead time
- Minimum order quantity
- Preferred vendor flag

Document Links

Connects documents to any related business record.

Fields:

- Document ID
- Related object type
- Related object ID

- Relationship type

Task Links

Connects tasks to one or more related business records.

Fields:

- Task ID
- Related object type
- Related object ID
- Relationship type

Compliance Links

Connects compliance requirements to products, vendors, batches, locations, documents, or company-level obligations.

Fields:

- Compliance Requirement ID
- Related object type
- Related object ID
- Applies from date
- Applies until date

Project Participants

Connects users to projects.

Fields:

- Project ID
- User ID
- Project role
- Responsibility

AI Source Links

Connects AI recommendations to the records used as source data.

Fields:

- AI Recommendation ID
- Source object type
- Source object ID
- Source relevance

5.17 Status Logic Across Objects

Each object should have clear statuses so dashboards and AI can reason about the business.

Example Customer Statuses

- Lead
- Prospect
- Active
- Inactive
- At Risk
- Lost
- Do Not Contact

Example Quote Statuses

- Draft
- Needs Approval
- Sent
- Accepted
- Rejected
- Expired
- Converted to Order

Example Sales Order Statuses

- Draft
- Confirmed
- Awaiting Inventory
- Awaiting Production
- In Production
- Ready to Fulfill
- Fulfilled
- Invoiced
- Paid
- Cancelled

Example Purchase Order Statuses

- Draft
- Needs Approval
- Sent
- Partially Received
- Received
- Closed
- Cancelled
- Delayed

Example Work Order Statuses

- Draft
- Scheduled
- Waiting on Materials
- In Progress
- QC Hold
- Completed
- Cancelled

Example Batch Statuses

- Planned
- In Progress
- Completed
- QC Hold
- Released
- Failed
- Archived

Example Task Statuses

- Open
- In Progress
- Blocked
- Waiting on Approval
- Completed
- Cancelled
- Deferred

Example Document Statuses

- Draft
- Active
- Under Review
- Expired
- Superseded
- Archived

Example Alert Statuses

- Open
- Assigned
- In Progress
- Resolved
- Dismissed
- Escalated

Example Approval Statuses

- Pending
- Approved
- Rejected
- Changes Requested
- Escalated
- Cancelled

5.18 Required Cross-Module Connections

The following cross-module connections are essential for the system to function correctly.

Customer and Sales

- Customer must connect to quotes, orders, invoices, communications, documents, and tasks.

Orders and Inventory

- Sales orders should connect to inventory availability and reservations.

Orders and Production

- Sales orders should connect to work orders or batches when production is required.

Production and Inventory

- Batches must consume materials and create finished goods inventory.

Purchasing and Inventory

- Purchase orders should increase inventory when received.

Finance and Sales

- Invoices should connect to customers and sales orders.

Finance and Purchasing

- Bills/expenses should connect to vendors and purchase orders when applicable.

Documents and Everything

- Documents should link to the records they support.

Tasks and Everything

- Tasks should link to the records they affect.

AI and Everything

- AI recommendations should link to the records and source data used.

Audit Logs and Everything

- Important changes and actions should create activity logs.

5.19 Data Ownership Rules

Each major record should have a clear owner.

Ownership Principles

1. Every customer should have an account owner.
2. Every vendor should have a relationship owner or department owner.
3. Every task must have an owner.
4. Every document should have an owner.
5. Every compliance requirement must have an owner.
6. Every project must have an owner.
7. Every alert should be assigned to an owner when action is required.
8. Every approval request must have an approver.
9. Every AI recommendation should have a review status or owner when action is needed.

Ownership makes accountability possible.

5.20 Data Quality Rules

AI and reporting are only as good as the underlying data.

The system should enforce minimum data quality rules.

Required Data Quality Standards

- Required fields must be completed before key status changes.
- Critical records should not be left without owners.
- Duplicate customers, vendors, products, and materials should be prevented where possible.
- Status changes should be logged.
- Important records should have timestamps.
- Important records should have created by and last updated by fields.
- Financial values should use consistent currency formatting.
- Units of measure should be standardized.
- Inventory changes should be recorded through transactions rather than manual overwrites where possible.
- AI-generated content should be labeled as AI-generated.

- Human approvals should be recorded before controlled actions are executed.

5.21 Reporting Relationship Requirements

The data model should support reporting across modules.

Reports the Data Model Must Support

- Daily leadership briefing
- Customer follow-up report
- Sales pipeline report
- Quote status report
- Open order report
- Orders at risk report
- Low inventory report
- Purchase order status report
- Production readiness report
- Batch performance report
- Receivables report
- Payables report
- Cash-flow forecast
- Document expiration report
- Compliance deadline report
- Overdue task report
- AI recommendation report
- Approval queue report
- Activity/audit report

These reports should be possible because records are properly linked.

5.22 AI Relationship Requirements

For AI to be trustworthy and useful, it must understand relationships between records.

AI Must Be Able To Trace

- Customer to orders
- Orders to inventory
- Orders to production
- Production to materials
- Materials to vendors
- Vendors to purchase orders
- Orders to invoices
- Invoices to payments

- Documents to obligations
- Alerts to source records
- Tasks to owners
- Recommendations to source data
- Approvals to decisions
- Activity logs to record history

AI Output Requirements

When AI gives an important recommendation, it should identify:

- What it noticed
- Why it matters
- Which records were used
- What action is recommended
- Who should own the action
- Whether approval is required
- What risk exists if no action is taken

5.23 MVP Data Model Scope

The MVP should include enough relationship structure to support daily management, sales follow-up, inventory awareness, purchasing, tasks, documents, alerts, approvals, and AI recommendations.

MVP Required Relationships

- Company to Users
- Users to Roles
- Customers to Contacts
- Customers to Quotes
- Customers to Sales Orders
- Customers to Tasks
- Customers to Documents
- Vendors to Contacts
- Vendors to Purchase Orders
- Vendors to Documents
- Products to Materials
- Products to Sales Orders
- Products to Inventory Items
- Materials to Vendors
- Materials to Inventory Items
- Materials to Purchase Orders
- Sales Orders to Customers
- Sales Orders to Products
- Sales Orders to Tasks
- Purchase Orders to Vendors

- Purchase Orders to Materials/Products/Services
- Tasks to Users
- Tasks to Related Records
- Documents to Related Records
- Alerts to Related Records
- AI Recommendations to Related Records
- Approval Requests to Related Records
- Activity Logs to Users and Records

MVP Deferred Relationships

- Work Orders to Batches
- Batches to Input/Output Inventory Transactions
- Payments to Bank Transactions
- Compliance Requirements to Jurisdictions
- Projects to Budgets
- Reports to KPI Snapshots
- Labor Records to Production
- Equipment Records to Production
- Shipment/Fulfillment Records to Orders

These deferred relationships should be planned for but do not need to be fully built in the first version.

5.24 Data Model Design Rule

The system should be designed so that every important record can answer:

1. What is this?
2. Who owns it?
3. What is its current status?
4. What is it connected to?
5. What tasks are attached?
6. What documents support it?
7. What alerts or risks exist?
8. What approvals are pending?
9. What did AI recommend?
10. What changed over time?

If the data model supports these questions, the platform can scale into a powerful AI-assisted business operating system.

Step 6 — User Roles and Permission Model

6.1 Purpose of This Section

This section defines who can use the AI Business Command Center, what each user can access, what actions each user can perform, what actions require approval, and how AI must respect permission boundaries.

The permission model is critical because the system may contain sensitive information, including:

- Financial records
- Customer records
- Vendor pricing
- Product specifications
- Production information
- Legal documents
- Compliance documents
- Patent/IP information
- Employee/task accountability data
- Strategic business plans
- AI-generated recommendations

The system must be useful, but it must also be controlled, auditable, and safe.

6.2 Permission Model Philosophy

The system should follow a role-based access control model.

This means users receive permissions based on their role, department, authority level, and business need.

The guiding principle is:

Users should have enough access to do their job, but not more access than necessary.

AI must follow the same rule.

If a user does not have permission to view financial records, they should not be able to ask AI to summarize financial records. If a user cannot approve purchase orders, AI should not allow them to approve a purchase order through chat or automation.

6.3 Permission Layers

The system should support multiple permission layers.

1. Module Access

Controls which areas of the system a user can access.

Examples:

- Command Center
- Customers and Sales
- Orders and Quotes
- Inventory and Purchasing
- Production and Batches
- Finance and Cash Flow
- Compliance, Legal, and IP
- Documents
- Tasks
- Reports
- Settings

2. Record Access

Controls which specific records a user can see.

Examples:

- A sales user may see only their assigned customers.
- A production user may see only production and inventory records.
- An outside advisor may see only selected documents or projects.

3. Action Access

Controls what a user can do with records.

Examples:

- View
- Create
- Edit
- Delete/archive
- Approve
- Export
- Assign
- Send
- Upload
- Download
- Override

4. Field-Level Access

Controls access to sensitive fields within a record.

Examples:

- Product cost
- Gross margin
- Customer credit limit
- Bank information
- Legal notes
- Patent strategy notes
- Employee compensation
- Sensitive compliance details

5. AI Access

Controls what AI can use, summarize, recommend, draft, or automate for each user.

AI access should be derived from the user's own permissions.

6. Approval Authority

Controls what users can approve and at what dollar amount or risk level.

Examples:

- Purchase orders up to a certain dollar amount
- Discounts above a threshold
- Vendor payments
- Customer credits
- Compliance status changes
- Document approvals
- Production specification changes

6.4 Standard User Roles

The initial system should support the following standard roles.

Owner / CEO

Purpose

Full executive access to the business.

Typical Access

- Full command center
- Sales
- Orders
- Inventory
- Production
- Finance
- Compliance
- Documents
- Tasks
- Projects
- Reports
- AI assistant
- Settings visibility

Typical Authority

- Approve high-value purchase orders
- Approve major discounts
- Approve financial decisions
- Approve legal/compliance actions
- Approve role and permission changes
- Override certain workflows
- View all AI recommendations

Restrictions

Few restrictions, but all actions should still be logged.

System Administrator

Purpose

Manages system configuration, users, permissions, settings, integrations, and technical controls.

Typical Access

- User management
- Role management
- Settings
- Audit logs
- Integrations
- Automation rules
- System health

Typical Authority

- Create users
- Modify roles
- Configure permissions
- Manage integrations
- Configure system settings
- View audit logs

Restrictions

A system administrator does not automatically need full access to business-sensitive financial, legal, or IP content unless separately granted.

Operations Manager

Purpose

Manages production, inventory, purchasing coordination, work orders, batches, and operational execution.

Typical Access

- Command Center operational view
- Inventory and Purchasing
- Production and Batches
- Products and Materials
- Vendors
- Tasks
- Operational documents
- Selected reports
- AI operational insights

Typical Authority

- Create and update work orders
- Update production status
- Manage batch records
- Manage inventory status
- Create purchase requests
- Approve low-risk operational tasks
- Assign operational tasks

Restrictions

May not access full financial reports, legal/IP documents, or company-wide strategic records unless granted.

Sales Manager

Purpose

Manages leads, customers, opportunities, quotes, sales orders, customer communications, and follow-ups.

Typical Access

- Customers and Sales
- Quotes
- Sales Orders
- Customer documents
- Customer tasks
- Sales reports
- AI sales support

Typical Authority

- Create leads/customers
- Create quotes
- Send approved communications
- Manage sales follow-ups
- Assign sales tasks
- Request quote approval
- View customer revenue summaries where permitted

Restrictions

May not view full company financials, vendor cost data, production formulas, legal/IP documents, or sensitive compliance content unless granted.

Finance Manager

Purpose

Manages receivables, payables, invoices, payments, cash-flow visibility, financial reporting, and financial risk.

Typical Access

- Finance and Cash Flow
- Invoices
- Payments
- Bills/Expenses
- Customer financial status
- Vendor payment status
- Financial reports

- Related documents
- AI financial insights

Typical Authority

- Update invoice/payment status
- Create financial summaries
- Approve or request payment workflows
- Flag collection priorities
- Manage finance-related tasks
- View financial audit logs

Restrictions

May not edit production specs, compliance approvals, or system-level permissions unless granted.

Compliance / Legal / IP Manager

Purpose

Manages compliance requirements, contracts, SOPs, COAs, licenses, legal documents, patent/IP matters, obligations, and deadlines.

Typical Access

- Compliance, Legal, and IP
- Documents
- Products related to compliance
- Batches related to compliance
- Vendor/customer compliance documents
- Compliance reports
- AI document intelligence

Typical Authority

- Create compliance requirements
- Upload and classify documents
- Review obligations
- Assign compliance tasks
- Approve document status changes
- Flag legal/compliance risk

Restrictions

May not modify financial records, production records, or sales records except where they relate to compliance/legal review.

Inventory / Warehouse User

Purpose

Updates inventory counts, receives materials, performs inventory movements, and supports purchasing/fulfillment.

Typical Access

- Inventory items
- Materials
- Products
- Purchase orders related to receiving
- Inventory tasks
- Inventory documents

Typical Authority

- Receive inventory
- Update counts where permitted
- Transfer stock
- Flag low stock
- Upload receiving documents
- Complete inventory tasks

Restrictions

May not view financial margins, customer pricing, legal/IP documents, or company-wide reports unless granted.

Production Operator

Purpose

Performs and updates assigned production work.

Typical Access

- Assigned work orders
- Assigned batches
- Production instructions
- Related SOPs
- Required materials
- Production tasks
- Quality checklist items where permitted

Typical Authority

- Update production status
- Record production notes
- Record material usage
- Report issues/deviations
- Complete assigned tasks

Restrictions

May not access full customer records, financial data, vendor pricing, strategic reports, or system settings unless granted.

Employee / General User

Purpose

Completes assigned work and views relevant records needed to do their job.

Typical Access

- Assigned tasks
- Assigned records
- Relevant documents
- Limited dashboards
- Limited AI assistance

Typical Authority

- Complete assigned tasks
- Add notes
- Upload allowed documents
- Request approvals
- Update allowed fields

Restrictions

No broad access by default.

Outside Advisor

Purpose

Allows controlled access for attorneys, accountants, consultants, advisors, investors, auditors, or technical partners.

Typical Access

- Specific documents
- Specific projects
- Specific reports
- Specific tasks
- Limited AI summaries of permitted data

Typical Authority

- View permitted records
- Comment where allowed
- Upload documents where allowed
- Complete assigned advisory tasks

Restrictions

No default access to full system data. Access should be intentionally granted, time-limited when appropriate, and logged.

Read-Only User

Purpose

Allows visibility without editing authority.

Typical Access

- Selected dashboards
- Selected reports
- Selected records
- Selected documents

Typical Authority

- View only
- Export only if explicitly granted

Restrictions

Cannot create, edit, delete, approve, or trigger AI actions that modify records.

6.5 Permission Actions

The system should define permissions at the action level.

Common Actions

- View
- Create
- Edit
- Delete
- Archive
- Restore
- Assign owner
- Change status
- Upload document
- Download document
- Export data
- Approve
- Reject
- Send communication
- Generate report
- Create AI recommendation
- Convert AI recommendation to task
- Run automation
- Configure automation
- Manage users
- Manage permissions

Sensitive Actions

The following should require stronger permissions and logging:

- Delete or archive records
- Export customer data
- Export financial data
- Send customer-facing communication
- Issue quote
- Approve purchase order
- Approve vendor payment
- Change product specification
- Change production formula/specification
- Change compliance status
- Change legal/IP record status
- Grant user permissions
- Modify system settings
- Run bulk updates
- Override alerts or controls

6.6 Approval Authority Levels

The system should support approval authority levels.

Level 0 — No Approval Authority

Can request actions but cannot approve controlled actions.

Level 1 — Low-Risk Operational Approval

Can approve low-risk internal actions, such as routine task completion, minor inventory adjustments, or non-sensitive document updates.

Level 2 — Department Approval

Can approve actions within a department, such as sales quotes, production changes, or purchasing requests below defined thresholds.

Level 3 — Financial / Compliance Approval

Can approve higher-risk actions involving money, compliance, legal documentation, or customer commitments.

Level 4 — Executive Approval

Can approve high-value, high-risk, strategic, financial, legal, or system-level actions.

Level 5 — Administrative/System Approval

Can approve user access, permission changes, integrations, system settings, and automation rules.

6.7 Approval Rules by Action Type

Customer Communication

AI or users may draft customer communications, but sending should require approval unless the user has permission to send directly.

Quotes

Quotes should require approval when:

- Discount exceeds threshold
- Margin is below target
- Order size exceeds threshold

- Customer has overdue balance
- Non-standard terms are used

Purchase Orders

Purchase orders should require approval when:

- Dollar amount exceeds threshold
- Vendor is new or unapproved
- Item is outside normal purchasing category
- Purchase is not tied to demand or approved project
- Cash-flow risk exists

Inventory Adjustments

Inventory adjustments should require approval when:

- Quantity adjustment exceeds threshold
- Value adjustment exceeds threshold
- Adjustment affects regulated/sensitive materials
- Adjustment reverses prior transaction

Production Changes

Production changes should require approval when:

- Product specification changes
- Formula or process changes
- Batch is released from QC hold
- Material substitution occurs
- Deviation is closed

Financial Records

Financial actions should require approval when:

- Payment is issued
- Customer credit is applied
- Invoice is voided
- Bad debt/write-off is recorded
- Terms are changed

Compliance / Legal / IP

Compliance and legal actions should require approval when:

- Compliance status is changed
- Contract is marked approved

- Legal obligation is closed
- IP deadline is modified
- Sensitive document is archived

User Permissions

Permission changes should always be logged and should require administrative authority.

6.8 AI Permission Rules

AI must operate under strict permission boundaries.

Rule 1 — AI Inherits User Permissions

AI can only access, summarize, analyze, or act on information that the requesting user is allowed to access.

Rule 2 — AI Cannot Bypass Approval Rules

If an action requires approval in the system, AI cannot complete it without the required approval.

Rule 3 — AI Must Identify Source Data

Important AI answers should identify the records or data categories used to generate the answer.

Rule 4 — AI-Generated Content Must Be Labeled

Tasks, recommendations, drafts, summaries, and reports created by AI should be marked as AI-generated.

Rule 5 — AI Actions Must Be Logged

AI-created recommendations, drafts, tasks, alerts, reports, and automated actions should create activity log entries.

Rule 6 — AI Should Not Make Binding Commitments Without Approval

AI should not independently send customer emails, issue quotes, approve purchases, make financial commitments, alter legal/compliance status, or change production specifications without human approval.

Rule 7 — AI Must Respect Sensitive Fields

If a user cannot view a field, AI should not reveal or infer that field in a response.

Rule 8 — AI Should Explain Risk

When AI recommends an action, it should explain the business risk if no action is taken.

6.9 Sensitive Data Categories

The system should classify sensitive data so access can be controlled.

Financial Sensitive Data

- Cash balances
- Bank information
- Profitability
- Gross margin
- Cost of goods
- Vendor pricing
- Customer credit limits
- Payables
- Receivables
- Payroll or compensation

Legal / IP Sensitive Data

- Patent strategy
- Draft patent filings
- Attorney communications
- Legal disputes
- Contracts
- NDAs
- Settlement discussions
- Confidential inventions

Compliance Sensitive Data

- Regulatory filings
- License information
- COAs
- SOP deviations
- Quality issues
- Audit findings
- Corrective actions

Operational Sensitive Data

- Product formulas/specifications
- Production processes
- Batch issues
- Vendor sourcing strategy
- Internal pricing strategy

Customer Sensitive Data

- Contact information
- Payment terms
- Credit status
- Customer-specific pricing
- Contract terms
- Complaint history

6.10 Access Levels

The system should support standard access levels.

No Access

User cannot see the module, record, or field.

View Only

User can view but not modify.

Comment / Note

User can add notes or comments but cannot change core fields.

Create

User can create new records.

Edit

User can modify records.

Manage

User can create, edit, assign, archive, and manage records within a module.

Approve

User can approve controlled actions within defined authority limits.

Admin

User can configure settings, permissions, and system behavior.

6.11 Default Role Matrix

The following is the recommended default permission model. Final permissions can be adjusted during implementation.

Module	CEO	Admin	Ops Manager	Sales Manager	Finance Manager	Compliance Manager	Inventory User	Production User	Outsi Advis
Command Center	Full	Config	Operational	Sales	Financial	Compliance	Limited	Limited	Limit
Customers & Sales	Full	Config	View	Manage	Financial View	Limited	None	None	Limit
Orders & Quotes	Full	Config	Operational View	Manage	Financial View	Limited	Limited	Limited	Limit
Production & Batches	Full	Config	Manage	Limited	Cost View	Compliance View	Limited	Assigned	Limit
Inventory & Purchasing	Full	Config	Manage	Limited	Financial View	Compliance View	Manage Limited	Limited	Limit
Finance & Cash Flow	Full	Config Limited	Limited	Limited	Manage	Limited	None	None	Limit
Compliance, Legal & IP	Full	Config Limited	Limited	None	Limited	Manage	None	Limited	Assig
Documents	Full	Config	Operational	Sales Docs	Financial Docs	Compliance Docs	Assigned	Assigned	Assig
Tasks	Full	Config	Manage Dept	Manage Dept	Manage Dept	Manage Dept	Assigned	Assigned	Assig
Projects	Full	Config	Assigned	Assigned	Assigned	Assigned	Assigned	Assigned	Assig
Reports	Full	Config	Operational	Sales	Financial	Compliance	Limited	Limited	Assig
Settings	Full	Manage	None	None	None	None	None	None	None
Audit Logs	Full	Manage	Limited	Limited	Financial	Compliance	None	None	None

6.12 Record Ownership Rules

Record ownership controls accountability and may affect visibility.

Recommended Ownership Rules

- Customers should have an account owner.

- Vendors should have a relationship owner.
- Products should have a product owner or department owner.
- Materials should have an inventory/purchasing owner.
- Sales orders should have an order owner.
- Work orders should have an operations owner.
- Batches should have a production owner.
- Invoices should have a finance owner.
- Documents should have a document owner.
- Compliance requirements should have a compliance owner.
- Projects should have a project owner.
- Tasks must always have an owner.
- Alerts should have an owner when action is required.
- AI recommendations should have a reviewer when action is required.

6.13 Permission Rules for Documents

Documents require special permission handling because a single document may contain sensitive information.

Document Access Controls

Each document should include:

- Document type
- Owner
- Related records
- Sensitivity level
- View permissions
- Download permissions
- Edit permissions
- Share permissions
- Expiration/review status

Document Sensitivity Levels

- Public/Internal General
- Internal Restricted
- Financial Confidential
- Legal Confidential
- Compliance Sensitive
- IP/Patent Confidential
- Executive Only

AI Document Rules

AI may summarize or answer questions about a document only if the user has permission to view that document.

AI should not reveal restricted document content through summaries, reports, or cross-record analysis to users without access.

6.14 Permission Rules for Reports

Reports may combine data from multiple modules, so access must be carefully controlled.

Report Access Rules

- A user may only view a report if they have access to the underlying data.
- Sensitive fields should be hidden or aggregated when appropriate.
- Executive reports should be restricted to leadership.
- Financial reports should be restricted to authorized finance/executive users.
- Compliance/legal reports should be restricted to authorized users.
- AI-generated reports should follow the same restrictions.

6.15 Permission Rules for Exports

Exporting data creates additional risk.

Export Controls

Exports should be restricted for:

- Customer lists
- Vendor lists
- Financial reports
- Product costs
- Inventory value
- Legal/IP documents
- Compliance records
- Audit logs
- Employee/user data

Exports should create activity log entries.

High-sensitivity exports may require approval.

6.16 Audit and Logging Requirements

The system should log important permission-sensitive events.

Events to Log

- User login/logout
- Failed login attempts
- Permission changes
- Role changes
- Sensitive record views
- Record creation
- Record edits
- Record deletion/archive
- Document uploads/downloads
- Data exports
- Approval decisions
- AI recommendations
- AI-generated drafts
- AI-created tasks
- Automation execution
- System setting changes

Log Fields

- User/system/AI actor
- Action type
- Timestamp
- Related record
- Previous value where applicable
- New value where applicable
- Source IP/device where applicable
- Approval reference where applicable

6.17 MVP Permission Scope

The MVP permission model should be simple but strong enough to protect sensitive information.

MVP Roles

The MVP should include:

1. Owner / CEO
2. System Administrator
3. Operations Manager
4. Sales Manager

5. Finance Manager
6. Compliance Manager
7. Inventory / Warehouse User
8. Production Operator
9. Outside Advisor
10. Read-Only User

MVP Permission Features

The MVP should include:

- Role-based module access
- Record ownership
- Basic action permissions
- Document sensitivity levels
- Approval authority levels
- Activity logs
- AI permission inheritance
- AI-generated content labels
- Approval requests for sensitive actions

Deferred Advanced Permission Features

Future versions may include:

- Advanced field-level permissions
- Time-limited external advisor access
- Attribute-based permissions
- IP/device-based access controls
- Multi-step approvals
- Department-specific dashboards
- Advanced anomaly detection for user behavior
- Automated periodic permission reviews

6.18 Permission Design Rule

Every permission decision should answer:

1. Who is the user?
2. What role do they have?
3. What record are they trying to access?
4. What action are they trying to perform?
5. Does the record contain sensitive data?
6. Does the action require approval?
7. Is AI involved?
8. Should this action be logged?

9. Should the user see the full data, limited data, or no data?
10. Who is responsible if the action creates risk?

This rule ensures that the system remains useful, controlled, auditable, and scalable.

Step 7 — Key Workflows

7.1 Purpose of This Section

This section defines the major workflows the AI Business Command Center must support.

A workflow is the step-by-step path that business activity follows through the system.

The purpose of defining workflows is to make sure the system does not merely store records, but actually helps the company operate.

Each workflow should define:

- What starts the workflow
- What records are created or updated
- Who is responsible
- What approvals are required
- What AI can assist with
- What alerts may be triggered
- What tasks are created
- What documents may be required
- What the successful end state looks like

The system should make work visible, structured, accountable, and auditable.

7.2 Workflow Design Principles

Every workflow should follow these principles.

1. Clear Trigger

Every workflow should have a clear starting event.

Examples:

- New lead received
- Customer requests quote
- Quote approved

- Sales order confirmed
- Inventory falls below reorder point
- Purchase order received
- Production is needed
- Invoice becomes overdue
- Document is uploaded
- Compliance deadline approaches
- AI detects risk

2. Clear Owner

Every workflow should have a responsible owner or department.

3. Clear Status

Each workflow should move through defined statuses so the business can see progress.

4. Clear Approval Rules

Sensitive actions should require approval before completion.

5. Clear AI Role

AI should support workflows by summarizing, recommending, drafting, monitoring, or creating tasks, but should not bypass approval rules.

6. Clear Audit Trail

Important workflow actions should create activity log entries.

7. Clear Completion Criteria

Each workflow should have a known finish point.

7.3 Workflow 1 — Lead to Customer

Purpose

Convert new leads into qualified prospects and active customers while ensuring follow-up does not get missed.

Trigger

- New lead entered manually
- Lead imported from form, email, event, referral, website, CRM, or spreadsheet
- AI identifies a potential lead from communication or document

Primary Records

- Customer
- Contact
- Note / Communication
- Task
- Document
- AI Recommendation

Workflow Steps

1. Lead is created.
2. Contact information is added.
3. Lead source is recorded.
4. Sales owner is assigned.
5. Lead status is set.
6. Initial follow-up task is created.
7. AI may summarize available context.
8. Sales owner qualifies the lead.
9. Lead becomes prospect, active customer, disqualified, or lost.
10. Follow-up history is recorded.

Statuses

- New Lead
- Contacted
- Qualified
- Prospect
- Active Customer
- Disqualified
- Lost
- Do Not Contact

AI Support

AI can:

- Draft first outreach email
- Summarize lead source/context
- Recommend follow-up timing
- Identify stale leads
- Suggest qualification questions
- Create follow-up tasks

Approval Rules

- Sending outbound communication may require approval depending on user permissions.
- Creating a customer record does not usually require approval.

Alerts

- Lead not contacted within target timeframe
- Follow-up overdue
- Prospect inactive too long
- Missing contact information

Completion Criteria

The workflow is complete when the lead becomes an active customer, is disqualified, or is marked lost/do not contact.

7.4 Workflow 2 — Quote to Order

Purpose

Convert customer interest into approved quotes and confirmed sales orders while protecting pricing, margin, and customer commitments.

Trigger

- Customer requests pricing
- Sales user creates quote
- AI recommends quote creation from customer communication

Primary Records

- Customer
- Contact
- Quote
- Product
- Task
- Approval Request
- Document
- AI Recommendation

Workflow Steps

1. Quote is created for customer.
2. Products/services, quantities, pricing, and terms are added.
3. System calculates estimated margin where cost data is available.
4. AI checks for missing information, low margin, unusual discount, or customer risk.
5. Quote is submitted for approval if required.
6. Approver reviews quote.
7. Quote is approved, rejected, or changes are requested.
8. Approved quote is sent to customer.
9. Customer accepts, rejects, or requests changes.

10. Accepted quote converts to sales order.

Statuses

- Draft
- Needs Approval
- Approved
- Sent
- Accepted
- Rejected
- Expired
- Converted to Order

AI Support

AI can:

- Draft quote language
- Flag low-margin quotes
- Flag unusual terms
- Summarize customer history
- Suggest pricing review
- Draft customer follow-up
- Identify expiring quotes

Approval Rules

Quote approval should be required if:

- Discount exceeds threshold
- Estimated margin is below target
- Customer has overdue balance
- Non-standard payment terms are used
- Quote exceeds dollar threshold
- Product requires compliance/legal review

Alerts

- Quote waiting for approval
- Quote expiring soon
- Quote accepted but not converted
- Quote sent with no follow-up
- Quote margin below target

Completion Criteria

The workflow is complete when the quote is converted to a sales order, rejected, expired, or cancelled.

7.5 Workflow 3 — Sales Order to Fulfillment

Purpose

Manage confirmed customer orders through inventory allocation, production needs, fulfillment, invoicing, and payment.

Trigger

- Accepted quote converts to sales order
- Sales order is created directly

Primary Records

- Customer
- Sales Order
- Product
- Inventory Item
- Work Order
- Batch
- Invoice
- Payment
- Task
- Alert
- Document

Workflow Steps

1. Sales order is created.
2. Products, quantities, due dates, terms, and shipping/delivery requirements are confirmed.
3. System checks available inventory.
4. Inventory is reserved if available.
5. If inventory is not available, system flags production or purchasing need.
6. Work order is created if production is required.
7. Purchase order is recommended if materials are missing.
8. Order status is updated throughout fulfillment.
9. Customer updates are sent as needed.
10. Order is fulfilled or marked ready for shipment/delivery.
11. Invoice is created.
12. Payment is tracked.

Statuses

- Draft
- Confirmed
- Awaiting Inventory
- Awaiting Production
- In Production

- Ready to Fulfill
- Fulfilled
- Invoiced
- Paid
- Cancelled

AI Support

AI can:

- Summarize order readiness
- Identify inventory gaps
- Recommend production or purchasing actions
- Draft customer status updates
- Flag orders at risk
- Prioritize orders by due date, value, or risk
- Create tasks for fulfillment blockers

Approval Rules

Approval may be required for:

- Non-standard order terms
- Customer credit risk
- Inventory override
- Expedited production
- Large order commitments
- Customer-facing updates if user lacks sending permission

Alerts

- Order due soon
- Order overdue
- Inventory unavailable
- Production delayed
- Invoice not created after fulfillment
- Payment overdue

Completion Criteria

The workflow is complete when the order is fulfilled, invoiced, paid, and all related tasks are closed.

7.6 Workflow 4 — Order to Production

Purpose

Convert customer demand or internal demand into production work that is scheduled, completed, quality-checked, and converted into finished goods.

Trigger

- Sales order requires unavailable product
- Forecasted demand requires production
- Internal stock replenishment need
- AI detects production need based on inventory and order demand

Primary Records

- Sales Order
- Product
- Material
- Inventory Item
- Work Order
- Batch
- Document
- Task
- Alert
- AI Recommendation

Workflow Steps

1. Production need is identified.
2. Work order is created.
3. Product, target quantity, due date, and priority are defined.
4. Required materials are checked.
5. Required SOPs/specifications are attached.
6. Missing materials trigger purchase recommendation or alert.
7. Work order is scheduled.
8. Batch or production run is created.
9. Materials are consumed.
10. Production output is recorded.
11. Yield, waste, and deviations are recorded.
12. Quality-control status is updated.
13. Finished goods inventory is created.
14. Related sales order is updated.

Statuses

- Draft
- Waiting on Materials

- Scheduled
- In Progress
- QC Hold
- Completed
- Released
- Cancelled

AI Support

AI can:

- Recommend production priority
- Check material readiness
- Identify missing documents/SOPs
- Summarize production status
- Detect yield or waste issues
- Create corrective action tasks
- Flag delayed batches
- Recommend purchase orders for missing inputs

Approval Rules

Approval may be required for:

- Formula/specification changes
- Material substitutions
- QC release
- Deviation closure
- Production priority override
- Batch cancellation

Alerts

- Missing materials
- Missing SOP/specification
- Production delayed
- Batch on QC hold
- Yield below expected range
- Waste above expected range
- Finished goods not created after production

Completion Criteria

The workflow is complete when finished goods inventory is created, quality status is resolved, linked sales order is updated, and all production tasks are closed.

7.7 Workflow 5 — Purchasing and Receiving

Purpose

Ensure materials, packaging, products, services, and supplies are purchased on time, approved when required, received properly, and connected to inventory or expense records.

Trigger

- Inventory falls below reorder point
- Production requires missing material
- User creates purchase request
- AI recommends purchase
- Vendor quote/proposal is received

Primary Records

- Vendor
- Material
- Product
- Purchase Order
- Inventory Item
- Expense/Bill
- Approval Request
- Task
- Document
- AI Recommendation

Workflow Steps

1. Purchasing need is identified.
2. System checks preferred vendors and lead times.
3. Purchase order draft is created.
4. AI may recommend vendor, quantity, and timing.
5. Purchase order is submitted for approval if required.
6. Approved purchase order is sent to vendor.
7. Expected delivery date is tracked.
8. Goods/services are received.
9. Inventory is updated for received goods.
10. Vendor bill/expense is recorded if applicable.
11. Documents such as receipts, COAs, or packing slips are attached.
12. Purchase order is closed.

Statuses

- Draft
- Needs Approval
- Approved

- Sent
- Partially Received
- Received
- Closed
- Delayed
- Cancelled

AI Support

AI can:

- Recommend reorder quantity
- Recommend vendor based on lead time/cost
- Draft purchase orders
- Flag late purchase orders
- Identify purchase orders tied to production needs
- Summarize vendor performance
- Flag missing receiving documents

Approval Rules

Purchase order approval should be required if:

- Dollar value exceeds threshold
- Vendor is new
- Item is unusual or non-standard
- Purchase is not tied to demand/project
- Cash-flow concern exists
- User lacks purchasing authority

Alerts

- Material below reorder point
- Purchase order waiting approval
- Purchase order not sent after approval
- Expected delivery date missed
- Partial receipt unresolved
- Missing receiving documents

Completion Criteria

The workflow is complete when the order is fully received, inventory or expense records are updated, required documents are attached, and related tasks are closed.

7.8 Workflow 6 — Inventory Adjustment and Movement

Purpose

Track changes to inventory accurately and maintain traceability for stock levels, locations, lots, batches, waste, transfers, and adjustments.

Trigger

- Goods received
- Materials consumed in production
- Finished goods produced
- Goods shipped/fulfilled
- Inventory count adjustment
- Transfer between locations
- Waste/scrap recorded
- Return processed

Primary Records

- Inventory Item
- Inventory Transaction
- Material
- Product
- Purchase Order
- Batch
- Sales Order
- Task
- Approval Request
- Activity Log

Workflow Steps

1. Inventory change is requested or triggered.
2. Transaction type is selected.
3. Related record is linked where possible.
4. Quantity, unit, location, lot, and reason are recorded.
5. Approval is requested if required.
6. Inventory quantity is updated.
7. Activity log is created.
8. Alert is triggered if stock falls below threshold or adjustment is unusual.

Transaction Types

- Receive
- Consume
- Produce
- Reserve

- Release Reservation
- Transfer
- Adjust
- Scrap/Waste
- Return
- Ship

AI Support

AI can:

- Explain inventory changes
- Flag unusual adjustments
- Identify causes of shortages
- Detect repeated waste patterns
- Recommend reorder tasks
- Summarize inventory movement history

Approval Rules

Approval may be required for:

- Large quantity adjustments
- High-value adjustments
- Regulated/sensitive items
- Scrap/waste above threshold
- Negative inventory overrides
- Lot/batch corrections

Alerts

- Inventory below reorder point
- Negative inventory attempted
- Adjustment above threshold
- Inventory discrepancy
- Expiring lot
- Reserved stock conflict

Completion Criteria

The workflow is complete when inventory is updated, the transaction is logged, approvals are recorded if required, and any resulting alerts/tasks are assigned.

7.9 Workflow 7 — Invoice to Payment

Purpose

Track money owed to the company and ensure invoices are created, sent, monitored, followed up on, and paid.

Trigger

- Sales order fulfilled
- Milestone reached
- Manual invoice creation
- Recurring billing event

Primary Records

- Customer
- Sales Order
- Invoice
- Payment
- Task
- Alert
- Document
- AI Recommendation

Workflow Steps

1. Invoice is created from sales order or billing event.
2. Invoice is reviewed and approved if required.
3. Invoice is sent to customer.
4. Due date is tracked.
5. Payment status is updated.
6. Overdue invoices trigger alerts.
7. AI may recommend collections priority.
8. Payment reminder is drafted or sent with approval.
9. Payment is recorded.
10. Invoice is closed.

Statuses

- Draft
- Approved
- Sent
- Partially Paid
- Paid
- Overdue
- Disputed
- Void

- Written Off

AI Support

AI can:

- Summarize receivables
- Prioritize collections
- Draft payment reminders
- Identify slow-paying customers
- Forecast cash impact
- Flag customers with overdue balances before new quotes/orders

Approval Rules

Approval may be required for:

- Sending invoice if user lacks permission
- Voiding invoice
- Applying credit
- Writing off balance
- Changing payment terms
- Sending sensitive collections communication

Alerts

- Invoice not sent
- Invoice due soon
- Invoice overdue
- Partial payment unresolved
- Customer has overdue balance with new order/quote

Completion Criteria

The workflow is complete when invoice is fully paid, disputed/voided/written off with approval, and related tasks are closed.

7.10 Workflow 8 — Vendor Bill to Payment

Purpose

Track money owed by the company and manage vendor bills, approvals, due dates, and payment status.

Trigger

- Vendor bill received
- Purchase order received

- Recurring expense due
- Manual expense entry

Primary Records

- Vendor
- Purchase Order
- Expense/Bill
- Payment
- Document
- Approval Request
- Task
- Alert

Workflow Steps

1. Bill or expense is created.
2. Bill is linked to vendor and purchase order where applicable.
3. Supporting document is attached.
4. Bill is reviewed for accuracy.
5. Approval is requested if required.
6. Due date is tracked.
7. Payment is scheduled or recorded.
8. Cash-flow forecast is updated.
9. Bill is closed.

Statuses

- Draft
- Needs Review
- Needs Approval
- Approved
- Scheduled for Payment
- Paid
- Disputed
- Cancelled

AI Support

AI can:

- Summarize upcoming payables
- Flag bills missing supporting documents
- Compare bill to purchase order
- Detect unusual expenses
- Forecast cash impact
- Recommend payment priority

Approval Rules

Approval may be required for:

- Bills over threshold
- Unmatched vendor bills
- Bills without purchase orders
- New vendor payments
- Early payments
- Disputed charges

Alerts

- Bill due soon
- Bill overdue
- Bill missing document
- Bill exceeds purchase order
- Bill waiting approval

Completion Criteria

The workflow is complete when the bill is approved, paid or otherwise resolved, and all supporting documents are attached.

7.11 Workflow 9 — Document to Obligation

Purpose

Turn important documents into searchable, structured, actionable business records.

Trigger

- Document uploaded
- Document received from vendor/customer/advisor
- Document created internally
- AI identifies important document in communication

Primary Records

- Document
- Customer
- Vendor
- Product
- Batch
- Compliance Requirement
- Task
- Alert

- AI Recommendation
- Approval Request

Workflow Steps

1. Document is uploaded or created.
2. Document type is selected.
3. Document owner is assigned.
4. Document is linked to relevant records.
5. AI summarizes the document.
6. AI extracts key dates, obligations, risks, parties, and required actions.
7. User reviews AI extraction.
8. Tasks, alerts, or compliance requirements are created as needed.
9. Document status is set.
10. Review/expiration dates are tracked.

Statuses

- Uploaded
- Needs Classification
- Needs Review
- Active
- Expiring Soon
- Expired
- Superseded
- Archived

AI Support

AI can:

- Summarize document contents
- Extract obligations
- Extract dates/deadlines
- Identify missing signatures or missing information
- Suggest related records
- Create tasks from obligations
- Flag document risk

Approval Rules

Approval may be required for:

- Marking legal/compliance document active
- Closing an obligation
- Archiving sensitive document
- Accepting AI-extracted contract obligations
- Publishing SOP/policy documents

Alerts

- Document unclassified
- Document needs review
- Document expiring soon
- Extracted deadline approaching
- Missing required linked record
- Missing signature or required field

Completion Criteria

The workflow is complete when document is classified, linked, reviewed, summarized, and all extracted obligations/tasks/deadlines are assigned or resolved.

7.12 Workflow 10 — Compliance Deadline Management

Purpose

Track compliance requirements, deadlines, responsible owners, required evidence, and completion status.

Trigger

- Compliance requirement created manually
- AI extracts requirement from document
- Recurring compliance schedule
- License/COA/SOP/contract deadline approaching

Primary Records

- Compliance Requirement
- Document
- Product
- Batch
- Vendor
- Customer
- Task
- Alert
- Approval Request

Workflow Steps

1. Compliance requirement is created.
2. Requirement type, jurisdiction, owner, due date, and evidence required are defined.
3. Requirement is linked to relevant products, batches, vendors, customers, or company records.
4. Reminder schedule is created.
5. Tasks are assigned.
6. Required evidence is uploaded.

7. Requirement is reviewed.
8. Approval is recorded if needed.
9. Requirement is marked complete, renewed, or escalated.
10. Next due date is scheduled if recurring.

Statuses

- Draft
- Active
- Upcoming
- In Progress
- Waiting on Evidence
- Waiting on Approval
- Complete
- Overdue
- Escalated
- Archived

AI Support

AI can:

- Create checklists
- Summarize requirements
- Identify missing evidence
- Extract deadlines from documents
- Recommend responsible owner
- Generate reminders
- Flag overdue compliance items

Approval Rules

Approval may be required for:

- Marking requirement complete
- Changing due date
- Closing legal/compliance obligation
- Accepting substituted evidence
- Archiving compliance record

Alerts

- Deadline approaching
- Deadline overdue
- Missing evidence
- Owner not assigned
- Requirement waiting approval
- Related document expired

Completion Criteria

The workflow is complete when evidence is attached, approval is recorded if required, status is updated, and next review/renewal date is scheduled.

7.13 Workflow 11 — Meeting or Email to Task

Purpose

Convert conversations, meetings, emails, notes, and communications into accountable action items.

Trigger

- Meeting note added
- Email imported or summarized
- User asks AI to extract action items
- Communication contains follow-up language
- Internal note creates commitment

Primary Records

- Note / Communication
- Task
- Customer
- Vendor
- Project
- Sales Order
- Document
- AI Recommendation

Workflow Steps

1. Communication or note is captured.
2. AI summarizes key points.
3. AI identifies action items, owners, due dates, and related records.
4. User reviews suggested tasks.
5. Approved tasks are created.
6. Tasks are assigned to owners.
7. Tasks are tracked until completion.
8. Communication history remains linked to related records.

Statuses

- Captured
- Needs Review
- Tasks Suggested
- Tasks Approved

- Tasks Created
- Closed

AI Support

AI can:

- Summarize meetings/emails
- Extract action items
- Recommend owners
- Recommend due dates
- Link tasks to records
- Draft follow-up messages
- Identify unresolved commitments

Approval Rules

Approval may be required before AI-created tasks are assigned to others, depending on role settings.

Customer-facing follow-up messages should follow communication approval rules.

Alerts

- Communication contains unresolved action items
- Suggested tasks not reviewed
- Follow-up due
- Commitment overdue

Completion Criteria

The workflow is complete when relevant tasks are created, assigned, linked to records, and communication history is saved.

7.14 Workflow 12 — AI Recommendation to Action

Purpose

Ensure AI recommendations become structured, reviewable, and accountable business actions instead of disappearing as chat suggestions.

Trigger

- AI detects risk
- AI identifies opportunity
- AI generates report insight
- AI summarizes document obligation

- AI recommends purchase, follow-up, task, approval, or review

Primary Records

- AI Recommendation
- Task
- Alert
- Approval Request
- Related business record
- Activity Log

Workflow Steps

1. AI generates recommendation.
2. Recommendation is linked to source records.
3. Recommendation is classified by type and risk level.
4. User reviews recommendation.
5. User accepts, rejects, modifies, or defers.
6. Accepted recommendation becomes task, alert, draft, report item, or approval request.
7. Action is assigned to owner.
8. Outcome is tracked.
9. Human decision is logged.

Statuses

- New
- Needs Review
- Accepted
- Modified
- Rejected
- Deferred
- Converted to Task
- Converted to Approval Request
- Closed

AI Support

AI creates the recommendation and may assist with:

- Explaining why it matters
- Showing source records
- Estimating risk
- Suggesting owner
- Drafting task or approval language
- Monitoring completion

Approval Rules

Approval is required if the recommendation would result in a sensitive action, such as:

- Sending communication
- Issuing quote
- Creating purchase order
- Changing financial record
- Changing compliance status
- Changing production specification
- Deleting/archiving records

Alerts

- High-risk AI recommendation waiting review
- Recommendation accepted but no task created
- Recommendation ignored past due date
- Repeated rejected recommendations in same category

Completion Criteria

The workflow is complete when the recommendation is accepted/rejected/deferred, follow-up action is created if needed, and outcome is logged.

7.15 Workflow 13 — Alert to Resolution

Purpose

Ensure alerts become managed issues with ownership, priority, action, and resolution history.

Trigger

- System rule detects condition
- AI detects risk
- User manually creates alert
- Deadline approaches
- Status changes trigger exception

Primary Records

- Alert
- Task
- Related business record
- AI Recommendation
- Approval Request
- Activity Log

Workflow Steps

1. Alert is created.
2. Alert is linked to source record.
3. Severity is assigned.
4. Owner is assigned automatically or manually.
5. AI may recommend next action.
6. Alert is converted to task or approval request if action is needed.
7. Owner resolves, dismisses, escalates, or defers alert.
8. Resolution notes are recorded.
9. Activity log is updated.

Statuses

- Open
- Assigned
- In Progress
- Waiting on Approval
- Resolved
- Dismissed
- Escalated
- Deferred

AI Support

AI can:

- Explain alert cause
- Recommend next action
- Identify related records
- Prioritize severity
- Draft resolution task
- Summarize repeated alert patterns

Approval Rules

Approval may be required to:

- Dismiss high-severity alerts
- Override inventory/cash/compliance warnings
- Mark compliance/legal alerts resolved
- Close alerts tied to sensitive records

Alerts

This workflow manages alerts, but unresolved alerts can also escalate.

Escalation triggers may include:

- Alert remains open too long
- Owner fails to respond
- Alert severity increases
- Related deadline approaches

Completion Criteria

The workflow is complete when the alert is resolved, dismissed with justification, escalated, or converted into a tracked task/approval workflow.

7.16 Workflow 14 — Daily Leadership Briefing

Purpose

Create a daily operating rhythm for leadership by summarizing the current state of the business, top risks, priorities, decisions, and recommended actions.

Trigger

- Scheduled daily generation
- Manual user request
- Major system changes since last briefing

Primary Records

- Report
- Alert
- Task
- Customer
- Sales Order
- Purchase Order
- Inventory Item
- Invoice
- Document
- Compliance Requirement
- AI Recommendation

Workflow Steps

1. System gathers current business data.
2. AI summarizes key changes and risks.
3. Briefing is generated.
4. Briefing highlights top priorities.
5. AI recommends actions.
6. User reviews briefing.

7. User converts items into tasks or approvals where needed.
8. Report is archived.

Briefing Sections

- Executive summary
- Today’s top priorities
- Overdue tasks
- Sales/customer follow-ups
- Open orders at risk
- Inventory and purchasing risks
- Production issues
- Cash-flow/receivables concerns
- Compliance/document deadlines
- Approvals waiting
- AI recommendations

AI Support

AI is central to this workflow and should:

- Summarize data
- Prioritize risks
- Recommend actions
- Explain changes
- Create suggested tasks
- Identify decisions needed from leadership

Approval Rules

The briefing itself does not require approval.

Actions taken from the briefing must follow normal approval rules.

Alerts

- High-priority item not reviewed
- Repeated risk appears across multiple briefings
- Leadership approval item overdue

Completion Criteria

The workflow is complete when briefing is generated, reviewed, and any required follow-up actions are assigned.

7.17 Workflow 15 — Weekly Management Review

Purpose

Provide a recurring operating review of sales, finance, operations, inventory, compliance, projects, and team accountability.

Trigger

- Scheduled weekly generation
- Manual leadership request

Primary Records

- Report
- Tasks
- Alerts
- Sales data
- Finance data
- Inventory data
- Production data
- Compliance data
- Project data
- AI Recommendations

Workflow Steps

1. System gathers weekly performance data.
2. AI identifies trends, issues, changes, and repeated risks.
3. Weekly report is generated.
4. Leadership reviews results.
5. Decisions and follow-up tasks are recorded.
6. Projects, risks, and KPIs are updated.
7. Report is archived.

Report Sections

- Weekly executive summary
- Sales pipeline and follow-up performance
- Orders and fulfillment status
- Inventory and purchasing status
- Production performance
- Receivables/payables/cash indicators
- Compliance and document status
- Overdue and blocked tasks
- Project progress
- Key decisions needed
- AI recommendations

AI Support

AI can:

- Explain what changed from last week
- Identify trends
- Flag repeated issues
- Recommend management priorities
- Draft meeting agenda
- Convert review decisions into tasks

Approval Rules

Reports do not require approval, but actions taken from reports may require approval.

Alerts

- Weekly report not reviewed
- Repeated unresolved risks
- KPI outside acceptable range
- Task owner repeatedly overdue

Completion Criteria

The workflow is complete when report is reviewed, decisions are recorded, and follow-up tasks are assigned.

7.18 Workflow 16 — Project / Strategic Initiative Management

Purpose

Manage larger business initiatives with milestones, owners, deadlines, budgets, risks, documents, and executive visibility.

Trigger

- Leadership creates strategic initiative
- AI identifies need for project from repeated tasks/risks
- New product, legal, operational, or growth initiative begins

Primary Records

- Project
- Task
- Document
- Budget/Finance records
- User

- Alert
- AI Recommendation
- Report

Workflow Steps

1. Project is created.
2. Objective, owner, priority, and target date are defined.
3. Milestones are added.
4. Tasks are assigned.
5. Documents are linked.
6. Budget or resource needs are added where applicable.
7. Risks are tracked.
8. Project status is updated regularly.
9. AI summarizes progress and risks.
10. Project is completed, cancelled, paused, or archived.

Statuses

- Proposed
- Approved
- Planning
- Active
- At Risk
- On Hold
- Completed
- Cancelled
- Archived

AI Support

AI can:

- Draft project plans
- Create milestone task lists
- Summarize project progress
- Identify blocked work
- Flag schedule risk
- Prepare leadership updates
- Suggest next actions

Approval Rules

Approval may be required for:

- Project creation above budget threshold
- Budget changes
- Strategic priority changes

- External commitments
- Project cancellation

Alerts

- Milestone overdue
- Project at risk
- Budget concern
- Owner not assigned
- Repeated blockers

Completion Criteria

The workflow is complete when the project reaches its final status, all required documentation is attached, and unresolved tasks/risks are closed or transferred.

7.19 Workflow Status and Escalation Rules

The system should include escalation rules so workflows do not stall silently.

Escalation Triggers

- Task overdue
- Approval overdue
- Alert unresolved
- Quote not followed up
- Order due date approaching with unresolved blocker
- Purchase order late
- Invoice overdue
- Compliance deadline approaching
- Document waiting review too long
- Project milestone overdue

Escalation Actions

- Notify owner
- Notify manager
- Increase priority
- Create alert
- Add to command center
- Add to daily briefing
- Request leadership review
- Create AI recommendation

7.20 Workflow Automation Rules

Automation should be introduced carefully.

Safe Automations

The following are generally safe to automate:

- Create reminders
- Create draft tasks
- Generate reports
- Flag stale records
- Generate AI summaries
- Suggest follow-ups
- Create alerts
- Detect missing documents
- Notify owners of overdue items

Controlled Automations

The following should require review or approval:

- Sending customer emails
- Sending vendor emails
- Issuing quotes
- Creating purchase orders
- Updating financial status
- Closing compliance requirements
- Releasing production batches
- Changing product specifications
- Archiving sensitive records

Prohibited Without Explicit Approval

AI should not automatically:

- Make financial commitments
- Modify legal status
- Delete sensitive records
- Change permissions
- Send binding customer commitments
- Change formulas/specifications
- Close legal/compliance obligations

7.21 Workflow MVP Scope

The MVP should include the workflows that create immediate management value.

Required MVP Workflows

1. Lead to Customer
2. Quote to Order
3. Sales Order to Fulfillment
4. Purchasing and Receiving
5. Inventory Adjustment and Movement
6. Invoice to Payment
7. Document to Obligation
8. Meeting/Email to Task
9. AI Recommendation to Action
10. Alert to Resolution
11. Daily Leadership Briefing

Phase 2 Workflows

1. Order to Production
2. Vendor Bill to Payment
3. Compliance Deadline Management
4. Weekly Management Review
5. Project / Strategic Initiative Management
6. Full batch traceability and QC workflows
7. Advanced cash-flow forecasting workflow

Reasoning

The MVP should focus on visibility, accountability, customer follow-up, purchasing, inventory, documents, alerts, AI recommendations, and daily leadership control.

Production, deeper compliance, project management, and advanced finance workflows can be layered in after the core system is stable.

7.22 Workflow Design Rule

Every workflow should answer:

1. What starts the workflow?
2. What records are created or updated?
3. Who owns the workflow?
4. What statuses does it move through?
5. What documents are required?
6. What tasks are created?

7. What alerts can be triggered?
8. What can AI assist with?
9. What actions require approval?
10. What does completion look like?

If every workflow follows this structure, the system will remain consistent, scalable, and manageable as the company grows.

Step 8 — AI Roles and Capabilities

8.1 Purpose of This Section

This section defines how AI should function inside the AI Business Command Center.

AI should not be treated as a generic chatbot added on top of the system. It should be designed as an operating layer that supports management, analysis, monitoring, drafting, recommendations, and controlled automation across the entire business.

The purpose of AI in this system is to help leadership and team members:

- Understand business status faster
- Identify risks earlier
- Make better decisions
- Reduce manual reporting
- Draft communications and documents
- Convert information into tasks
- Monitor deadlines and thresholds
- Recommend next actions
- Improve accountability
- Reduce missed follow-ups
- Support scalable management

AI should increase speed and visibility while keeping humans in control of sensitive decisions.

8.2 AI Design Philosophy

The AI layer should follow this philosophy:

AI assists, analyzes, monitors, recommends, drafts, and prepares actions. Humans approve, decide, and remain accountable for high-impact outcomes.

AI should be powerful enough to reduce management workload, but controlled enough to avoid uncontrolled business, financial, legal, compliance, or customer-facing actions.

AI should help the company move from reactive management to proactive management.

8.3 Core AI Roles

The AI layer should include four primary roles:

1. AI Analyst
2. AI Assistant
3. AI Monitor
4. AI Operator

Each role has a different purpose and risk level.

8.4 AI Role 1 — AI Analyst

Purpose

The AI Analyst explains business data, trends, risks, performance, and relationships.

It helps users understand what is happening and why it matters.

Example Capabilities

- Analyze sales pipeline
- Explain revenue changes
- Identify margin issues
- Summarize overdue invoices
- Compare vendor performance
- Analyze inventory movement
- Explain production delays
- Identify repeated task bottlenecks
- Summarize customer profitability
- Analyze document/compliance risk
- Explain weekly business performance

Example User Questions

- What are the biggest risks this week?
- Which customers need attention?
- Which orders are at risk and why?
- What inventory items are most urgent to reorder?
- Why did gross margin drop this month?
- Which vendors are causing the most delays?
- What tasks are repeatedly overdue?
- What changed since last week?

Outputs

- Plain-English analysis
- Ranked lists
- KPI explanations
- Risk summaries
- Trend summaries
- Root-cause hypotheses
- Recommended next questions
- Linked source records

Human Control Level

Low to medium risk.

AI Analyst outputs are advisory and do not directly change business records unless the user converts insights into tasks or actions.

8.5 AI Role 2 — AI Assistant

Purpose

The AI Assistant helps users create, summarize, organize, draft, and prepare work.

It reduces manual writing, data entry, and administrative burden.

Example Capabilities

- Draft customer emails
- Draft vendor emails
- Draft quote notes
- Summarize customer history
- Summarize documents
- Summarize meetings
- Extract action items
- Draft task descriptions
- Draft weekly reports
- Draft SOP summaries
- Draft payment reminders
- Draft project updates
- Draft internal announcements

Example User Requests

- Draft a follow-up email to this customer.
- Summarize this vendor contract.

- Turn this meeting note into tasks.
- Write a weekly operations update.
- Create a customer status summary.
- Prepare a payment reminder email.
- Summarize this product document.

Outputs

- Draft messages
- Summaries
- Task suggestions
- Document summaries
- Meeting notes
- Report drafts
- Checklists
- Action item lists

Human Control Level

Medium risk.

Most AI Assistant outputs should be treated as drafts until reviewed by a human. Customer-facing, vendor-facing, legal, compliance, or financial communications should require approval before sending.

8.6 AI Role 3 — AI Monitor

Purpose

The AI Monitor watches business data for deadlines, thresholds, exceptions, stale activity, missing information, and risk signals.

It helps the company catch problems before they become urgent.

Example Capabilities

- Detect stale customer follow-ups
- Detect overdue tasks
- Detect overdue invoices
- Detect low inventory
- Detect purchase order delays
- Detect missing documents
- Detect compliance deadlines
- Detect production delays
- Detect margin concerns
- Detect unusual inventory adjustments
- Detect unresolved alerts

- Detect repeated bottlenecks

Example Monitoring Rules

- Alert when a lead has not been contacted within 3 business days.
- Alert when inventory falls below reorder point.
- Alert when an invoice is more than 7 days overdue.
- Alert when a purchase order is past expected delivery date.
- Alert when a document expires within 30 days.
- Alert when a task is overdue by more than 2 days.
- Alert when a quote is about to expire.
- Alert when a production batch is stuck in QC hold.

Outputs

- Alerts
- Risk flags
- Suggested tasks
- Escalation notices
- Daily briefing items
- Weekly trend summaries
- Exception reports

Human Control Level

Medium risk.

AI Monitor can create alerts and suggested tasks automatically, but controlled actions still require human review or approval.

8.7 AI Role 4 — AI Operator

Purpose

The AI Operator prepares or performs structured actions inside the system.

This is the highest-risk AI role and should be carefully controlled.

Example Capabilities

- Create draft tasks
- Create draft purchase orders
- Create draft quotes
- Prepare approval requests
- Generate reports
- Update non-sensitive statuses where permitted

- Prepare customer email drafts
- Prepare vendor email drafts
- Create document review tasks
- Create compliance checklist tasks
- Convert accepted recommendations into tasks

Example User Requests

- Create a task for the sales manager to follow up with this customer.
- Prepare a purchase order draft for the low-stock materials.
- Generate the weekly leadership report.
- Create approval requests for quotes below target margin.
- Turn this AI recommendation into a task.
- Create document review tasks for all expiring documents.

Outputs

- Draft records
- Tasks
- Approval requests
- Reports
- Draft communications
- Draft purchase orders
- Draft quotes
- Status updates where permitted

Human Control Level

High risk.

AI Operator actions must be limited by user permissions, approval rules, audit logs, and system settings. Sensitive actions should remain human-approved.

8.8 AI Capability Categories

The AI system should support capabilities across the following categories.

1. Search and Question Answering

AI should answer questions using business data, documents, and linked records.

Examples:

- What did we promise this customer?
- Which orders are waiting on inventory?
- What documents are missing for this vendor?

- What tasks are overdue for this project?

2. Summarization

AI should summarize records, documents, customers, projects, workflows, and reports.

Examples:

- Customer history summary
- Vendor performance summary
- Contract summary
- Daily briefing
- Weekly leadership report
- Batch issue summary

3. Recommendation

AI should recommend actions based on business context.

Examples:

- Follow up with this customer.
- Reorder this material.
- Review this low-margin quote.
- Escalate this overdue task.
- Investigate this production delay.

4. Drafting

AI should draft messages, documents, reports, notes, and approval summaries.

Examples:

- Customer email
- Vendor follow-up
- Payment reminder
- Quote note
- SOP summary
- Project update

5. Monitoring

AI should watch for risks, deadlines, stale activity, and exceptions.

Examples:

- Low stock
- Overdue invoice

- Expiring document
- Stale lead
- Late purchase order
- Missing COA

6. Forecasting

AI should support forecasting where enough data exists.

Examples:

- Cash forecast
- Inventory demand forecast
- Sales forecast
- Production workload forecast
- Purchasing needs forecast

7. Classification

AI should classify records and documents.

Examples:

- Document type
- Customer status
- Task category
- Risk level
- Communication intent
- Compliance category

8. Extraction

AI should extract structured information from unstructured sources.

Examples:

- Contract obligations
- Due dates
- Action items
- Parties involved
- Payment terms
- Product requirements
- Compliance evidence needs

9. Workflow Assistance

AI should help move workflows forward.

Examples:

- Suggest next workflow step
- Create missing tasks
- Identify blockers
- Prepare approval requests
- Recommend status updates

10. Reporting

AI should generate plain-English business reports.

Examples:

- Daily briefing
- Weekly operating report
- Monthly performance report
- Sales summary
- Inventory risk report
- Compliance status report

8.9 AI Recommendation Structure

Every important AI recommendation should be stored as a structured record.

Required Recommendation Fields

- Recommendation ID
- Recommendation type
- Title
- Summary
- Related module
- Related record
- Source records used
- Reason for recommendation
- Risk level
- Business impact
- Suggested action
- Suggested owner
- Approval required flag
- Confidence level
- Status
- Created date
- Reviewed by
- Human decision
- Outcome

Recommendation Statuses

- New
- Needs Review
- Accepted
- Modified
- Rejected
- Deferred
- Converted to Task
- Converted to Approval Request
- Closed

Example Recommendation

Title: Reorder Packaging Material A

Summary: Inventory is below reorder point and upcoming sales orders require this material within 10 days.

Source Records:

- Inventory Item: Packaging Material A
- Sales Orders: Open orders requiring Material A
- Vendor: Preferred supplier
- Purchase History: Average lead time

Suggested Action:

Create purchase order draft for recommended quantity.

Approval Required:

Yes, if order value exceeds purchasing threshold.

8.10 AI Trust and Review Model

The system should be designed so users can trust AI outputs without blindly relying on them.

Trust Requirements

AI should show:

- What records were used
- What assumption was made
- What risk was detected
- Why the recommendation matters
- What action is suggested

- Whether approval is required
- What confidence level applies
- What could be wrong or missing

Human Review Options

Users should be able to:

- Accept
- Reject
- Modify
- Defer
- Ask for explanation
- Convert to task
- Convert to approval request
- Dismiss with reason
- Mark as incorrect

Feedback Loop

The system should track whether AI recommendations are accepted, modified, rejected, or ignored.

Over time, this feedback can improve AI usefulness, reporting, and automation confidence.

8.11 AI Data Access Rules

AI access must be controlled by permissions.

Rules

1. AI can only access records the user is allowed to access.
2. AI can only summarize documents the user is allowed to view.
3. AI cannot reveal restricted fields through summaries or reports.
4. AI-generated reports must respect the viewer's permissions.
5. AI cannot perform actions the user lacks permission to perform.
6. AI cannot bypass approval rules.
7. AI source records should be visible when possible.
8. AI should label generated content clearly.
9. AI actions should be logged.
10. AI should not use sensitive records for unrelated outputs.

8.12 AI Approval Boundaries

AI may assist with sensitive work, but should not complete sensitive actions without approval.

AI May Draft Without Final Approval

- Customer emails
- Vendor emails
- Quote language
- Purchase order drafts
- Payment reminder drafts
- Compliance checklists
- SOP summaries
- Project updates
- Internal reports

AI May Create Automatically When Allowed

- Low-risk alerts
- Suggested tasks
- Draft reports
- Document summaries
- Risk flags
- Follow-up reminders
- Data quality warnings

AI Requires Approval Before

- Sending customer-facing communication
- Sending vendor-facing communication
- Issuing quotes
- Creating purchase orders above threshold
- Changing financial records
- Changing compliance status
- Changing product specifications
- Changing production formulas/specifications
- Closing legal/IP obligations
- Releasing QC hold
- Deleting or archiving sensitive records
- Granting permissions
- Running bulk updates

AI Should Never Do Without Explicit Authorization

- Make binding legal commitments
- Make financial commitments
- Alter legal/IP status
- Override compliance controls
- Delete critical records
- Change user permissions
- Hide or suppress audit logs
- Fabricate source data

- Present guesses as facts

8.13 AI Interaction Modes

Users should be able to interact with AI in multiple ways.

Chat Interface

Users can ask questions and request help in natural language.

Examples:

- What needs my attention today?
- Summarize this customer.
- Which inventory items are at risk?
- Draft a follow-up email.

Embedded Record Assistant

Each major record should have AI support directly on the record page.

Examples:

- Summarize this customer.
- Explain this order status.
- What tasks are related to this document?
- What risks exist for this batch?

Dashboard AI

The Command Center should include AI-generated summaries, risk rankings, and recommended actions.

Report AI

Reports should include AI-generated explanations of trends, changes, risks, and recommended next actions.

Workflow AI

AI should appear inside workflows to suggest next steps, prepare drafts, detect missing information, and create tasks.

8.14 AI Prompt and Action Templates

The system should use structured templates for common AI requests so outputs are consistent and safe.

Daily Briefing Template

AI should summarize:

- Top priorities
- Open risks
- Overdue tasks
- Sales follow-ups
- Orders at risk
- Inventory/purchasing issues
- Finance/cash concerns
- Compliance/document deadlines
- Approvals waiting
- Recommended actions

Customer Summary Template

AI should summarize:

- Customer status
- Recent communication
- Open opportunities
- Open orders
- Open invoices
- Related tasks
- Risks
- Recommended next action

Inventory Risk Template

AI should summarize:

- Item status
- Quantity on hand
- Reorder point
- Open demand
- Vendor lead time
- Purchase order status
- Risk level
- Recommended action

Document Review Template

AI should summarize:

- Document type
- Parties involved
- Key terms
- Dates/deadlines
- Obligations
- Missing information
- Related records
- Recommended tasks

Approval Summary Template

AI should summarize:

- Requested action
- Reason for request
- Related record
- Business impact
- Risks
- Supporting data
- Recommended decision

8.15 AI Output Quality Standards

AI outputs should meet quality standards before being useful in the business.

Standards

AI should be:

- Clear
- Actionable
- Sourced to records where possible
- Permission-aware
- Specific
- Honest about uncertainty
- Labeled when generated
- Easy to convert into tasks
- Structured enough for review
- Consistent across modules

AI Should Avoid

- Unsupported conclusions
- Hidden assumptions
- Overconfident recommendations
- Revealing restricted data
- Making commitments without approval
- Creating duplicate tasks
- Creating vague tasks
- Ignoring business context
- Recommending actions without explaining why

8.16 AI Memory and Business Context

The AI layer should maintain business context through structured records, not through vague conversational memory.

Context Should Come From

- Customer records
- Vendor records
- Product records
- Inventory records
- Orders
- Invoices
- Tasks
- Documents
- Compliance requirements
- Projects
- Reports
- Activity logs
- User roles and permissions

Context Should Not Depend On

- Unverified chat history alone
- User memory
- Scattered notes
- Unsupported assumptions
- Unlinked documents

The system should prefer structured, linked, current business data over loose conversation history.

8.17 AI Error Handling

The system should handle AI uncertainty and possible mistakes transparently.

When AI Is Uncertain

AI should say what is uncertain and why.

Examples:

- Missing source data
- Conflicting records
- Outdated documents
- No linked invoice
- No vendor lead time available
- Inventory count not recently verified

User Options

Users should be able to:

- Correct AI output
- Mark recommendation wrong
- Provide missing data
- Request re-analysis
- Add source records
- Convert correction into data quality task

Data Quality Feedback

If AI cannot answer because data is missing, the system should identify what data is needed.

Example:

AI cannot forecast reorder timing because vendor lead time is missing for this material.

Suggested task: Add vendor lead time for Material A.

8.18 AI Auditing and Logging

All meaningful AI activity should be logged.

AI Events to Log

- AI recommendation generated

- AI summary generated
- AI draft created
- AI-created task
- AI-created alert
- AI-created approval request
- AI report generated
- AI output accepted
- AI output rejected
- AI output modified
- AI output used to trigger action

Log Fields

- User requesting AI output
- AI function used
- Related records
- Source records
- Output type
- Timestamp
- Human decision
- Follow-up action

This creates accountability and allows the company to evaluate AI usefulness over time.

8.19 AI Risk Levels

AI functions should be categorized by risk.

Low-Risk AI

Examples:

- Summarizing non-sensitive records
- Creating internal task suggestions
- Generating daily summaries
- Flagging stale follow-ups
- Identifying missing data

Typical control:

- Logging
- User review optional

Medium-Risk AI

Examples:

- Drafting customer/vendor emails
- Drafting quote notes
- Summarizing contracts
- Extracting obligations
- Recommending purchases
- Prioritizing collections

Typical control:

- Human review recommended or required
- Source records shown
- Logging required

High-Risk AI

Examples:

- Issuing quotes
- Creating purchase orders
- Updating financial records
- Changing compliance status
- Changing production specifications
- Releasing QC holds
- Handling legal/IP matters

Typical control:

- Human approval required
- Permission checks required
- Audit log required
- Source data required

8.20 AI MVP Capabilities

The MVP should include AI capabilities that create immediate value without excessive risk.

Required AI MVP Capabilities

1. Daily leadership briefing
2. Customer summary
3. Stale follow-up detection
4. Task summary and overdue task detection

5. Document summarization
6. Action item extraction from notes/documents
7. Low inventory alerts
8. Purchase recommendation drafts
9. Quote risk review
10. Order risk summary
11. AI recommendation records
12. Convert AI recommendation to task
13. Approval request drafting
14. Basic report summaries
15. AI activity logging

Deferred Advanced AI Capabilities

1. Advanced cash-flow forecasting
2. Production schedule optimization
3. Vendor pricing optimization
4. Predictive customer churn scoring
5. Automated contract risk scoring
6. Full compliance rule monitoring
7. Automated quote generation from conversation
8. Automated purchase order execution
9. Advanced anomaly detection
10. Multi-agent workflow automation

Reasoning

The MVP should focus on visibility, summarization, risk detection, task creation, document intelligence, and controlled recommendations.

More advanced automation should be added only after the data model, permissions, approval rules, and audit logs are reliable.

8.21 AI Design Rule

Every AI capability should answer:

1. What business problem does this solve?
2. What records does AI need access to?
3. Is the user allowed to access those records?
4. What output will AI produce?
5. Is the output advisory, draft, monitoring, or action-oriented?
6. What risk level applies?
7. Is human approval required?
8. How will the output be logged?
9. Can the output become a task, alert, report, or approval request?

10. How will users correct AI if it is wrong?

This rule ensures AI remains useful, controlled, trustworthy, and scalable.

Step 9 — Human Approval and Control Rules

9.1 Purpose of This Section

This section defines how the system keeps humans in control while still allowing AI and automation to improve speed, visibility, and execution.

The AI Business Command Center should not operate as an uncontrolled automation system. It should operate as an intelligent management system where AI can prepare, suggest, monitor, and draft, but sensitive decisions remain governed by human approval.

The control model protects the business from:

- Unauthorized commitments
- Incorrect customer communications
- Unapproved purchases
- Financial mistakes
- Compliance errors
- Legal/IP risk
- Production specification mistakes
- Improper data access
- Untracked automation
- Accidental deletion or modification of important records

The goal is to create a system that is fast, useful, accountable, and safe.

9.2 Control Philosophy

The system should follow this principle:

AI may assist with work, but humans remain accountable for business decisions and controlled actions.

AI can help identify what should happen next, but the system must define when a human must review, approve, reject, modify, or override that action.

The system should make approval easy, but not optional for high-risk actions.

9.3 Action Risk Levels

Every important system action should be classified by risk level.

Level 1 — Low Risk

Low-risk actions are informational or internal and do not create external commitments, financial exposure, legal obligations, or sensitive changes.

Examples:

- Generate internal summary
- Create draft task
- Flag stale follow-up
- Generate internal report
- Suggest inventory review
- Summarize customer history
- Summarize non-sensitive document
- Create low-priority alert

Default control:

- AI may perform or suggest automatically
- User review optional
- Activity logging recommended

Level 2 — Moderate Risk

Moderate-risk actions may affect business workflow, customer/vendor communication drafts, internal priorities, or operational execution, but do not create final external commitments without approval.

Examples:

- Draft customer email
- Draft vendor email
- Draft quote
- Draft purchase order
- Create task assigned to another user
- Create alert requiring action
- Extract contract obligation for review
- Recommend purchase
- Recommend collections action
- Recommend production priority

Default control:

- AI may draft or recommend
- Human review required before external or binding action
- Activity logging required

Level 3 — High Risk

High-risk actions can create financial, legal, compliance, production, customer, or operational consequences.

Examples:

- Send customer-facing communication
- Send vendor-facing communication
- Issue quote
- Approve purchase order
- Update invoice/payment status
- Change product cost
- Change customer payment terms
- Change production batch status
- Close compliance requirement
- Mark legal document approved
- Archive sensitive document

Default control:

- Permission required
- Human approval required
- Activity logging required
- Source data should be shown
- Approval history must be stored

Level 4 — Critical Risk

Critical-risk actions could materially affect the company, its legal rights, financial obligations, compliance position, intellectual property, production integrity, or system security.

Examples:

- Change user permissions
- Delete critical records
- Modify audit logs
- Change production formula/specification
- Release batch from QC hold

- File or close legal/IP matter
- Make large financial commitment
- Approve high-value purchase order
- Write off customer balance
- Change compliance/legal status
- Execute bulk data update
- Configure automation that can perform controlled actions

Default control:

- Executive or administrative approval required
- Multi-step approval recommended for some cases
- Full audit logging required
- Source data required
- Human justification required
- AI cannot execute independently

9.4 Action Categories and Default Control Rules

Informational Actions

Examples:

- Summaries
- Reports
- Analysis
- KPI explanations
- Risk explanations

Default control:

- AI may generate automatically if user has access.
- No approval required.
- Logging recommended for important outputs.

Internal Task Actions

Examples:

- Create task
- Assign task
- Suggest task
- Mark task complete
- Escalate overdue task

Default control:

- AI may suggest or create draft tasks.
- Assigning tasks to other users may require review depending on role.
- Completing tasks should generally require the owner or manager.

Communication Actions

Examples:

- Draft email
- Send email
- Send quote message
- Send payment reminder
- Send vendor follow-up

Default control:

- AI may draft.
- Sending external communication requires user permission.
- Sensitive messages may require approval.

Financial Actions

Examples:

- Create invoice
- Void invoice
- Apply payment
- Approve bill
- Schedule payment
- Write off balance
- Change terms

Default control:

- AI may summarize, flag, and draft.
- Financial record changes require permission.
- High-impact financial actions require approval.

Purchasing Actions

Examples:

- Draft purchase order
- Submit purchase order
- Approve purchase order
- Send purchase order to vendor

Default control:

- AI may draft PO.
- Sending or approving PO requires permission and approval rules.

Production Actions

Examples:

- Create work order
- Update batch status
- Record material consumption
- Release QC hold
- Change specification

Default control:

- AI may recommend and prepare.
- Production changes require role permissions.
- Formula/specification/QC release changes require approval.

Compliance / Legal / IP Actions

Examples:

- Extract obligation
- Create compliance task
- Mark requirement complete
- Close legal deadline
- Archive legal document
- Modify IP record

Default control:

- AI may extract and recommend.
- Human review required for legal/compliance conclusions.
- Controlled status changes require approval.

System / Permission Actions

Examples:

- Add user
- Change role
- Modify permissions
- Configure integrations
- Change automation rules

Default control:

- AI may explain or draft recommendations.
- Only authorized administrators may perform.
- Sensitive changes require logging and sometimes executive approval.

9.5 Human Approval Workflow

Approval workflows should be consistent across modules.

Approval Workflow Steps

1. Action is requested by user, workflow, alert, or AI.
2. System checks whether approval is required.
3. Approval request is created if required.
4. Request is assigned to authorized approver.
5. Approver receives context, source data, risk level, and recommended decision.
6. Approver chooses approve, reject, request changes, escalate, or defer.
7. Decision is logged.
8. Approved action proceeds.
9. Rejected action is closed or returned for revision.
10. Follow-up tasks are created when needed.

Approval Request Should Include

- Requested action
- Requesting user or AI source
- Related record
- Business reason
- Risk level
- Supporting data
- AI recommendation if available
- Deadline
- Approver
- Approval history
- Final decision
- Notes/justification

9.6 Approval Queue

The system should include an approval queue so users can manage pending approvals efficiently.

Approval Queue Should Show

- Pending approval requests
- Request type
- Related record
- Requested by
- Assigned approver
- Risk level
- Amount/value where applicable
- Due date
- Age of request
- AI recommendation
- Status

Queue Filters

- By approver
- By department
- By risk level
- By due date
- By request type
- By overdue status
- By AI-generated request

AI Support

AI can summarize the approval queue and recommend which approvals need immediate attention.

Example:

“Three approvals need attention today: one purchase order tied to production, one quote below target margin, and one expiring compliance document review.”

9.7 Approval Statuses

Approval requests should use clear statuses.

Recommended Statuses

- Draft
- Pending
- Under Review
- Approved
- Rejected
- Changes Requested
- Escalated

- Deferred
- Cancelled
- Expired

Status Rules

- Pending means action is waiting for approver.
- Under Review means approver has opened or engaged with the request.
- Approved means the controlled action may proceed.
- Rejected means the controlled action may not proceed.
- Changes Requested means requester must revise and resubmit.
- Escalated means higher authority is required.
- Deferred means the request is intentionally paused.
- Expired means the request was not handled in time.

9.8 Approval Authority Rules

Approval authority should be based on role, department, risk level, and dollar threshold.

Authority Factors

- User role
- Department
- Approval level
- Dollar amount
- Risk level
- Record type
- Sensitive data category
- Customer/vendor status
- Compliance/legal impact

Example Rules

- Sales Manager may approve standard quotes within margin limits.
- CEO must approve quotes below margin threshold or above dollar threshold.
- Operations Manager may approve routine work orders.
- Compliance Manager must approve compliance status changes.
- Finance Manager may approve invoice/payment status changes.
- CEO or executive approver must approve write-offs, high-value purchases, and strategic commitments.
- System Administrator may manage users, but executive approval may be required for high-level permissions.

9.9 Escalation Rules

Approval requests and controlled workflows should not stall silently.

Escalation Triggers

- Approval pending too long
- High-risk request not reviewed
- Due date approaching
- Approval rejected repeatedly
- Required approver unavailable
- Alert severity increases
- Compliance/legal deadline approaching
- Cash-flow risk connected to pending approval
- Customer/order risk connected to pending approval

Escalation Actions

- Notify original approver
- Notify backup approver
- Notify department manager
- Notify executive
- Increase priority
- Add to command center
- Add to daily briefing
- Create escalation task
- Mark as overdue

Escalation Timing Examples

- Low-risk approval: escalate after 3 business days
- Moderate-risk approval: escalate after 2 business days
- High-risk approval: escalate after 1 business day
- Critical-risk approval: escalate same day or immediately

These timing values should be configurable.

9.10 Human Override Rules

Authorized users should be able to override certain system recommendations or alerts, but overrides must be controlled.

Override May Be Allowed For

- Dismissing alert
- Changing priority

- Changing owner
- Extending due date
- Approving exception
- Reopening closed item
- Bypassing non-critical workflow step

Override Should Require Justification For

- High-risk alert dismissal
- Compliance/legal item closure
- Inventory negative adjustment
- Production QC release
- Low-margin quote approval
- High-value purchase approval
- Write-off or financial adjustment
- Permission changes

Override Should Not Be Allowed For

- Hiding audit logs
- Editing historical approval decisions without trace
- Removing required compliance evidence without logging
- Deleting critical financial/legal records without archive trail
- Allowing AI to bypass human approval for critical actions

9.11 AI Control Rules

AI must follow strict control boundaries.

AI Can Automatically

- Generate summaries
- Create draft internal tasks
- Create low-risk alerts
- Identify missing information
- Detect stale records
- Suggest follow-ups
- Draft internal reports
- Extract tentative action items
- Suggest owners
- Recommend next steps

AI Can Prepare But Not Execute Without Review

- Customer emails
- Vendor emails
- Quotes

- Purchase orders
- Payment reminders
- Approval requests
- Compliance checklists
- Contract obligation summaries
- Production recommendations
- Financial recommendations

AI Cannot Execute Without Approval

- Send external communications
- Issue quotes
- Approve purchases
- Change financial records
- Change compliance/legal status
- Change product specifications
- Change production formulas/processes
- Release QC holds
- Delete/archive sensitive records
- Change user permissions
- Run bulk updates

AI Must Never

- Hide source uncertainty
- Fabricate data
- Present guesses as confirmed facts
- Reveal restricted information
- Bypass role permissions
- Suppress audit logs
- Make binding commitments without approval

9.12 Safe Automation Boundaries

Automation should be introduced in tiers.

Tier 1 — Informational Automation

Low risk.

Examples:

- Daily briefing generation
- Weekly report generation
- Overdue task summaries
- Low-stock detection

- Stale lead detection
- Expiring document detection

Control:

- No approval required
- Logging recommended

Tier 2 — Draft Automation

Moderate risk.

Examples:

- Draft follow-up emails
- Draft purchase orders
- Draft quotes
- Draft payment reminders
- Draft compliance checklists
- Draft project updates

Control:

- Human review required before external/binding use
- Logging required

Tier 3 — Workflow Automation

Moderate to high risk.

Examples:

- Convert accepted recommendation to task
- Create approval request
- Assign alert to owner
- Escalate overdue approvals
- Generate recurring compliance tasks

Control:

- Permission checks required
- Logging required
- Approval required where configured

Tier 4 — Controlled Execution Automation

High risk.

Examples:

- Send approved email
- Send approved quote
- Send approved purchase order
- Update status after approval
- Apply approved workflow transition

Control:

- Prior approval required
- Audit log required
- Role permission required

Tier 5 — Critical Automation

Critical risk.

Examples:

- Financial commitments
- Legal status changes
- Compliance closure
- Production release
- Permission changes
- Bulk updates

Control:

- Executive/admin approval required
- Multi-step approval recommended
- Full audit trail required
- AI cannot independently execute

9.13 Approval Rules by Module

Customers and Sales

Approval required for:

- Sending certain customer communications
- Creating quotes below margin target
- Approving non-standard terms
- Approving large quotes
- Reopening lost accounts
- Changing customer credit/payment terms

Orders and Quotes

Approval required for:

- Quote issuance above threshold
- Discounts above threshold
- Orders with credit risk
- Expedited commitments
- Order cancellation above threshold

Inventory and Purchasing

Approval required for:

- Purchase orders above threshold
- New vendor purchases
- Inventory adjustments above threshold
- Negative inventory overrides
- Scrapping high-value inventory

Production and Batches

Approval required for:

- Work order priority overrides
- Formula/specification changes
- Material substitutions
- QC releases
- Batch deviation closures
- Batch cancellations

Finance and Cash Flow

Approval required for:

- Invoice voids
- Payment scheduling
- Customer credits
- Write-offs
- Term changes
- Bill approvals above threshold

Compliance, Legal, and IP

Approval required for:

- Closing compliance requirements
- Changing legal/IP status

- Archiving sensitive documents
- Accepting AI-extracted obligations as final
- Modifying deadlines

Documents

Approval required for:

- Publishing SOPs
- Marking legal documents active
- Archiving sensitive documents
- Changing document sensitivity level
- Sharing restricted documents externally

Tasks and Accountability

Approval may be required for:

- AI assigning tasks to other users
- Closing executive-level tasks
- Deferring critical tasks
- Dismissing escalated tasks

Settings and Permissions

Approval required for:

- Creating admin users
- Changing roles
- Granting sensitive permissions
- Configuring controlled automations
- Changing approval thresholds

9.14 Approval Threshold Examples

Actual threshold values should be configured by company policy, but the system should support rules like these:

Quote Approval Thresholds

- Discount over defined percentage
- Margin below defined percentage
- Quote value over defined dollar amount
- Customer has overdue invoice
- Custom terms included

Purchase Approval Thresholds

- Purchase over defined dollar amount
- New vendor
- Non-standard material
- Emergency purchase
- Purchase not linked to sales order, production need, or project

Inventory Adjustment Thresholds

- Quantity change above defined amount
- Value change above defined amount
- Adjustment affects tracked lot/batch
- Adjustment creates negative inventory

Finance Approval Thresholds

- Invoice void
- Customer credit above threshold
- Write-off above threshold
- Bill/payment above threshold
- Change in payment terms

Compliance Approval Thresholds

- Requirement marked complete
- Deadline changed
- Evidence substituted
- Legal document archived
- Compliance status overridden

9.15 Approval Notifications

The system should notify users when action is needed.

Notification Types

- New approval assigned
- Approval due soon
- Approval overdue
- Approval escalated
- Approval approved
- Approval rejected
- Changes requested
- Comment added
- Approved action completed

Notification Channels

Future implementation may support:

- In-app notifications
- Email notifications
- Mobile push notifications
- Daily briefing inclusion
- Weekly approval summary

9.16 Audit Requirements for Controlled Actions

Controlled actions require strong audit trails.

Audit Log Should Capture

- Requested action
- Requesting user or AI source
- Approver
- Approval decision
- Timestamp
- Related record
- Supporting data
- AI recommendation if applicable
- Before/after values
- Justification notes
- Final executed action

Controlled Action Audit Examples

- Quote approved below target margin
- Purchase order approved above threshold
- Compliance deadline changed
- Sensitive document archived
- Inventory adjustment approved
- Customer credit applied
- Production batch released
- User permission changed

9.17 Human Review Interface Requirements

Approvers should have enough context to make decisions quickly.

Review Screen Should Show

- Requested action
- Risk level
- Related record details
- Business impact
- Supporting documents
- Source data
- AI recommendation
- Relevant history
- Prior related approvals
- Approval options
- Comment box

Approval Options

- Approve
- Reject
- Request Changes
- Escalate
- Defer
- Reassign

AI Support in Review Screen

AI may summarize:

- Why approval is needed
- What could go wrong
- What records support the request
- Similar prior approvals
- Recommended decision

9.18 Control Failure Handling

The system should handle attempted unauthorized or unsafe actions clearly.

When User Lacks Permission

System should:

- Block action
- Explain permission issue
- Offer to request approval if appropriate
- Log attempt if sensitive

When Approval Is Required

System should:

- Prevent execution
- Create approval request
- Notify approver
- Track status

When AI Is Uncertain

System should:

- Show uncertainty
- Identify missing data
- Avoid executing action
- Suggest data correction task

When Data Conflicts

System should:

- Flag conflict
- Prevent high-risk automated action
- Ask for human review
- Create data quality task if needed

9.19 MVP Control Scope

The MVP should include enough control to safely support AI-assisted workflows.

MVP Required Controls

1. Role-based permissions
2. Approval request object
3. Approval queue
4. Approval statuses
5. Action risk levels
6. AI-generated content labels
7. Activity logs
8. Human approval for controlled actions
9. Document sensitivity levels
10. Basic escalation rules
11. Approval notifications
12. Convert AI recommendation to task or approval request

Deferred Advanced Controls

Future versions may include:

- Multi-step approvals
- Conditional approval routing
- Advanced field-level controls
- External advisor time-limited access
- Automated permission review
- Behavioral anomaly detection
- Digital signature workflows
- Advanced compliance-specific approval routing
- Automated audit report generation

9.20 Control Design Rule

Every controlled action should answer:

1. What action is being attempted?
2. Who or what requested it?
3. What record does it affect?
4. What risk level applies?
5. Does the user have permission?
6. Does the action require approval?
7. Who is authorized to approve it?
8. What source data supports it?
9. What happens if it is approved?
10. What gets logged?

This rule ensures the system remains safe, accountable, auditable, and scalable.

Step 10 — Dashboards and Reporting

10.1 Purpose of This Section

This section defines how information should be presented to leadership and users through dashboards, briefings, reports, alerts, and analytics.

Dashboards and reports are the visibility layer of the AI Business Command Center.

The purpose is not just to show data. The purpose is to help users understand:

- What is happening

- What changed
- What is at risk
- What needs action
- Who is responsible
- What is overdue
- What requires approval
- What AI recommends doing next

The system should turn business data into clear management visibility and actionable decisions.

10.2 Dashboard and Reporting Philosophy

Dashboards should not become cluttered walls of charts.

The system should prioritize attention.

The guiding principle is:

Show the right person the right information at the right time, with the next action clearly visible.

Every dashboard should answer:

1. What should I know?
2. What should I do?
3. What is urgent?
4. What is at risk?
5. What changed?
6. Who owns the next step?

10.3 Dashboard Types

The system should include several dashboard types.

Executive Dashboards

Designed for leadership and ownership.

Examples:

- Command Center
- Daily Leadership Briefing
- Weekly Management Review
- Financial Overview
- Risk Overview

Department Dashboards

Designed for department managers.

Examples:

- Sales Dashboard
- Operations Dashboard
- Inventory Dashboard
- Finance Dashboard
- Compliance Dashboard
- Project Dashboard

User Dashboards

Designed for individual users.

Examples:

- My Tasks
- My Approvals
- My Customers
- My Projects
- My Alerts

AI Dashboards

Designed to review AI outputs and recommendations.

Examples:

- AI Recommendations
- AI-Created Tasks
- AI-Detected Risks
- AI Drafts Awaiting Review
- AI Accuracy/Feedback Summary

10.4 Main Dashboard — Command Center

Purpose

The Command Center is the primary leadership dashboard and daily starting point for managing the business.

It should present the most important information across the entire company in one place.

Primary Users

- Owner / CEO
- Executives
- Senior managers
- Authorized advisors with limited views

Main Sections

1. Executive Summary

Plain-English AI-generated overview of the current state of the business.

Should include:

- Business health summary
- Most important changes
- Top risks
- Top priorities
- Decisions needed
- Recommended next actions

2. Today's Priorities

A ranked list of the most important items needing attention.

Should include:

- Priority title
- Related record
- Owner
- Due date
- Risk level
- Recommended action
- Approval needed flag

3. Alerts and Risks

Shows open risks across modules.

Should include:

- High-severity alerts
- Overdue alerts
- Escalated alerts
- Repeated alert patterns
- AI-detected risks

4. Approval Queue

Shows items waiting for leadership or manager approval.

Should include:

- Pending approvals
- Overdue approvals
- High-risk approvals
- Dollar values where applicable
- AI recommendation summary

5. Sales and Customer Snapshot

Shows sales health and customer follow-up needs.

Should include:

- Active leads
- Open opportunities
- Stale opportunities
- Quotes pending approval
- Quotes expiring soon
- Customer follow-ups due
- Orders at risk

6. Operations and Production Snapshot

Shows operating activity and production risk.

Should include:

- Active work orders
- Active batches
- Production delays
- QC holds
- Missing materials
- Orders waiting on production

7. Inventory and Purchasing Snapshot

Shows inventory and purchasing risk.

Should include:

- Low-stock items
- Items below reorder point

- Expiring inventory
- Purchase orders late
- Purchase orders awaiting approval
- Materials required for upcoming production

8. Finance Snapshot

Shows management-level financial indicators.

Should include:

- Cash position where integrated/available
- Open receivables
- Overdue invoices
- Open payables
- Payables due soon
- Cash-flow warnings
- Margin concerns

9. Compliance, Legal, and Document Snapshot

Shows deadline and document risk.

Should include:

- Upcoming compliance deadlines
- Expiring documents
- Missing documents
- Legal/IP reminders
- Documents waiting review
- Contract obligations needing action

10. Team Accountability Snapshot

Shows execution health.

Should include:

- Overdue tasks
- Blocked tasks
- Tasks by owner
- Tasks waiting on leadership
- Repeated overdue patterns

11. AI Recommendations

Shows the most important AI-generated recommendations.

Should include:

- Recommendation title
- Related record
- Reason
- Suggested action
- Risk level
- Status
- Convert to task/approval option

10.5 Daily Leadership Briefing

Purpose

The Daily Leadership Briefing is a plain-English report generated each day to help leadership quickly understand what requires attention.

This should become the daily operating rhythm of the company.

Timing

Generated daily, preferably before the business day begins.

Primary Audience

- Owner / CEO
- Executives
- Department managers as permitted

Briefing Sections

1. Executive Overview

A concise summary of the business state.

Example structure:

- Overall status
- Main risks
- Main opportunities
- Key decisions needed

2. Top 5 Priorities Today

Ranked list of the most important actions.

Each priority should include:

- What needs attention
- Why it matters
- Owner
- Deadline
- Suggested action

3. Overnight / Recent Changes

Shows what changed since the previous briefing.

Examples:

- New orders
- New leads
- New overdue invoices
- Inventory changes
- Completed tasks
- New alerts
- Updated documents

4. Sales and Customer Follow-Ups

Shows customer-facing activity requiring attention.

Examples:

- Leads needing contact
- Stale opportunities
- Quotes waiting response
- Customers needing follow-up
- Customer issues

5. Orders and Fulfillment

Shows order status and risks.

Examples:

- Orders due soon
- Orders overdue
- Orders waiting on inventory
- Orders waiting on production
- Orders ready to invoice

6. Inventory and Purchasing

Shows materials, inventory, and vendor concerns.

Examples:

- Low-stock items
- Purchase orders late
- Purchase orders needing approval
- Expiring inventory
- Materials needed soon

7. Finance and Cash

Shows management-level finance concerns.

Examples:

- Overdue receivables
- Payables due soon
- Cash concerns
- Margin warnings
- Customer balance concerns

8. Compliance, Legal, and Documents

Shows document and deadline concerns.

Examples:

- Expiring documents
- Missing documents
- Compliance deadlines
- Legal/IP tasks
- Documents waiting review

9. Team Accountability

Shows execution issues.

Examples:

- Overdue tasks
- Blocked tasks
- Tasks waiting on leadership
- High-priority tasks due today

10. AI Recommended Actions

AI-generated recommended actions for leadership review.

Examples:

- Approve purchase order draft
- Follow up with customer
- Review overdue invoice
- Escalate blocked task
- Assign document review

User Actions From Briefing

Users should be able to:

- Convert item to task
- Assign owner
- Approve action
- Dismiss item
- Defer item
- Ask AI for explanation
- Open related record

10.6 Weekly Management Report

Purpose

The Weekly Management Report provides a structured operating review of the business.

It should help leadership understand performance, trends, risks, accountability, and decisions needed.

Timing

Generated weekly.

Primary Audience

- Owner / CEO
- Leadership team
- Department managers

Report Sections

1. Weekly Executive Summary

Plain-English summary of the week.

Should include:

- What went well
- What worsened
- What changed
- What requires attention
- What decisions are needed

2. Sales Performance

Should include:

- New leads
- Active opportunities
- Quotes created
- Quotes accepted/rejected
- Pipeline value
- Follow-up performance
- Stale opportunities

3. Customer and Order Status

Should include:

- New orders
- Fulfilled orders
- Orders at risk
- Orders overdue
- Customer issues
- Customer follow-ups due

4. Inventory and Purchasing

Should include:

- Low-stock items
- Stockouts
- Open purchase orders
- Late purchase orders
- Inventory value
- Expiring/aging inventory

5. Production and Operations

Should include:

- Active work orders
- Completed batches
- Delayed batches
- QC issues
- Yield/waste concerns
- Operational bottlenecks

6. Finance and Cash Flow

Should include:

- Receivables
- Payables
- Overdue invoices
- Cash concerns
- Revenue trend
- Margin concerns
- Collections priorities

7. Compliance, Legal, and Documents

Should include:

- Upcoming deadlines
- Expired/missing documents
- Documents reviewed
- Open legal/IP tasks
- Compliance risks

8. Tasks and Accountability

Should include:

- Tasks completed
- Tasks overdue
- Blocked tasks
- Tasks by owner
- Repeated accountability issues

9. Projects and Strategic Initiatives

Should include:

- Project status

- Milestones completed
- Milestones overdue
- Strategic risks
- Resource needs

10. AI Insights and Recommendations

Should include:

- Important AI recommendations
- Accepted/rejected recommendations
- Repeated risks detected
- Suggested leadership decisions

10.7 Monthly Performance Report

Purpose

The Monthly Performance Report provides a deeper business performance review.

It should focus on trends, profitability, operational performance, customer behavior, and strategic progress.

Report Sections

- Executive summary
- Revenue trend
- Sales pipeline trend
- Customer growth/loss
- Product performance
- Gross margin by product/customer/order
- Inventory value and turnover
- Purchasing performance
- Production efficiency
- Receivables/payables trend
- Compliance/document status
- Task/accountability performance
- Project progress
- Top risks entering next month
- AI recommendations for improvement

Key Questions Answered

- What improved this month?
- What got worse?
- Where did we make money?
- Where did we lose money, time, or efficiency?

- Which customers/products deserve more focus?
- What risks are building?
- What should leadership prioritize next month?

10.8 Quarterly Strategic Report

Purpose

The Quarterly Strategic Report helps leadership step back from daily operations and evaluate business direction.

Report Sections

- Strategic summary
- Revenue and profitability trend
- Customer concentration
- Product line performance
- Operational capacity
- Inventory and vendor risk
- Cash-flow direction
- Compliance/legal/IP status
- Team accountability trends
- Strategic project progress
- Growth opportunities
- System improvement opportunities
- AI automation opportunities

Key Questions Answered

- Are we moving in the right direction?
- What is constraining growth?
- What risks could affect the next quarter?
- Which projects should be accelerated, paused, or cancelled?
- What should we automate next?
- What decisions should leadership make now?

10.9 Department Dashboards

Each department should have its own dashboard focused on relevant work.

Sales Dashboard

Key Metrics

- New leads
- Active opportunities
- Pipeline value
- Quotes pending
- Quotes accepted/rejected
- Follow-ups due
- Stale opportunities
- Revenue by customer
- Customer issues

AI Support

- Prioritize leads
- Suggest next follow-up
- Draft customer emails
- Summarize customer history
- Identify at-risk opportunities

Operations Dashboard

Key Metrics

- Active work orders
- Active batches
- Orders waiting on production
- Production delays
- QC holds
- Missing materials
- Yield/waste indicators
- Operational blockers

AI Support

- Recommend production priorities
- Identify bottlenecks
- Summarize batch issues
- Create corrective action tasks
- Flag material shortages

Inventory and Purchasing Dashboard

Key Metrics

- Low-stock items
- Items below reorder point
- Stockouts
- Purchase orders pending
- Purchase orders late
- Inventory value
- Expiring inventory
- Vendor lead-time performance

AI Support

- Recommend purchases
- Draft purchase orders
- Identify vendor risk
- Forecast material needs
- Explain inventory movement

Finance Dashboard

Key Metrics

- Cash position
- Receivables
- Payables
- Overdue invoices
- Payment status
- Revenue trend
- Margin trend
- Collections priorities
- Cash-flow forecast

AI Support

- Summarize financial status
- Prioritize collections
- Draft payment reminders
- Explain margin changes
- Forecast cash risk

Compliance / Legal / IP Dashboard

Key Metrics

- Upcoming deadlines
- Overdue compliance tasks
- Expired documents
- Missing documents
- Contracts under review
- SOP review status
- COA status
- Legal/IP task status

AI Support

- Summarize obligations
- Extract deadlines
- Create compliance tasks
- Flag missing evidence
- Summarize IP/legal status

Task and Accountability Dashboard

Key Metrics

- Open tasks
- Overdue tasks
- Blocked tasks
- Tasks by owner
- Tasks by department
- High-priority tasks
- Tasks waiting on approval
- AI-created tasks awaiting review

AI Support

- Summarize overdue work
- Identify bottlenecks
- Suggest reassignment
- Create accountability report
- Detect repeated missed deadlines

10.10 User Dashboard — My Work

Purpose

Each user should have a personal dashboard showing the work they need to complete.

Sections

- My tasks due today
- My overdue tasks
- My approvals
- My assigned alerts
- My assigned customers/vendors/projects
- My recently updated records
- AI-suggested next actions
- Notifications

Key Actions

Users should be able to:

- Complete task
- Update status
- Add note
- Request approval
- Open related record
- Ask AI for help
- Defer or escalate with reason

10.11 Approval Queue Dashboard

Purpose

The Approval Queue gives authorized users one place to review and act on pending approvals.

Sections

- Pending approvals
- Overdue approvals
- High-risk approvals
- Approvals by module
- Approvals by requester
- Approvals requiring executive review
- AI-generated approval requests

Approval Record Should Show

- Requested action
- Related record
- Risk level
- Business impact
- Supporting data
- AI recommendation
- Requester
- Approver
- Due date
- Approval history

Actions

Approvers should be able to:

- Approve
- Reject
- Request changes
- Escalate
- Defer
- Reassign
- Ask AI for explanation

10.12 Alert Dashboard

Purpose

The Alert Dashboard gives users visibility into risks, exceptions, and unresolved issues.

Sections

- Open alerts
- High-severity alerts
- Alerts by module
- Alerts by owner
- Overdue alerts
- Escalated alerts
- Repeated alert patterns
- Dismissed alerts

Alert Record Should Show

- Alert title
- Alert type

- Severity
- Related record
- Trigger condition
- Owner
- Status
- Suggested action
- AI explanation
- Due date
- Resolution notes

Actions

Users should be able to:

- Assign owner
- Convert to task
- Convert to approval request
- Resolve
- Dismiss with reason
- Escalate
- Ask AI for explanation

10.13 AI Recommendations Dashboard

Purpose

The AI Recommendations Dashboard allows users to review, accept, reject, modify, or act on AI-generated recommendations.

Sections

- New recommendations
- High-risk recommendations
- Recommendations needing review
- Accepted recommendations
- Rejected recommendations
- Recommendations converted to tasks
- Recommendations converted to approval requests
- Repeated recommendation categories

Recommendation Record Should Show

- Recommendation title
- Type
- Related record
- Source records

- Reason
- Risk level
- Suggested action
- Suggested owner
- Approval required flag
- Confidence level
- Status
- Human decision

Actions

Users should be able to:

- Accept
- Reject
- Modify
- Defer
- Convert to task
- Convert to approval request
- Mark incorrect
- Ask AI for explanation

10.14 KPI Dashboard

Purpose

The KPI Dashboard provides measurable business performance indicators.

KPI Categories

Sales KPIs

- Active leads
- Open opportunities
- Pipeline value
- Quote volume
- Quote conversion rate
- Customer follow-ups due
- Stale opportunities
- Revenue by customer

Finance KPIs

- Receivables
- Payables
- Overdue invoices

- Cash position
- Revenue trend
- Gross margin
- Product profitability
- 30/60/90-day cash forecast

Operations KPIs

- Active orders
- Orders at risk
- Work orders open
- Batches completed
- Production delays
- Yield
- Waste
- QC holds

Inventory KPIs

- Inventory value
- Low-stock items
- Stockouts
- Reorder alerts
- Purchase orders late
- Expiring inventory
- Vendor lead-time performance

Compliance KPIs

- Upcoming deadlines
- Overdue compliance tasks
- Expired documents
- Missing documents
- SOPs due for review
- COAs missing
- Legal/IP deadlines

Accountability KPIs

- Open tasks
- Overdue tasks
- Blocked tasks
- Tasks by owner
- Completion rate
- Average completion time
- Approvals pending

10.15 Report Generation Rules

Reports should be generated from structured records and linked data.

Report Requirements

Each report should include:

- Report title
- Date/period covered
- Generated by
- Data sources used
- KPI summary
- AI summary
- Risks
- Recommended actions
- Linked records
- Tasks created from report
- Approval items if applicable

AI-Generated Report Standards

AI reports should:

- Identify source modules/data
- Explain important changes
- Highlight risks
- Avoid unsupported claims
- Distinguish facts from recommendations
- Respect permissions
- Allow items to convert into tasks

10.16 Dashboard Filtering and Views

Users should be able to filter dashboards based on role and need.

Common Filters

- Date range
- Module
- Owner
- Department
- Status
- Priority
- Risk level

- Customer
- Vendor
- Product
- Project
- Approval status
- AI-generated flag

Saved Views

Users should be able to save useful views, such as:

- My overdue tasks
- High-risk alerts
- Quotes needing approval
- Orders at risk
- Low inventory
- Documents expiring soon
- Customers needing follow-up
- Purchase orders late

10.17 Notifications and Digest Reports

The system should support notifications and digest reports.

Notification Types

- Task assigned
- Task overdue
- Approval requested
- Approval overdue
- Alert created
- Alert escalated
- Document expiring
- Inventory low
- Quote expiring
- Invoice overdue
- AI recommendation ready
- Report generated

Digest Types

- Daily leadership briefing
- Daily personal task digest
- Weekly management report
- Weekly department report
- Weekly approval summary

- Monthly performance report

10.18 Permission Rules for Dashboards and Reports

Dashboards and reports must respect permissions.

Rules

1. Users should only see modules they have access to.
2. Users should only see records they have permission to view.
3. Sensitive fields should be hidden if the user lacks access.
4. AI-generated summaries should not reveal restricted data.
5. Reports combining multiple modules should filter based on viewer permissions.
6. Exporting reports should require export permission.
7. Viewing sensitive reports should be logged when appropriate.

10.19 MVP Dashboard and Reporting Scope

The MVP should include the dashboards and reports that provide immediate management value.

Required MVP Dashboards

1. Command Center
2. Daily Leadership Briefing
3. My Work Dashboard
4. Task and Accountability Dashboard
5. Customer/Sales Dashboard
6. Inventory and Purchasing Dashboard
7. Document Dashboard
8. Alert Dashboard
9. Approval Queue
10. AI Recommendations Dashboard

Required MVP Reports

1. Daily Leadership Briefing
2. Weekly Management Report
3. Customer Follow-Up Report
4. Open Orders Report
5. Low Inventory Report
6. Open Purchase Orders Report
7. Overdue Tasks Report
8. Document Review/Expiration Report
9. AI Recommendations Report

10. Approval Queue Report

Deferred Advanced Dashboards and Reports

1. Advanced Finance Dashboard
2. Production Efficiency Dashboard
3. Batch Performance Dashboard
4. Compliance Dashboard
5. Project Portfolio Dashboard
6. Monthly Performance Report
7. Quarterly Strategic Report
8. Predictive Cash-Flow Report
9. Product Profitability Dashboard
10. Vendor Scorecard

10.20 Dashboard Design Rule

Every dashboard or report should answer:

1. Who is this for?
2. What decision does it support?
3. What action should the user take from it?
4. What records does it use?
5. What risks does it highlight?
6. What tasks or approvals are connected?
7. What AI summary or recommendation is useful?
8. What permissions apply?
9. What can be filtered or drilled into?
10. What should be logged?

This rule ensures dashboards remain practical, actionable, and aligned with business management needs.

Step 11 — Technical Architecture

11.1 Purpose of This Section

This section defines the technical foundation for building the AI Business Command Center.

The goal is to create a scalable, secure, modular system that can begin as a focused MVP and grow into a full business operating platform.

The architecture must support:

- Structured business records

- Cross-module relationships
- Role-based permissions
- Human approval workflows
- AI-assisted analysis and recommendations
- Document storage and intelligence
- Dashboards and reporting
- Activity logs and audit trails
- Future integrations with accounting, email, calendar, inventory, banking, shipping, and other systems

The system should be designed for reliability, extensibility, and data integrity from the beginning.

11.2 Architecture Philosophy

The system should be built as a modular platform with shared core services.

Each module should share the same underlying foundation for:

- Users and roles
- Permissions
- Records
- Tasks
- Documents
- Alerts
- Approvals
- AI recommendations
- Activity logs
- Reporting

This prevents each module from becoming a separate silo.

The core philosophy is:

Build one connected business platform, not many disconnected apps.

11.3 High-Level System Layers

The technical architecture should include the following layers:

1. Frontend Application Layer
2. Backend API Layer
3. Database Layer
4. Workflow Engine
5. AI Service Layer

6. Document and File Storage Layer
7. Integration Layer
8. Permissions and Security Layer
9. Audit and Logging Layer
10. Reporting and Analytics Layer
11. Notification Layer
12. Infrastructure and Deployment Layer

Each layer should be independently understandable, but designed to work together.

11.4 Frontend Application Layer

Purpose

The frontend is the user-facing application where leadership and team members interact with the system.

It should be clean, fast, role-aware, and action-oriented.

Core Frontend Areas

- Login and user access
- Command Center
- Module dashboards
- Record pages
- Task views
- Approval queue
- Alert center
- AI assistant interface
- Document viewer
- Reports
- Settings

Recommended Frontend Characteristics

- Web-based application
- Responsive design for desktop and tablet
- Mobile-friendly for task updates and approvals
- Role-based navigation
- Global search
- Contextual AI assistant on record pages
- Clear status badges
- Action buttons tied to permissions
- Audit/history views where needed

Important UI Principles

- Show next actions clearly.

- Avoid cluttered dashboards.
- Make related records easy to open.
- Make AI recommendations reviewable.
- Make approvals fast but controlled.
- Make tasks easy to assign and complete.

11.5 Backend API Layer

Purpose

The backend API controls business logic, database access, permissions, workflow execution, AI calls, integrations, and audit logging.

The backend should be the source of truth for business rules.

Core Backend Responsibilities

- User authentication
- Role and permission enforcement
- CRUD operations for core records
- Workflow status transitions
- Approval rules
- Alert generation
- Task creation and assignment
- AI recommendation storage
- Document metadata management
- Reporting queries
- Integration sync jobs
- Activity logging

API Design Principles

- All sensitive actions must go through the backend.
- Permission checks should be enforced server-side.
- AI should not directly modify protected records without backend validation.
- Every controlled action should create an audit log.
- Business rules should not live only in the frontend.

11.6 Database Layer

Purpose

The database stores structured business records and relationships.

It should support strong relationships, transactional integrity, reporting, permissions, and future scale.

Recommended Database Type

A relational database should be used as the primary data store.

Recommended option:

- PostgreSQL

PostgreSQL is well-suited because it supports relational modeling, indexing, JSON fields where needed, full-text search options, strong constraints, and scalability.

Core Database Tables

The MVP database should include tables for:

- companies
- users
- roles
- permissions
- customers
- contacts
- vendors
- products
- materials
- inventory_items
- quotes
- quote_line_items
- sales_orders
- sales_order_line_items
- purchase_orders
- purchase_order_line_items
- tasks
- documents
- document_links
- alerts
- approval_requests
- ai_recommendations
- ai_source_links
- activity_logs
- notes_communications

Phase 2 Tables

Future tables may include:

- inventory_transactions

- work_orders
- batches
- batch_inputs
- batch_outputs
- invoices
- payments
- expenses_bills
- compliance_requirements
- compliance_links
- projects
- project_participants
- reports
- report_snapshots
- notifications
- integrations
- automation_rules

Database Design Rules

- Every major table should include company_id.
- Every major table should include created_at and updated_at.
- Every major table should include created_by and updated_by where possible.
- Important records should include owner_id.
- Important records should include status.
- Sensitive records should support access control metadata.
- Deletions should usually be soft deletes or archival, not hard deletes.
- Many-to-many relationships should use link tables.
- Status transitions should be validated by backend rules.

11.7 Shared Core Services

The system should include shared services used by every module.

User Service

Manages users, roles, permissions, and user status.

Record Service

Provides common behavior for business records, such as ownership, status, tags, notes, related documents, and activity history.

Task Service

Creates, assigns, updates, escalates, and tracks tasks.

Document Service

Manages document metadata, storage references, links, review dates, expiration dates, and AI summaries.

Alert Service

Creates, assigns, resolves, dismisses, and escalates alerts.

Approval Service

Manages approval requests, queues, decisions, authority checks, and audit history.

AI Recommendation Service

Stores AI outputs, source data references, human decisions, and conversion to tasks or approvals.

Audit Service

Creates activity log entries for important system actions.

Notification Service

Sends in-app, email, or future mobile notifications.

Reporting Service

Generates dashboards, KPI summaries, recurring reports, and AI-assisted business summaries.

11.8 Workflow Engine

Purpose

The workflow engine controls business processes, status changes, approval requirements, alerts, and automation rules.

It ensures workflows behave consistently across modules.

Workflow Engine Responsibilities

- Define workflow states
- Validate status transitions
- Trigger approval requirements
- Trigger alerts
- Create tasks
- Escalate overdue items

- Run scheduled checks
- Execute safe automations
- Log workflow events

Example Workflow Rules

- A quote below target margin cannot be sent until approved.
- A purchase order above threshold cannot be issued until approved.
- An overdue invoice triggers an alert and collection task.
- A low inventory item triggers reorder recommendation.
- A document expiring within 30 days creates a review task.
- A task overdue by 2 days escalates to manager.

Workflow Design Rule

Workflows should be configurable where possible, but core safety rules should be enforced by backend logic.

11.9 AI Service Layer

Purpose

The AI Service Layer connects business data, documents, prompts, AI models, recommendation records, and human review workflows.

AI should not operate independently from the system. It should operate through controlled services that enforce permissions, source tracking, and approval rules.

AI Service Responsibilities

- Receive AI requests from users or scheduled jobs
- Check user permissions
- Retrieve relevant source records
- Retrieve relevant documents where permitted
- Generate summaries, recommendations, drafts, alerts, and reports
- Store AI outputs as structured records
- Link AI outputs to source data
- Label AI-generated content
- Route outputs to tasks or approval requests
- Log AI activity

AI Capabilities Supported

- Record summarization
- Document summarization
- Question answering over business records
- Daily briefing generation

- Weekly report generation
- Risk detection
- Task extraction
- Recommendation generation
- Draft communication generation
- Approval summary generation
- Data quality issue detection

AI Architecture Components

Prompt Templates

Reusable templates for daily briefings, customer summaries, document reviews, approval summaries, and risk recommendations.

Retrieval Layer

Finds relevant records and documents for AI to use.

Permission Filter

Ensures AI can only access data the user is allowed to access.

Model Call Layer

Sends structured context to AI models.

Output Parser

Converts AI output into structured records such as recommendations, tasks, alerts, or reports.

Review Layer

Allows humans to accept, reject, modify, or convert AI output.

Logging Layer

Stores AI activity, source records, and human decisions.

11.10 Retrieval and Knowledge Architecture

Purpose

AI must be able to retrieve the correct business context before generating answers or recommendations.

Structured Data Retrieval

AI should retrieve structured records from the database, such as:

- Customer records
- Orders
- Quotes
- Inventory items
- Tasks
- Approvals
- Alerts
- Documents metadata
- Activity logs

Document Retrieval

For uploaded documents, the system should support document indexing and search.

Potential components:

- Document text extraction
- Chunking
- Metadata tagging
- Embeddings/vector search
- Permission-aware retrieval
- Source citations/links back to document records

Retrieval Rules

- Prefer structured records over loose text when available.
- Use linked records to narrow context.
- Respect permissions before retrieval.
- Include source references with AI outputs.
- Avoid using outdated documents without showing version/status.
- Flag missing or conflicting data.

11.11 Document and File Storage Layer

Purpose

The system needs secure storage for uploaded files and generated documents.

File Storage Requirements

- Store uploaded files securely
- Store file metadata in database

- Support document versioning
- Support document links to business records
- Support access permissions
- Support expiration/review dates
- Support AI summaries and extracted obligations
- Support previews where practical
- Support audit logs for upload, download, and deletion/archive

Storage Approach

Use object storage for files and relational database records for metadata.

Potential object storage options:

- AWS S3
- Google Cloud Storage
- Azure Blob Storage
- Supabase Storage

Document Processing Pipeline

1. User uploads document.
2. File is stored securely.
3. Metadata record is created.
4. Text is extracted if supported.
5. AI summary/extraction is generated if requested or configured.
6. Document is linked to related records.
7. Review/expiration dates are tracked.
8. Activity log is created.

11.12 Integration Layer

Purpose

The integration layer connects the platform to outside business systems.

The MVP may begin with limited integrations, but the architecture should support future integrations.

Potential Integrations

Accounting

- QuickBooks
- Xero
- NetSuite
- Other accounting platforms

Use cases:

- Invoices
- Payments
- Bills
- Expenses
- Customers
- Vendors
- Financial summaries

Banking / Cash Visibility

- Bank feeds through secure providers
- Cash balance visibility
- Cash-flow forecasting

Email

- Gmail
- Microsoft Outlook

Use cases:

- Customer communication history
- Follow-up detection
- Draft replies
- Action item extraction

Calendar

- Google Calendar
- Microsoft Outlook Calendar

Use cases:

- Meeting-to-task workflows
- Follow-up scheduling
- Deadline reminders

Cloud Documents

- Google Drive
- OneDrive
- Dropbox
- Box

Use cases:

- Document storage/linking

- Contract review
- SOP management

Shipping / Fulfillment

- ShipStation
- UPS/FedEx tools
- E-commerce platforms

Use cases:

- Shipment status
- Fulfillment tracking

Inventory / Barcode

- Barcode scanners
- Inventory labels
- Warehouse tools

Use cases:

- Receiving
- Transfers
- Stock counts

Integration Design Rules

- Integrations should be permission-aware.
- Sync history should be logged.
- External IDs should be stored.
- Data conflicts should be flagged.
- Critical external updates should not overwrite internal records without rules.
- Integration failures should create alerts.

11.13 Permissions and Security Layer

Purpose

The security layer protects records, documents, AI outputs, reports, and controlled actions.

Security Requirements

- Secure authentication
- Role-based access control
- Module permissions
- Record permissions

- Field-level permissions where needed
- Document sensitivity levels
- Approval authority enforcement
- AI permission inheritance
- Audit logging
- Secure file access
- Soft deletion/archive controls

Security Design Rules

- Enforce permissions server-side.
- Do not rely on frontend hiding alone.
- AI must never access records outside user permissions.
- Sensitive downloads and exports should be logged.
- Admin actions should be logged.
- High-risk actions should require approval.

11.14 Audit and Logging Layer

Purpose

The audit layer records important activity for accountability, security, compliance, and debugging.

Events to Log

- User login/logout
- Record creation
- Record update
- Record archive/delete
- Status changes
- Permission changes
- Approval requests
- Approval decisions
- AI outputs
- AI recommendations
- Task completion
- Alert resolution
- Document upload/download
- Report generation
- Data export
- Integration sync events
- Automation execution

Audit Log Fields

- Actor type: user, AI, system, integration

- Actor ID
- Action type
- Related object type
- Related object ID
- Timestamp
- Previous value where applicable
- New value where applicable
- IP/device where applicable
- Source system where applicable
- Approval reference where applicable

Audit Design Rule

Every high-risk action must be reconstructable later.

The system should be able to answer:

- Who did it?
- What changed?
- When did it happen?
- Why was it approved?
- What data supported the decision?
- Was AI involved?

11.15 Reporting and Analytics Layer

Purpose

The reporting layer transforms structured data into dashboards, KPI views, recurring reports, and AI-generated summaries.

Reporting Components

- Dashboard queries
- KPI calculations
- Report templates
- Scheduled report generation
- AI report summarization
- Export controls
- Report snapshots
- Drill-down links to source records

Reporting Design Rules

- Reports should be based on structured data.
- Reports should respect permissions.

- Reports should link back to source records.
- AI-generated commentary should distinguish facts from recommendations.
- Reports should allow creation of tasks or approval requests from findings.

11.16 Notification Layer

Purpose

The notification layer makes sure users know when attention or action is required.

Notification Types

- Task assigned
- Task overdue
- Approval requested
- Approval overdue
- Alert created
- Alert escalated
- Document expiring
- Inventory low
- Quote expiring
- Invoice overdue
- AI recommendation ready
- Report generated

Notification Channels

MVP:

- In-app notifications
- Email notifications

Future:

- Mobile push notifications
- SMS for critical alerts
- Slack/Microsoft Teams integration

Notification Design Rules

- Notifications should be actionable.
- Notifications should link to related records.
- Users should control preferences where appropriate.
- Critical notifications should not be easy to miss.
- Notifications should avoid overwhelming users.

11.17 Infrastructure and Deployment Layer

Purpose

The infrastructure layer hosts and operates the application securely and reliably.

MVP Infrastructure Requirements

- Web application hosting
- Backend API hosting
- Managed PostgreSQL database
- Object storage for files
- Secure environment variables/secrets
- Background job processing
- Scheduled task runner
- Logging/monitoring
- Backup strategy
- Basic disaster recovery

Possible Deployment Options

Option A — Supabase + Modern Web App

Components:

- Supabase PostgreSQL
- Supabase Auth
- Supabase Storage
- Web frontend hosted on Vercel/Netlify
- Backend/API server or serverless functions

Good for:

- Fast MVP
- Relational database
- Built-in auth/storage
- Lower initial complexity

Option B — AWS-Based Architecture

Components:

- AWS RDS PostgreSQL
- S3 storage
- ECS/Lambda/API Gateway
- Cognito/Auth provider
- CloudWatch logging

Good for:

- Enterprise-grade scale
- More infrastructure control
- Strong long-term extensibility

Option C — Custom Cloud Stack

Components:

- PostgreSQL managed by cloud provider
- Object storage
- Containerized backend
- Separate auth provider
- Custom AI service

Good for:

- Maximum flexibility
- More engineering control

Recommended MVP Direction

For a first build, a managed stack is preferred so development effort focuses on business logic rather than infrastructure complexity.

A practical MVP stack could use:

- React or Next.js frontend
- Node.js/TypeScript or Python backend
- PostgreSQL database
- Object storage for documents
- Background worker for scheduled jobs
- AI service layer connected to selected model provider
- Role-based permissions built into backend

11.18 Background Jobs and Scheduled Processes

Purpose

Many system functions should run automatically on a schedule or in response to events.

Examples

- Generate daily briefing
- Generate weekly report

- Check overdue tasks
- Check approval escalations
- Check low inventory
- Check expiring documents
- Check overdue invoices
- Sync integrations
- Process uploaded documents
- Create recurring tasks
- Update KPI snapshots

Requirements

- Job queue
- Scheduled job runner
- Retry logic
- Error logging
- Failure alerts
- Job execution history

11.19 Event System

Purpose

The system should use events to trigger workflows, alerts, logs, and automations.

Example Events

- customer.created
- quote.submitted_for_approval
- quote.approved
- sales_order.created
- inventory.low_stock_detected
- purchase_order.delayed
- task.overdue
- document.uploaded
- document.expiring_soon
- approval.requested
- approval.approved
- ai.recommendation_created
- alert.escalated

Event System Benefits

- Keeps modules loosely connected
- Makes workflows easier to extend
- Supports automation

- Supports audit logging
- Supports notifications

11.20 Data Import and Migration

Purpose

The system should support importing existing business data.

Possible Import Sources

- Spreadsheets
- CSV files
- Existing CRM exports
- Accounting exports
- Inventory lists
- Customer lists
- Vendor lists
- Product lists
- Document folders

Import Requirements

- Field mapping
- Duplicate detection
- Validation rules
- Import preview
- Error reporting
- Import logs
- Rollback or archive option

AI Support

AI may help classify imported records, identify missing fields, summarize documents, and suggest cleanup tasks.

11.21 API and Integration Readiness

The system should be designed with API readiness from the beginning.

API Requirements

- Authentication
- Permission checks
- CRUD endpoints for core objects

- Search endpoints
- Reporting endpoints
- AI request endpoints
- Webhook endpoints for integrations
- Audit logging for controlled endpoints

Future API Use Cases

- Mobile app
- Barcode scanning tool
- External accounting sync
- Customer portal
- Vendor portal
- E-commerce integration
- Reporting tools

11.22 MVP Technical Scope

The MVP should focus on the technical foundation required to make the system usable and scalable.

Required MVP Technical Components

1. Web application frontend
2. Secure login/authentication
3. Role-based permissions
4. PostgreSQL database
5. Core object tables
6. Task system
7. Document upload and metadata
8. Alert system
9. Approval request system
10. AI recommendation system
11. Activity log system
12. Command Center dashboard
13. Daily briefing generation
14. Basic reporting
15. Background jobs for scheduled checks
16. Basic notification system
17. Soft delete/archive behavior
18. Data import support for core records

MVP AI Technical Components

1. Prompt templates
2. Permission-aware data retrieval
3. AI summary generation

4. AI recommendation generation
5. AI output storage
6. AI source links
7. AI-to-task conversion
8. AI-to-approval conversion
9. AI activity logging

Deferred Technical Components

1. Advanced forecasting engine
2. Full accounting sync
3. Bank feed integration
4. Full email/calendar integration
5. Mobile app
6. Barcode scanning
7. Advanced production scheduling
8. Advanced compliance rule engine
9. Multi-step approvals
10. Advanced data warehouse
11. Customer/vendor portals
12. Advanced analytics/BI layer

11.23 Technical Risks and Considerations

Data Quality Risk

AI and reports will only be useful if records are accurate and updated.

Mitigation:

- Required fields
- Data validation
- Data quality alerts
- Import cleanup workflows

Permission Complexity

Permissions can become complicated as the system grows.

Mitigation:

- Start with simple roles
- Enforce backend permission checks
- Add field-level controls later where needed

AI Trust Risk

Users may overtrust or undertrust AI.

Mitigation:

- Show source records
- Label AI outputs
- Require approval for sensitive actions
- Track user feedback

Integration Complexity

External systems can introduce data conflicts.

Mitigation:

- Start with limited integrations
- Store external IDs
- Log sync history
- Flag conflicts for review

Scope Creep

The system could become too large too quickly.

Mitigation:

- Build MVP foundation first
- Prioritize daily management value
- Add modules in phases

11.24 Technical Architecture Design Rule

Every technical decision should answer:

1. Does this support the core business objects?
2. Does this preserve clean relationships between records?
3. Does this enforce permissions?
4. Does this support AI safely?
5. Does this create an audit trail?
6. Does this make workflows easier to manage?
7. Does this scale to future modules?
8. Does this reduce operational complexity?
9. Does this support reporting and visibility?
10. Does this help the MVP deliver real management value?

This rule keeps the technical architecture aligned with the business purpose of the system.

Step 12 — MVP Scope

12.1 Purpose of This Section

This section defines the first buildable version of the AI Business Command Center.

The MVP should not attempt to include every feature, workflow, integration, or advanced AI capability described in the full blueprint.

The purpose of the MVP is to create a strong operating foundation that delivers immediate business value while proving the core system design.

The MVP should establish:

- Centralized business records
- Daily leadership visibility
- Customer and sales follow-up tracking
- Inventory and purchasing awareness
- Task accountability
- Document intelligence
- Alerts and risks
- Approval controls
- AI-generated recommendations
- Activity logging

The MVP should make the business easier to manage from day one.

12.2 MVP Objective

The MVP objective is:

Build a secure, AI-assisted business management platform that gives leadership one place to manage customers, sales opportunities, orders, vendors, inventory, purchasing, documents, tasks, alerts, approvals, and daily business priorities.

The MVP should answer the most important daily management questions:

- What needs my attention today?
- Which customer follow-ups are due or stale?
- Which quotes or orders need action?
- What inventory or purchasing risks exist?

- What tasks are overdue or blocked?
- What documents need review?
- What approvals are waiting?
- What has AI identified as important?

12.3 MVP Guiding Principle

The MVP should focus on **visibility, accountability, and controlled AI assistance**.

It should not attempt to fully automate the company.

The first version should help leadership see the business clearly, assign responsibility, monitor risks, and take action faster.

Advanced automation, deep integrations, and predictive modeling should come later.

12.4 MVP Required Modules

The MVP should include the following modules.

1. Command Center

Provides the main leadership dashboard.

Includes:

- Daily briefing
- Top priorities
- Alerts
- Overdue tasks
- Approvals waiting
- Sales follow-ups
- Order risks
- Inventory/purchasing risks
- Document review items
- AI recommendations

2. Customers and Sales

Manages customers, contacts, leads, follow-ups, and sales opportunities.

Includes:

- Customer records

- Contact records
- Lead/customer status
- Sales stage
- Last contact date
- Next follow-up date
- Customer tasks
- Customer documents
- AI customer summary
- Stale follow-up detection

3. Quotes and Sales Orders

Manages quote and order visibility.

Includes:

- Quote records
- Quote line items
- Quote status
- Sales order records
- Sales order line items
- Order status
- Due dates
- Related customer/product records
- Quote/order tasks
- Quote risk flags

4. Vendors and Purchasing

Manages vendors and purchase orders.

Includes:

- Vendor records
- Vendor contacts
- Purchase orders
- Purchase order line items
- Purchase order status
- Expected delivery dates
- Purchasing tasks
- Late PO alerts
- PO approval workflow

5. Products, Materials, and Inventory

Provides basic inventory awareness.

Includes:

- Product records
- Material records
- Inventory item records
- Quantity on hand
- Reorder point
- Low-stock alerts
- Vendor/material relationships
- Inventory-related tasks

6. Tasks and Accountability

Creates the execution layer of the system.

Includes:

- Task creation
- Task ownership
- Due dates
- Priorities
- Statuses
- Related records
- Overdue task detection
- Blocked task tracking
- AI-to-task conversion

7. Documents and Knowledge Base

Stores and connects important documents.

Includes:

- Document upload
- Document metadata
- Document type
- Document owner
- Related records
- Expiration/review dates
- AI summaries
- AI-extracted dates/action items
- Document review tasks

8. Alerts and Risks

Tracks business exceptions and risks.

Includes:

- Alert creation
- Alert severity
- Alert owner
- Related record
- Status
- Conversion to task
- Conversion to approval request
- AI-generated alerts

9. Approvals and Controls

Provides human review for controlled actions.

Includes:

- Approval request records
- Approval queue
- Approver assignment
- Approval statuses
- Approve/reject/request changes
- Approval activity logs
- AI approval summaries

10. AI Assistant and Recommendations

Provides AI support across the MVP.

Includes:

- Daily briefing generation
- Customer summaries
- Document summaries
- Task extraction
- Risk recommendations
- Purchase recommendations
- Quote risk review
- AI recommendation records
- Convert AI output to task or approval request

11. Settings, Roles, and Audit Logs

Provides security and traceability.

Includes:

- User accounts

- Roles
- Basic permissions
- Activity logs
- AI-generated content labels
- Soft delete/archive behavior

12.5 MVP Required Objects

The MVP should include these core objects.

Company and Access Objects

- Company
- User
- Role
- Permission
- Activity Log

Sales Objects

- Customer
- Contact
- Quote
- Quote Line Item
- Sales Order
- Sales Order Line Item

Vendor and Purchasing Objects

- Vendor
- Purchase Order
- Purchase Order Line Item

Product and Inventory Objects

- Product
- Material
- Inventory Item
- Product Material Link
- Vendor Material Link

Execution Objects

- Task
- Alert
- Approval Request
- AI Recommendation

- AI Source Link

Document Objects

- Document
- Document Link
- Document Summary

Communication / Notes Objects

- Note / Communication

12.6 MVP Required Workflows

The MVP should support the following workflows.

1. Lead / Customer Follow-Up Workflow

- Create customer or lead
- Add contacts
- Assign owner
- Set next follow-up date
- AI detects stale follow-ups
- Tasks are created for follow-up

2. Quote Workflow

- Create quote
- Add line items
- Review pricing/margin where available
- Trigger approval if needed
- Send/update status manually
- Convert accepted quote to sales order

3. Sales Order Workflow

- Create sales order
- Add products and quantities
- Set due date
- Check related inventory manually or automatically where possible
- Flag order risks
- Create related tasks

4. Purchasing Workflow

- Create purchase order
- Add vendor and line items

- Submit for approval if needed
- Track expected delivery date
- Mark received/closed manually
- Alert on late purchase orders

5. Inventory Monitoring Workflow

- Create product/material/inventory records
- Track quantity on hand
- Set reorder point
- Generate low-stock alerts
- AI recommends purchasing task or PO draft

6. Task Workflow

- Create task manually or from AI recommendation
- Assign owner
- Set due date and priority
- Update status
- Detect overdue or blocked tasks
- Escalate important overdue tasks

7. Document Workflow

- Upload document
- Classify document type
- Link document to related record
- Generate AI summary
- Extract review/expiration dates when possible
- Create document review task

8. Alert Workflow

- Create alert manually, by rule, or by AI
- Assign severity
- Link to related record
- Assign owner
- Convert to task or approval request
- Resolve/dismiss/escalate

9. Approval Workflow

- Create approval request
- Assign approver
- Show supporting context
- Approver approves, rejects, requests changes, defers, or escalates
- Decision is logged

10. Daily Briefing Workflow

- System gathers current data
- AI generates daily leadership summary
- User reviews priorities, alerts, tasks, approvals, and recommendations
- User converts briefing items into tasks or approvals

12.7 MVP Required Dashboards

The MVP should include the following dashboards.

Command Center

Main leadership view.

Must show:

- Daily briefing
- Top priorities
- Open alerts
- Overdue tasks
- Pending approvals
- Sales follow-ups
- Quote/order risks
- Low inventory
- Late purchase orders
- Documents needing review
- AI recommendations

My Work Dashboard

Personal user dashboard.

Must show:

- My tasks
- My overdue tasks
- My approvals
- My alerts
- My recent records
- AI suggested actions

Customer/Sales Dashboard

Must show:

- Active leads/customers
- Follow-ups due
- Stale follow-ups
- Open quotes
- Quotes needing approval
- Open sales orders

Inventory and Purchasing Dashboard

Must show:

- Low-stock inventory
- Items below reorder point
- Open purchase orders
- Late purchase orders
- Purchase orders needing approval
- AI purchase recommendations

Document Dashboard

Must show:

- Recently uploaded documents
- Documents needing classification
- Documents needing review
- Expiring documents
- AI summaries available

Task and Accountability Dashboard

Must show:

- Tasks by owner
- Tasks by priority
- Overdue tasks
- Blocked tasks
- AI-created tasks

Approval Queue

Must show:

- Pending approvals
- Overdue approvals

- Approvals by type
- Approvals by risk level
- AI-generated approval summaries

AI Recommendations Dashboard

Must show:

- New recommendations
- High-risk recommendations
- Recommendations needing review
- Accepted/rejected recommendations
- Recommendations converted to tasks or approvals

12.8 MVP Required AI Capabilities

The MVP should include practical AI capabilities that support daily management.

Daily Leadership Briefing

AI generates a daily summary of business priorities, risks, alerts, overdue tasks, pending approvals, and recommended actions.

Customer Summary

AI summarizes customer status, recent notes, open quotes/orders, follow-ups, tasks, and risks.

Stale Follow-Up Detection

AI or rules identify leads/customers that have not been contacted within a defined timeframe.

Task Summary

AI summarizes overdue tasks, blocked tasks, and tasks needing leadership attention.

Document Summary

AI summarizes uploaded documents and extracts key dates, obligations, parties, or action items where possible.

Action Item Extraction

AI extracts tasks from notes, communications, or documents.

Low Inventory Recommendation

AI or rules identify low-stock items and recommend purchasing action.

Purchase Order Draft Support

AI helps prepare purchase order drafts from inventory need, vendor, and material data.

Quote Risk Review

AI flags potential quote issues such as missing data, unusual discount, low margin, or customer balance concern where data exists.

Order Risk Summary

AI identifies open orders that may be at risk due to due date, inventory, production, or unresolved tasks.

Approval Summary

AI summarizes why an approval request matters and what supporting data is relevant.

AI Recommendation Records

Every important AI recommendation is stored, reviewed, and convertible into a task or approval request.

12.9 MVP Required Reports

The MVP should include these basic reports.

Daily Leadership Briefing

Generated daily or on demand.

Weekly Management Report

Summarizes weekly sales, tasks, alerts, approvals, inventory, purchasing, documents, and AI recommendations.

Customer Follow-Up Report

Shows customers/leads needing contact and stale opportunities.

Open Quotes Report

Shows quotes by status, age, expiration date, and approval state.

Open Orders Report

Shows active sales orders, due dates, status, and risk flags.

Low Inventory Report

Shows inventory items below reorder point or marked as risk.

Open Purchase Orders Report

Shows purchase orders by vendor, status, expected delivery date, and approval status.

Overdue Tasks Report

Shows overdue tasks by owner, priority, and related record.

Document Review Report

Shows documents needing review, classification, or expiration attention.

Approval Queue Report

Shows pending and overdue approvals.

AI Recommendations Report

Shows AI recommendations by status, type, risk level, and outcome.

12.10 MVP Required Permissions and Controls

The MVP should include basic but strong controls.

Required Roles

- Owner / CEO
- System Administrator
- Operations Manager
- Sales Manager
- Finance Manager
- Compliance Manager
- Inventory / Warehouse User
- Production Operator
- Outside Advisor
- Read-Only User

Required Controls

- Secure login
- Role-based module access
- Basic record ownership
- Approval request workflow
- Document sensitivity levels
- AI permission inheritance
- AI-generated labels
- Activity logs
- Soft delete/archive
- Export restrictions where applicable

MVP Approval Rules

Approval should be required for:

- Quotes below margin or above threshold
- Purchase orders above threshold
- Sensitive document archive
- External communications where user lacks permission
- Role/permission changes
- High-risk AI recommendations

12.11 MVP Technical Components

The MVP technical build should include:

Application

- Web application
- Role-aware navigation
- Record pages
- Dashboards
- Search
- AI assistant panel

Backend

- API layer
- Business logic
- Permission checks
- Workflow/status rules
- Approval rules
- AI service integration

Database

- PostgreSQL database
- Core tables
- Link tables
- Status fields
- Ownership fields
- Activity logs

Documents

- Secure file storage
- Document metadata
- Document links
- Document AI summaries

AI

- Prompt templates
- Permission-aware retrieval
- AI output storage
- Source links
- AI-to-task conversion
- AI-to-approval conversion

Background Jobs

- Daily briefing generation
- Overdue task checks
- Low inventory checks
- Late purchase order checks
- Document review/expiration checks
- Approval escalation checks

Notifications

- In-app notifications
- Basic email notifications

12.12 Excluded From MVP

The following should be intentionally excluded from the MVP to prevent overbuilding.

Excluded Advanced Finance

- Full accounting sync
- Bank feeds

- Automated reconciliation
- Advanced cash-flow forecasting
- Tax reporting
- Payroll

Excluded Advanced Production

- Full batch traceability
- Equipment tracking
- Labor tracking
- Advanced production scheduling
- QC release workflow
- Formula/specification version control

Excluded Advanced Compliance

- Full compliance rule engine
- Jurisdiction-specific regulatory automation
- Digital signature workflows
- Automated legal/IP filing management

Excluded Advanced Integrations

- Full email sync
- Full calendar sync
- Barcode scanning
- Shipping integrations
- Customer portal
- Vendor portal
- E-commerce integration

Excluded Advanced AI

- Autonomous purchasing
- Autonomous customer communication
- Predictive customer churn scoring
- Advanced production optimization
- Automated legal risk scoring
- Multi-agent automation
- Unapproved controlled actions

These exclusions should remain visible so the team does not confuse future roadmap items with MVP requirements.

12.13 MVP Build Phases

The MVP can be built in internal milestones.

MVP Phase 1 — Foundation

Build:

- Authentication
- Users and roles
- Core database schema
- Customers
- Contacts
- Vendors
- Products
- Materials
- Inventory items
- Activity logs

Goal:

Create the core business record foundation.

MVP Phase 2 — Execution Layer

Build:

- Tasks
- Alerts
- Approval requests
- Notes/communications
- Record linking
- Basic notifications

Goal:

Create accountability and control.

MVP Phase 3 — Sales, Orders, and Purchasing

Build:

- Quotes
- Sales orders
- Purchase orders
- Line items
- Quote/order statuses
- PO statuses
- Basic risk flags

Goal:

Connect customer demand, orders, vendors, and purchasing.

MVP Phase 4 — Documents and AI

Build:

- Document upload
- Document metadata
- Document links
- AI summaries
- AI recommendations
- AI source links
- AI-to-task conversion
- AI-to-approval conversion

Goal:

Turn documents and AI insights into usable business actions.

MVP Phase 5 — Command Center and Reports

Build:

- Command Center dashboard
- Daily briefing
- My Work dashboard
- Approval queue
- Alert dashboard
- AI recommendations dashboard
- Weekly management report

Goal:

Give leadership one place to manage the business.

MVP Phase 6 — Hardening and Testing

Build/refine:

- Permission testing
- Workflow testing
- Data validation
- Activity log completeness
- AI output review
- Import testing
- User acceptance testing
- Performance review

Goal:

Prepare the MVP for real business use.

12.14 MVP Acceptance Criteria

The MVP should be considered ready when it can satisfy the following criteria.

Business Criteria

- Leadership can open the Command Center and see daily priorities.
- Customer follow-ups can be tracked and stale follow-ups are visible.
- Quotes and sales orders can be tracked by status.
- Inventory items can be monitored for low stock.
- Purchase orders can be created and tracked.
- Tasks can be assigned, completed, and monitored.
- Documents can be uploaded, summarized, and linked to records.
- Approvals can be requested and decided.
- AI recommendations can be reviewed and converted to tasks or approvals.
- Activity history is visible for important actions.

Technical Criteria

- Users can log in securely.
- Roles restrict access correctly.
- Database relationships work correctly.
- Core records can be created, edited, archived, and searched.
- Permission checks occur server-side.
- Activity logs are created for controlled actions.
- AI outputs are stored and labeled.
- Background jobs run reliably.
- Basic notifications work.
- Soft delete/archive behavior works.

AI Criteria

- Daily briefing generates useful summary.
- Customer summaries are accurate enough for review.
- Document summaries show key points and uncertainty where appropriate.
- AI recommendations identify related records and suggested actions.
- AI cannot access records outside user permissions.
- AI cannot execute controlled actions without approval.
- Users can accept, reject, modify, or convert AI outputs.

Control Criteria

- Approval requests can be created.

- Approval queue works.
- Approvers can approve/reject/request changes.
- Approval decisions are logged.
- High-risk actions are blocked without approval.
- Sensitive documents have access controls.

12.15 MVP Success Metrics

The MVP should be judged by whether it improves management execution.

Early Success Metrics

- Daily briefing reviewed regularly
- Number of overdue tasks reduced
- Number of stale customer follow-ups reduced
- Number of low inventory surprises reduced
- Number of documents classified and linked
- Number of AI recommendations reviewed
- Number of AI recommendations converted to tasks
- Time saved preparing weekly management summary
- Number of approvals handled through the system
- User adoption by role

Qualitative Success Signals

- Leadership feels more aware of business status.
- Managers know what needs action.
- Fewer items are forgotten.
- Customer follow-up is more consistent.
- Inventory risks are visible earlier.
- Documents are easier to find and understand.
- AI outputs are useful but controlled.

12.16 MVP Data Import Requirements

To make the MVP useful quickly, the system should support importing existing data.

Initial Imports

- Customer list
- Contact list
- Vendor list
- Product list
- Material list

- Inventory list
- Open quotes
- Open sales orders
- Open purchase orders
- Current tasks
- Important documents

Import Features

- CSV upload
- Field mapping
- Required field validation
- Duplicate detection
- Import preview
- Error report
- Import activity log

AI Import Assistance

AI may help:

- Classify records
- Identify missing fields
- Suggest cleanup tasks
- Summarize imported documents
- Identify duplicate-like records for review

12.17 MVP Launch Readiness Checklist

Before using the MVP in the business, the following should be completed.

System Setup

- Company profile configured
- Users created
- Roles assigned
- Permission defaults reviewed
- Approval thresholds configured
- Notification settings configured

Data Setup

- Customers imported
- Contacts imported
- Vendors imported
- Products/materials imported
- Inventory imported

- Open quotes/orders/POs imported where available
- Key documents uploaded

Workflow Setup

- Task statuses configured
- Quote statuses configured
- Sales order statuses configured
- Purchase order statuses configured
- Alert rules configured
- Approval rules configured

AI Setup

- AI prompts/templates configured
- Daily briefing tested
- Customer summary tested
- Document summary tested
- Recommendation workflow tested
- AI permission behavior tested

Team Setup

- Users trained on core workflows
- Managers trained on dashboards
- Approvers trained on approval queue
- Leadership trained on Command Center
- Feedback process established

12.18 MVP Design Rule

The MVP should answer:

1. Can leadership see what needs attention today?
2. Can users manage customers, quotes, orders, vendors, inventory, documents, tasks, alerts, and approvals?
3. Can AI summarize, recommend, and draft without bypassing human control?
4. Can recommendations become tasks or approvals?
5. Can sensitive actions be controlled and logged?
6. Can the system scale into production, finance, compliance, and advanced automation later?

If the MVP does these things well, it is a strong first version.

Step 13 — Future Roadmap

13.1 Purpose of This Section

This section defines how the AI Business Command Center should expand after the MVP.

The roadmap prevents the system from becoming scattered or overbuilt too early. It gives the company a phased path for adding deeper functionality, stronger integrations, advanced AI, and more automation after the core foundation is working.

The roadmap should be treated as a strategic guide, not a fixed schedule. Each phase should be prioritized based on business value, data readiness, user adoption, and operational need.

13.2 Roadmap Philosophy

The system should grow in layers.

The correct order is:

1. Build the foundation.
2. Prove daily usefulness.
3. Improve operational depth.
4. Add financial and compliance intelligence.
5. Add integrations.
6. Add advanced automation.
7. Scale to portals, mobile, analytics, and broader company operations.

The guiding principle is:

Do not automate chaos. Structure the business first, then automate the structured workflows.

13.3 Roadmap Overview

The long-term roadmap should include these phases:

Phase 1 — MVP Foundation

Establish core records, tasks, documents, alerts, approvals, AI recommendations, and the Command Center.

Phase 2 — Operations, Production, and Inventory Depth

Add deeper manufacturing, batch, inventory transaction, QC, and production planning capability.

Phase 3 — Finance and Cash-Flow Intelligence

Add deeper financial visibility, receivables/payables workflows, margin analysis, cash forecasting, and accounting integrations.

Phase 4 — Compliance, Legal, and IP Management

Add deeper compliance workflows, legal document controls, patent/IP tracking, evidence management, and recurring obligations.

Phase 5 — Integrations and Workflow Automation

Connect outside systems and automate repeatable workflows with approval controls.

Phase 6 — Advanced AI and Predictive Management

Add forecasting, optimization, anomaly detection, advanced recommendations, and semi-autonomous workflows.

Phase 7 — Portals, Mobile, and Scale

Add mobile access, customer/vendor portals, external collaboration, and scale features for larger operations.

13.4 Phase 1 — MVP Foundation

Objective

Create the first usable version of the business operating system.

Core Capabilities

- Command Center
- Daily briefing
- Customers and contacts
- Vendors
- Products and materials
- Basic inventory
- Quotes
- Sales orders
- Purchase orders
- Tasks
- Documents
- Alerts
- Approvals
- AI recommendations

- Activity logs
- Basic dashboards and reports

Business Value

- Creates one place to manage the business
- Improves visibility
- Reduces missed follow-ups
- Improves task accountability
- Identifies low inventory risks
- Makes documents more useful
- Gives leadership a daily management rhythm

Completion Signal

The MVP is successful when leadership uses the Command Center and daily briefing to manage real business activity.

13.5 Phase 2 — Operations, Production, and Inventory Depth

Objective

Expand the system from basic inventory and orders into deeper operational control.

New Capabilities

Work Orders

- Create work orders from sales orders, forecasts, or internal needs
- Assign production owner
- Track status
- Link required materials
- Link SOPs/specifications
- Connect to batches

Batch / Production Runs

- Batch records
- Input materials
- Output quantities
- Yield
- Waste
- Rework
- QC status
- Deviations
- Batch documents
- COAs/lab records

Inventory Transactions

- Receive
- Consume
- Produce
- Transfer
- Reserve
- Ship
- Adjust
- Scrap/waste
- Return

Production Planning

- Material readiness checks
- Production due dates
- Order-to-production visibility
- Workload planning
- Production priority queue

Quality and Traceability

- QC hold status
- Deviation tracking
- Corrective action tasks
- Lot/batch traceability
- COA linking

AI Enhancements

- Production readiness summary
- Missing material detection
- Batch issue summary
- Yield variance explanation
- Waste pattern detection
- Production priority recommendations
- Corrective action task suggestions

Business Value

- Better control over manufacturing
- Reduced production delays
- Improved traceability
- Better material planning
- Stronger quality documentation
- Clear connection between orders, batches, inventory, and fulfillment

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13.6 Phase 3 — Finance and Cash-Flow Intelligence

Objective

Expand the system into stronger management-level financial visibility.

New Capabilities

Receivables

- Customer invoices
- Due dates
- Payment status
- Aging reports
- Collections tasks
- Payment reminder workflows

Payables

- Vendor bills
- Expense records
- Due dates
- Approval workflows
- Payment scheduling status

Cash-Flow Forecasting

- 30-day forecast
- 60-day forecast
- 90-day forecast
- Expected inflows
- Expected outflows
- Scenario planning

Margin and Profitability

- Gross margin by product
- Gross margin by order
- Gross margin by customer
- Cost variance
- Product cost tracking
- Inventory cost basis

Accounting Integration

Potential integrations:

- QuickBooks

- Xero
- NetSuite
- Other accounting systems

AI Enhancements

- Collections prioritization
- Payment reminder drafting
- Cash-risk alerts
- Margin issue detection
- Expense anomaly detection
- Financial summary generation
- Customer profitability explanation

Business Value

- Better cash awareness
- Earlier detection of financial risk
- Stronger collections process
- Better margin visibility
- Reduced manual reporting
- Improved financial decision-making

13.7 Phase 4 — Compliance, Legal, and IP Management

Objective

Expand the system into a stronger compliance, legal, documentation, and intellectual property management platform.

New Capabilities

Compliance Requirements

- Requirement records
- Jurisdictions
- Owners
- Deadlines
- Evidence requirements
- Recurring schedules
- Status tracking
- Escalation rules

Contract and Obligation Management

- Contract summaries
- Extracted obligations

- Renewal dates
- Termination dates
- Responsible owners
- Related tasks
- Approval workflows

Legal/IP Tracking

- Patent matters
- Filing deadlines
- Attorney tasks
- Invention records
- Document packages
- IP milestone tracking
- Confidential document controls

SOP and Quality Documents

- SOP review schedules
- Version control
- Approval workflow
- Training acknowledgement links in future version

COA and Evidence Management

- COA linking to batches/products
- Evidence document tracking
- Missing evidence alerts
- Expiring document alerts

AI Enhancements

- Obligation extraction
- Legal/compliance deadline detection
- Contract review summaries
- IP deadline summaries
- Missing evidence detection
- Compliance checklist generation
- SOP change summaries

Business Value

- Fewer missed legal/compliance deadlines
- Better document control
- Stronger IP oversight
- Easier audits
- Better visibility into obligations
- Reduced legal and operational risk

13.8 Phase 5 — Integrations and Workflow Automation

Objective

Connect the AI Business Command Center to outside systems and automate repeatable workflows with human control.

Integration Priorities

Accounting Integration

- Sync customers
- Sync invoices
- Sync payments
- Sync bills/expenses
- Sync vendors
- Sync financial summaries

Email Integration

- Link customer/vendor emails to records
- Detect follow-up needs
- Extract action items
- Draft replies
- Create communication history

Calendar Integration

- Link meetings to customers/projects
- Extract meeting action items
- Create follow-up tasks
- Track deadlines

Cloud Document Integration

- Link existing file folders
- Sync document metadata
- Import documents into knowledge base
- Track review/expiration dates

Shipping / Fulfillment Integration

- Track shipments
- Update fulfillment status
- Link tracking numbers
- Notify customer/order owners

Barcode / Inventory Tools

- Scan receiving
- Scan transfers
- Scan consumption
- Scan finished goods
- Improve inventory accuracy

Automation Priorities

- Recurring tasks
- Overdue escalations
- Document review reminders
- Low-stock purchase requests
- Follow-up task creation
- Approval routing
- Weekly report generation
- AI recommendation routing

Business Value

- Less manual data entry
- Better system adoption
- Faster workflows
- Fewer missed updates
- More accurate records
- Stronger connection between daily tools and management system

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13.9 Phase 6 — Advanced AI and Predictive Management

Objective

Move from AI-assisted visibility to predictive and proactive management.

Advanced AI Capabilities

Cash Forecasting

- Predict cash position
- Model best/worst/base cases
- Identify upcoming cash gaps
- Recommend collections or purchasing actions

Demand Forecasting

- Forecast product demand
- Forecast material needs

- Identify seasonality
- Predict purchasing requirements

Production Optimization

- Recommend production schedule
- Balance orders, inventory, materials, deadlines, and labor
- Identify bottlenecks before they occur

Vendor Performance Intelligence

- Score vendor reliability
- Predict vendor delays
- Compare cost, lead time, and quality
- Recommend preferred vendor changes

Customer Risk and Opportunity Scoring

- Identify customers at risk of going cold
- Identify high-value opportunities
- Recommend next best actions
- Score customer profitability

Anomaly Detection

- Unusual expenses
- Unusual inventory adjustments
- Margin changes
- Production yield changes
- Repeated overdue tasks
- Unusual customer/order behavior

Advanced AI Agents

Specialized AI agents may be added for:

- Sales
- Finance
- Inventory
- Production
- Compliance
- Documents
- Executive reporting

Business Value

- Earlier risk detection
- Better planning
- Stronger forecasting

- More intelligent automation
- Reduced management burden
- Better strategic decisions

Control Requirement

Advanced AI should still follow the approval and audit rules defined earlier.

AI may recommend and prepare actions, but high-risk execution must remain human-controlled.

13.10 Phase 7 — Portals, Mobile, and Scale

Objective

Expand access, collaboration, and usability as the company grows.

Mobile App

Use cases:

- Approve requests on the go
- Complete tasks
- View daily briefing
- Receive alerts
- Scan inventory
- Upload documents/photos
- Update production status

Customer Portal

Potential features:

- View order status
- View documents
- Submit requests
- Approve quotes
- Upload documents
- View invoices/payment status where appropriate

Vendor Portal

Potential features:

- Confirm purchase orders
- Upload COAs/documents
- Update delivery status

- Submit invoices
- Maintain vendor information

Advisor Portal

Potential features:

- Controlled legal/accounting/consultant access
- Document review
- Project collaboration
- Task assignment
- Commenting

Enterprise Scale Features

- Multiple locations
- Multiple companies/entities
- Advanced department permissions
- Advanced reporting warehouse
- SSO
- Advanced audit controls
- Multi-step approval chains
- Data retention policies

Business Value

- Better collaboration
- Faster approvals
- Improved external communication
- More accurate operational data
- Scalable management infrastructure

13.11 Long-Term Product Vision

The long-term vision is to create a system that becomes the company's operating brain.

The system should eventually be able to:

- Understand the state of the business in real time
- Detect risks across departments
- Recommend decisions
- Prepare actions
- Track accountability
- Forecast needs
- Manage documents and obligations
- Coordinate workflows

- Support leadership strategy
- Improve over time through feedback

The goal is not to replace leadership.

The goal is to give leadership better visibility, faster decision-making, stronger control, and more leverage.

13.12 Roadmap Prioritization Criteria

Future features should be prioritized using clear criteria.

Prioritization Questions

1. Does this feature improve leadership visibility?
2. Does it reduce missed follow-ups or missed deadlines?
3. Does it improve cash, margin, or profitability awareness?
4. Does it reduce operational risk?
5. Does it improve customer experience?
6. Does it improve production or inventory control?
7. Does it reduce compliance/legal/IP risk?
8. Does it save meaningful time?
9. Does it improve data quality?
10. Does it support scalable growth?

Features that do not support these outcomes should not be prioritized.

13.13 Roadmap Dependencies

Some phases depend on earlier foundations.

Production Depth Depends On

- Products
- Materials
- Inventory
- Sales orders
- Workflows
- Documents
- Tasks

Finance Intelligence Depends On

- Customers
- Orders
- Invoices

- Payments
- Vendors
- Purchase orders
- Expenses
- Cost data

Compliance Depth Depends On

- Documents
- Products
- Batches
- Vendors
- Tasks
- Owners
- Deadlines

Advanced AI Depends On

- Clean structured data
- Reliable relationships
- User feedback
- Permissions
- Approval logs
- Historical records

Integrations Depend On

- Stable internal data model
- External ID storage
- Sync rules
- Conflict handling
- Audit logs

13.14 Roadmap Governance

The roadmap should be reviewed regularly.

Recommended Review Cadence

- Monthly during MVP build
- Monthly for first 90 days after MVP launch
- Quarterly after system adoption stabilizes

Review Questions

- What parts of the system are being used?
- What features are not being used?

- What data is missing or unreliable?
- What workflows are still happening outside the system?
- What reports are leadership actually using?
- Which AI recommendations are useful?
- Which automations are safe to expand?
- What should be built next?

13.15 Future Roadmap Design Rule

Every roadmap item should answer:

1. What business problem does this solve?
2. What existing data does it require?
3. What new records or workflows are needed?
4. What permissions or approvals are required?
5. What AI capabilities does it need?
6. What risks does it introduce?
7. What value will it create?
8. How will success be measured?
9. Does it depend on another phase?
10. Should it be built now or later?

This rule keeps the roadmap disciplined and aligned with business value.

Blueprint Completion Summary

The initial AI Business Management System Blueprint now includes:

1. Vision, Purpose, and Design Principles
2. Business Goals and Success Metrics
3. Core Business Areas / Modules
4. Core Business Objects
5. Data Model and Relationships
6. User Roles and Permission Model
7. Key Workflows
8. AI Roles and Capabilities
9. Human Approval and Control Rules
10. Dashboards and Reporting
11. Technical Architecture
12. MVP Scope
13. Future Roadmap

Recommended Next Work Products

With the blueprint foundation complete, the next practical documents to create are:

1. Database Schema Specification

A technical table-by-table schema for the MVP.

2. MVP Product Requirements Document

A build-ready PRD that defines features, user stories, acceptance criteria, and priorities.

3. Wireframe / Screen Map

A screen-by-screen layout of the application.

4. AI Prompt and Workflow Specification

Detailed prompts, structured outputs, and AI action rules.

5. Development Roadmap

A sprint or milestone-based implementation plan.

6. Data Import Plan

A plan for bringing existing customers, vendors, products, inventory, documents, and tasks into the system.